

Hidden liquidity, market quality, and order submission strategies

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Abstract

Using exogenous changes in hidden liquidity from the Tick Size Pilot Program, we show that hidden liquidity on lit venues has significant effects on various measures of market quality and order submission strategies after controlling for the impact of dark pools and stock attributes. We find that spreads, depths, trading volume, and trade size increase with hidden liquidity. The extent of informed trading and the price impact of a trade also increase with hidden liquidity on lit venues. Traders use more inside-the-quote limit orders, while algorithmic traders engage in more pinging activities and raise message traffic with more hidden liquidity.

JEL classification: G14, G18, G28

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1. Introduction

We analyze the role of hidden liquidity in lit markets using data from the Tick Size Pilot Program. Pre-trade opacity is a standard feature in most financial markets. For instance, pre-trade opacity is an essential element of dark pools and hidden limit orders on lit exchanges. Regulators and stock exchanges have expressed concerns about the adverse impact of dark trading on market quality.¹ Numerous studies have examined the role of dark pools in price discovery and their impact on liquidity and pricing efficiency.² By contrast, there is only limited empirical evidence regarding the effects of hidden liquidity on lit exchanges. In the present study, we analyze the impact of hidden liquidity on various measures of market quality, informed trading, trading activities, and order submission strategies using exogenous changes in hidden liquidity on lit exchanges. We also examine how hidden orders affect algorithmic traders' pinging activities and message traffic. To our best knowledge, this paper is the first to provide evidence regarding the causal effect of hidden liquidity on these market metrics.

While transparency is considered a desirable attribute of a market, most stock markets grant traders an option to hide or display their orders on the book. As a result, market participants can observe only a portion of available liquidity and make trading decisions based on this incomplete information. While hidden liquidity adds uncertainties to liquidity demanders' trading environments, it provides liquidity suppliers with more flexibility in order placement decisions. The pre-trade opacity associated with hidden liquidity may increase both information-based trading and total trading volume if traders are willing to supply more liquidity when they are allowed to use hidden orders than when they are not.

¹ <https://www.financemagnates.com/institutional-forex/regulation/sec-to-tighten-regulation-for-dark-pools/>.

² See, for example, Nimalendran and Ray (2014), Comerton-Forde and Putniņš (2015), Kwan et al. (2015), Foley and Putniņš (2016), Buti et al. (2017), and Comerton-Forde et al. (2019).

During our sample period of October 2015 through October 2017, 13 national exchanges in the United States allow at least one type of hidden orders. Of these, ten exchanges report trades against hidden orders. These ten exchanges collectively account for more than 97% of the market share. Furthermore, nearly 20% of the shares traded on the ten exchanges are executed against hidden orders, underscoring the importance of hidden liquidity on lit venues.³ Since hidden orders lose priority to displayed orders with the same price and fail to elicit trading interest from reactive traders (Harris, 1996), they suffer from a lower probability of execution and a longer expected time to execution.

Bessembinder et al. (2009, p. 382) show that hidden orders are used “primarily by traders without superior information on future security price movements,” while Yao (2017) reports that traders who use hidden orders are well informed. Moinas (2010), Boulatov and George (2013), and Buti and Rindi (2013) provide analytical models in which they examine the effect of hidden and reserve orders on liquidity and other market quality measures. In particular, Boulatov and George (2013) show analytically that pricing efficiency improves when hidden orders are allowed because they induce more competition among informed traders. On the other hand, Bloomfield et al. (2015) and Gozluclu (2016) provide the experimental result that pricing efficiency does not change when hidden orders are introduced.

We exploit the Tick Size Pilot Program to examine the effect of hidden liquidity on our variables of interest. The U.S. Securities and Exchange Commission (SEC) started the pilot program in October 2016, which randomly assigned 2,400 stocks to one of four groups.⁴ Stocks in

³ Internationally, hidden liquidity accounts for 16% to 44% of trading volume (see Gozluclu, 2016). The remaining three exchanges not included in our hidden liquidity data source (SEC MIDAS) offer at least one type of hidden order.

⁴ Congress passed the Jumpstart Our Business Startups Act (JOBS Act) in 2012 to address “The U.S. Listing Gap,” a phenomenon that fewer U.S. companies are listed on stock exchanges than companies in other countries. The JOBS Act provided the option to amend Rule 612, which set the minimum price variation to one cent for stocks trading

Test Group 1 are subject to the quote rule, which increases the quote tick size to five cents. Stocks in Test Group 2 are subject to both the quote rule and the trade rule, where the latter increases the trade tick size to five cents. Stocks in Test Group 3 are subject to the trade-at rule in addition to the quote and trade rules. The trade-at rule prohibits a trading center from executing a trade at the National Best Bid and Offer (NBBO) without displaying the orders. Lastly, stocks in the control group are not subject to any tick size change. Each test group includes 400 stocks, and the control group consists of 1,200 stocks.

We use the trade-at rule, which discourages hidden orders, as an instrument to examine the causal effect of hidden liquidity. As expected, we find significant decreases in hidden liquidity for stocks subject to the trade-at rule. Using the predicted hidden liquidity calculated from the first-stage regression, we find that hidden liquidity increases the quoted spread, effective spread, realized spread, and depth after controlling for the effect of dark pools. The increase in the quoted spread is consistent with some traders at the NBBO before the pilot program switching to hide trading intentions. The increase in the effective spread is due to the increase in the quoted spread, rather than a decrease in price improvement.

Following Bennett et al. (2020) and other studies (e.g., Chen et al., 2007; Morck et al., 2013), we use the probability of informed trading (PIN) and stock price nonsynchronicity as measures of informed trading. We find that an increase in hidden liquidity causes an increase in informed trading after controlling for the effect of dark pools. We find similar results when we use variance ratios, order imbalance predictability, and autocorrelation as additional measures of informed trading. These results are consistent with the idea that hidden liquidity on lit exchanges

above \$1, in the hope that a larger tick size may improve trading and information environments for emerging growth companies. The SEC implemented the Tick Size Pilot Program to assess the impact of larger tick sizes on trading and information environments.

encourages informed traders to trade more (Boulatov and George, 2013). As a result of more informed trading, price impact increases, consistent with Chung et al. (2005). We show that an increase in hidden liquidity is associated with increases in dollar trading volume, trade size, pinging activities, and message traffic.

Our study contributes to the literature in several dimensions. Foley et al. (2013) investigate the effects of hidden liquidity on spreads, depth, and trading volume. Our study complements their study using the treatment group of 400 stocks included in the Tick Size Pilot Program, providing new evidence regarding the effects of hidden liquidity and dark pool trading on order aggressiveness, various measures of informed trading and price efficiency, trade size and frequency, pinging activities, and message traffic. Degryse et al. (2021) show that trading in dark pools and hidden order trading on lit venues increase with market volatility and decrease with algorithmic trading. They also show that smart order routing increases trading in dark pools but reduces hidden order trading on lit exchanges. While Degryse et al. (2021) focus on the factors that affect trading in dark pools and hidden order trading on lit markets, our study focuses on the impact of hidden order trading on liquidity, informed trading, and order submission strategies after controlling for the effects of dark pools.

Our study underscores differences between dark pools and hidden orders on lit exchanges. Dark pools are alternative trading systems (ATSs) that maintain pre-trade opacity by not publicly revealing their best quotes (see SEC, 2010). Dark pools offer “new trading systems with potentially faster execution, greater anonymity, lower costs of trading, and more liquidity” (Kwan et al., 2015, p. 330). Foley and Putniņš (2016) show that dark limit order markets improve liquidity and pricing efficiency, while dark midpoint crossing systems do not affect market quality using the introduction of minimum price improvement rules for dark trading in Canada and Australia.

Comerton-Forde et al. (2019) show that liquidity and pricing efficiency increase with the market share of dark pools using data from the Tick Size Pilot Program. The pre-trade opacity feature of dark pools resembles hidden orders on lit venues. However, dark pools are not necessarily a substitute for hidden orders because the sub-penny tick size of dark pools is an important feature that attracts investors and affects exchange market shares (Kwan et al., 2015). Changes in the market share of dark pools naturally alter the degree of market fragmentation, which is shown to affect market quality (Comerton-Forde and Putniņš, 2015; Comerton-Forde et al., 2019). Therefore, it is plausible that at least part of the results from dark pool studies is driven by the sub-penny tick size and market fragmentation and not entirely by differences in pre-trade transparency.

Furthermore, competition between dark pools and displayed limit orders on lit venues is likely different from competition between displayed and hidden limit orders on lit exchanges. Hence, the effects of these competitive forces on the market quality of lit venues are likely different.⁵ Indeed, we find that hidden liquidity increases the quoted, effective, and realized spreads, while Foley and Putniņš (2016) and Comerton-Forde et al. (2019) show that the opposite is true for dark pools. We also find that informed trading on lit exchanges decreases with the market share of dark pools, which is the opposite of the positive relation between informed trading and hidden liquidity on lit exchanges. Although both dark pools and hidden liquidity on lit venues likely facilitate informed trading, whether informed trading on lit exchanges decreases or increases with the market share of dark pools depends on trader migration from lit venues to dark pools. Our finding of the inverse relation between informed trading on lit exchanges and the dark pool market share suggests that the migration of informed institutional traders to dark pools leaves fewer

⁵ Buti et al. (2017) develop a model in which they show that both market and limit orders in the lit market migrate to the dark venue when a dark pool is introduced and the nature of this migration determines the effect of a dark pool on market quality. However, their model does not consider hidden orders in the lit market.

informed traders on lit venues.⁶ This is another reason why the effect of dark pools could be different from the effect of hidden liquidity on the liquidity of lit venues.⁷ Consequently, market regulators should not treat these non-transparent constituents of the financial market as a homogenous group in regulatory considerations.

Our study contributes to the literature on the effects of the Tick Size Pilot Program. Comerton-Forde et al. (2019) study changes in the market share of inverted venues after the program implementation and show that inverted venue activities improve liquidity and pricing efficiency. Chung et al. (2020) report that liquidity for small orders has decreased through larger quoted and effective spreads, while liquidity for large orders has increased through larger cumulative depths and the reduced price impact of multiple trades. Our study shows that hidden liquidity is a channel through which the Tick Size Pilot Program affects market liquidity and trading, contributing to the policy debate on whether changing tick sizes and implementing the trade-at rule help close the so-called U.S. listing gap.⁸

Finally, our finding that hidden liquidity increases pinging activities and message traffic from algorithmic traders has implications for the algorithmic and high-frequency trading literature. While algorithmic trading and high-frequency trading could be beneficial to market quality (see Brogaard et al., 2014), an increase in message traffic with the sole purpose of finding hidden liquidity may not necessarily be beneficial to traders in general. Also, the increase in message

⁶ Institutional investors benefit from dark pool trading because they can make large trades without exposure. The SEC has stepped up its scrutiny of dark pools over complaints that institutional traders capitalize on the uptick in share prices by front-running customer orders.

⁷ During the Tick Size Pilot Program, stocks that are subject to the trade-at rule are also subject to the quote and trade rules, eliminating the tick size differences between lit venues and dark pools. Nevertheless, there still exist differences between hidden and dark trading activities. For example, informed traders in general prefer lit venues as dark pools are typically associated with a higher risk of nonexecution (see Zhu, 2014). Therefore, empirical results from dark pools may not necessarily be solely driven by pre-trade opacity even during the Tick Size Pilot Program.

⁸ The Tick Size Pilot Program was implemented in part to evaluate whether larger tick sizes could close the “U.S. listing gap,” a phenomenon that proportionally fewer U.S. companies are listed on stock exchanges than companies in other countries, by improving trading and information environments for small companies.

traffic will make monitoring market activities more difficult for regulators (see Jones, 2013).

The rest of the paper is organized as follows. We develop the hypotheses in Section 2. In Section 3, we explain the datasets and variables. We present the empirical results in Section 4. Concluding remarks are in Section 5.

2. Hypotheses development

In this section, we develop the hypotheses on how the Tick Size Pilot Program affects hidden liquidity and how hidden liquidity can influence market quality, informed trading, trading volume, and order submission strategies.

2.1. *Hidden liquidity*

In the absence of the trade-at rule, hidden limit orders would be executed according to the price, display, and time precedence rules. Under the trade-at rule, a hidden limit order will not execute unless it improves the NBBO quote. Hence, all things being equal, traders are less likely to use hidden orders under the trade-at rule, which leads to our first hypothesis:

Hypothesis 1: *The trade-at rule decreases hidden liquidity.*

2.2. *The quoted spread, effective spread, and quoted depth*

To establish the expected relation between hidden liquidity and the quoted spread, we consider two markets where one market allows hidden orders while the other does not. We conjecture that the allowance of hidden orders would increase the quoted spread to the extent that some traders switch to using hidden orders to minimize the probability of their orders being undercut by others (Buti and Rindi, 2013). For instance, suppose traders submitted limit buy orders

at \$20.15, \$20.10, and \$20.05 and limit sell orders at \$20.20, \$20.25, and \$20.30 with the bid-ask spread of \$0.05 ($= \$20.20 - \20.15) when hidden orders were not allowed. If hidden orders are permitted, and traders who submitted limit buy orders at \$20.15 and limit sell orders at \$20.20 switch to using hidden orders, the bid-ask spread would, all things being equal, increase to \$0.15 ($= \$20.25 - \20.10). To the extent that hidden liquidity increases the price impact of trades (see below), it could also increase the quoted spread by raising the adverse selection component of the spread.⁹ These considerations lead to our second hypothesis:

Hypothesis 2: *The quoted spread increases with hidden liquidity.*

The impact of hidden liquidity on the effective spread is less straightforward than its impact on the quoted spread. The effective spread is the difference between the quoted spread and price improvement. Thus, the impact of hidden liquidity on the effective spread depends on how it affects the quoted spread and price improvement. Hidden liquidity can influence price improvement in two ways. Hidden liquidity will likely increase price improvements if more undisplayed limit orders are ready to be executed at better prices than the NBBO. In this case, an increase in hidden liquidity could decrease the effective spread, although the rise in hidden liquidity increases the quoted spread.¹⁰ Alternatively, hidden liquidity may reduce price

⁹ We use the term “the quoted spread” to indicate the displayed spread. Prior research suggests that the quoted spread reflects the inventory, order processing, and adverse selection costs.

¹⁰ The effective spread can increase even when price improvements increase, provided that the increase in the quoted spread is large enough. For example, suppose there are two trading venues called X and Y, both without hidden liquidity. The current national best ask price is \$20.30 and the national best bid price is \$20.20, hence a quote midpoint of \$20.25, both quoted from venue X. Venue Y displays inferior quotes where the best ask is \$20.35 and the best bid is \$20.15. A market buy order will hit the best ask price at \$20.30, resulting in an effective spread of $2 \times (\$20.30 - \$20.25) = \$0.10$. Now, suppose hidden liquidity is allowed on both venues, and the trader who was quoting the best prices at venue X decides to hide all of his or her orders while keeping the same prices. Then, the (displayed) national best ask price is now \$20.35 and the best bid price is \$20.15, quoted through venue Y (assuming there was no other displayed quote from venue X). Suppose a trader decides to submit a hidden limit sell order at \$20.34 to venue Y, to undercut the widened (displayed) NBBO by a cent. Now, we see that the midpoint has not changed (whether we use the displayed or hidden best quotes), and the true NBBO has not changed either. However, a market buy order

improvements through its impact on order size. For example, suppose an increase in hidden liquidity leads to an increase in order size. In that case, more orders need to walk up (or down) the limit order book and be executed at less favorable prices than the NBBO, increasing the effective spread. Therefore, the impact of hidden liquidity on the effective spread depends on the extent to which hidden liquidity affects the quoted spread, price improvement, and order size.

The impact of hidden orders on the quoted depth depends on two competing forces. Insofar as the quoted spread increases with hidden liquidity, hidden liquidity would also increase the quoted depth because liquidity providers have a stronger incentive to quote large depths when quoted spreads are wider. On the other hand, if traders who submitted limit orders at or near the NBBO switch to using hidden orders as they become permissible, the quoted (displayed) depth at the NBBO may decrease. Hence, whether hidden orders increase or decrease the quoted depth at the NBBO depends on the relative strength of the two forces.

2.3. Informed trading, the price impact of trade, and the realized spread

Prior research is not unanimous on whether informed traders use hidden orders. Bessembinder et al. (2009) report evidence that hidden orders are primarily used by uninformed traders from a sample of Euronext-Paris stocks. In contrast, using Schedule 13D filings by hedge funds as informative corporate events, Yao (2017) shows that executed hidden order imbalances for Nasdaq stocks increase before the events, suggesting that informed traders use hidden orders. Boulatov and George (2013) show analytically that hidden orders improve price efficiency by increasing competition among informed traders. We conjecture that informed traders have greater

submitted to venue Y will be executed at \$20.34, resulting in an effective spread of $2 \times (\$20.34 - \$20.25) = \$0.18$. Note that price improvement increased from \$0 to \$0.02. Therefore, while the price improvement has increased from \$0 to \$0.02, the effective spread has increased as well since the displayed NBBO has changed, even though the true NBBO (including the hidden NBBO) has not changed.

incentives to hide their trading interests than liquidity traders. Informed investors are more likely to trade when allowed to hide their orders because it helps them take advantage of their private information more effectively. Hence, the trade-at rule, which discourages hidden orders, is likely to reduce informed trading, suggesting a positive relation between hidden liquidity and informed trading. To the extent that the price impact of trades is positively related to informed trading, hidden liquidity is also likely to increase the price impact of trades. Based on these considerations, we have the following hypothesis:

Hypothesis 3: *The extent of informed trading and the price impact of trades increase with hidden liquidity.*

The realized spread is the difference between the effective spread and the price impact of trades. As discussed above, the effective spread may decrease or increase with hidden liquidity. Therefore, even if the price impact of trades increases with hidden liquidity, the relation between the realized spread and hidden liquidity could not be predicted unconditionally.

2.4. Trading volume, trade size, and the number of trades

Moinas (2010) shows that liquidity suppliers use larger limit orders when hidden orders are allowed than when they are not. All things being equal, liquidity demanders are likely to use larger market orders when they expect liquidity providers to use larger limit orders (i.e., when there is more available liquidity) because larger limit orders imply lower execution costs.¹¹ These considerations lead to our next hypothesis:

Hypothesis 4: *Trade size increases with hidden liquidity.*

¹¹ Large limit orders reduce the execution cost of market orders because large limit orders reduce the probability that market orders execute at inferior prices (i.e., walk up (for buys) or down (sell) the limit order book).

Prior research does not establish a clear consensus on the effect of hidden liquidity on trading volume. Bloomfield et al. (2015) show no difference in trading volume regardless of whether a market allows hidden orders. Buti and Rindi (2013) show that trading volume decreases when reserve orders (i.e., partially hidden orders) are permitted. On the other hand, Moinas (2010) argues that hidden orders enable liquidity suppliers to submit large orders, which increases trading volume. Therefore, the relation between trading volume and hidden liquidity is likely to depend on which of the economic forces described in the two theoretical results is stronger. If both trading volume and trade size are positively related to hidden liquidity, the effect of hidden liquidity on the number of trades would depend on which of the two effects is stronger. If trading volume decreases while trade size increases with hidden liquidity, the number of trades would also decrease with hidden liquidity. Therefore, the effect of hidden liquidity on the number of trades depends on the direction and magnitude of the change in trading volume.

2.5. Order and trade aggressiveness

Hidden liquidity is likely to affect the trading strategies of investors. Bloomfield et al. (2015) and Gozluklu (2016) show that informed investors trade less aggressively (e.g., less use of market orders and marketable limit orders) when hidden orders are allowed because they could avoid information leakage by using hidden orders. In particular, informed traders may use more inside-the-quote limit orders and fewer market orders/marketable limit orders when hidden orders are allowed because it enables them to exploit their informational advantage by trading ahead of orders at the NBBO (i.e., high execution probability) but reduces implementation shortfall by submitting less aggressive orders (i.e., inside-the-quote limit orders) than market orders and

marketable limit orders.¹² In contrast, uninformed investors trade more aggressively because they can no longer patiently observe the limit order book and infer private information. Uninformed traders may submit more inside-the-quote limit orders and fewer at-the-quote orders and behind-the-quote orders because it increases execution probability without excessively increasing implementation shortfall by using less aggressive orders than market orders or marketable limit orders. These considerations lead to our next hypothesis:

Hypothesis 5: *An increase in hidden orders is associated with an increase in trades using inside-the-quote limit orders and a decrease in trades using market orders, marketable limit orders, at-the-quote limit orders, and behind-the-quote limit orders.*

2.6. Order cancellation, odd-lot trades, and order-to-trade ratio

In response to an increase in hidden liquidity, some algorithmic traders are likely to engage in more pinging activities and increase message traffic (Bloomfield et al., 2015). Because hidden orders are not visible, these algorithmic traders would send small orders quickly and frequently to discover any trading interest that has not been displayed in the limit order book and cancel the orders immediately if not executed to avoid being picked off by another trader. These considerations lead to the following hypothesis:

Hypothesis 6: *An increase in hidden liquidity leads to increases in cancellations of orders, odd-lot trades (those with a trade size of fewer than 100 shares), and order-to-trade ratio due to algorithmic traders' increasing message traffic to discover hidden liquidity.*

¹² Implementation shortfall is the difference between the actual transaction price and the midpoint of the bid and ask quotes at the time of trading decision.

3. Data sources and variable construction

3.1. Study sample and data sources

Our study sample consists of 2,395 stocks included in the Tick Size Pilot Program. The pilot program followed a staged rollout. The first batch of stocks in the program was assigned to their corresponding pilot program group on October 3, 2016, the subsequent batches on October 10, October 17, October 24, and the remaining stocks on October 31, 2016. Therefore, we use October 1, 2015 through September 23, 2016 as the pre-event period and October 31, 2016 through October 31, 2017 as the post-event period. For our month-level data, we drop October 2016 and use October 2015 through September 2016 as the pre-event period and November 2016 through October 2017 as the post-event period. We obtain the list of stocks and their corresponding group in the pilot program from FINRA's website: <http://www.finra.org/industry/tick-size-pilot-program>.

We apply the same screening process as in Chung et al. (2020). Starting with 2,395 stocks in the pilot program, we drop 12 test stocks created for the program, 63 stocks that changed their pilot program group or primary listing during the program, 159 stocks added after the program started or deleted before the end of our sample period, stock-day observations with less than 20 transactions, and 167 stocks that recorded less than 20 transactions for more than half of their daily observations, and 291 stocks that do not have a share code of 10 or 11 in the Center for Research in Security Prices (CRSP) database. A total of 1,703 stocks remain in our study sample after applying these filters. We also follow Rindi and Werner (2019) and drop the following stock-day observations: Stock symbols NRCIA and PBIP on November 28, 2016; HUBS on August 4, 2016; VCRA on October 28, 2016; and DTLK on December 19 to 21, 2016.

3.2. Variable measurement

3.2.1. Hidden liquidity

Market orders and marketable limit orders are more likely to execute against hidden limit orders when there are more competitive hidden limit orders (i.e., hidden liquidity) than displayed limit orders. SEC MIDAS distributes stock-day data on the number of transactions and shares traded against hidden orders and the total number of transactions and shares traded on the ten exchanges that report trades against hidden orders.¹³ With these four variables, we compute the percentage of transactions (*Hidden_PT*) and shares (*Hidden_PS*) traded against hidden orders and use them as our proxies for hidden liquidity.¹⁴ We use the trade-based hidden liquidity measure because it reflects only relevant (i.e., competitive) hidden limit orders.

As long as the percentage of market orders executed against hidden limit orders increases with the percentage of hidden limit orders, our proxies for hidden liquidity are likely to be reasonable measures of hidden liquidity. To see this point, we define Z_1 as the percentage of hidden limit orders before the trade-at rule (i.e., the ratio of the number of hidden limit orders to the number of hidden and displayed limit orders) and Z_2 as the corresponding figure after the implementation of the trade-at rule, where $Z_1 > Z_2$ (our Hypothesis 1). Likewise, E_1 denotes the percentage of market orders executed against hidden limit orders before the trade-at rule, and E_2 denotes the corresponding figure after its implementation. Now suppose that market orders are more likely to execute against hidden limit orders (than with displayed limit orders) when there are more hidden limit orders. For simplicity, let's assume that $E_1 = \alpha_1 + \beta_1 Z_1$ and $E_2 = \alpha_2 + \beta_2 Z_2$,

¹³ During our sample period, 10 out of the 13 U.S. national securities exchanges report trades against hidden orders. These ten exchanges collectively account for more than 97% of the lit venue market share. SEC MIDAS data can be retrieved from the SEC website: <https://www.sec.gov/marketstructure>.

¹⁴ Cox et al. (2019) also use the same variables constructed from SEC MIDAS to study the changes in hidden liquidity around the Tick Size Pilot Program.

where $\beta_1 > 0$ and $\beta_2 > 0$. Then, $E_2 - E_1 = (\alpha_2 - \alpha_1) + (\beta_2 Z_2 - \beta_1 Z_1)$. If the relation between hidden order execution and hidden liquidity is similar before and after the implementation of the trade-at rule (i.e., $\beta_2 \approx \beta_1 \equiv \beta$), it follows that $E_2 - E_1 = \alpha + \beta(Z_2 - Z_1)$, where $\alpha = \alpha_2 - \alpha_1$. The last equation shows that we can measure the effect of the trade-at rule on hidden liquidity (i.e., $Z_2 - Z_1$) by its effect on hidden liquidity execution (i.e., $E_2 - E_1$) because the latter is a linear transformation of the former.

3.2.2. Spreads, depth, and price impact

We use the Daily Trade and Quote (TAQ) data to estimate spreads, depth, and price impact. The quoted spread at time t ($QuotedSpread_t$) is defined as the national best ask minus the national best bid, scaled by the quote midpoint at time t . We compute $QuotedSpread_t$ at each time t when the NBBO updates, then estimate the time-weighted mean $QuotedSpread_d$ for each stock on day d . Similarly, we calculate the daily depth for each stock in thousand U.S. dollars ($Depth_d$) as the time-weighted mean quoted depth at the NBBO at each NBBO update.

We calculate the effective spread for trade m on day d as follows:

$$EffectiveSpread_{m,d} = \frac{2BuySell_m \times (Price_m - Midpoint_t)}{Midpoint_t}, \quad (1)$$

where $BuySell_m$ equals 1 (-1) if the trade is buyer-initiated (seller-initiated), $Price_m$ is the transaction price, and $Midpoint_t$ is the quote midpoint at time t when the trade occurred. We classify trades into buyer- or seller-initiated according to the method in Ellis et al. (2000). For each trade m , $EffectiveSpread_{m,d}$ is multiplied by the number of shares traded and divided by the total number of shares traded on day d for each stock, then summed over for all trades on day d .

We calculate the realized spread for trade m on day d as follows:

$$RealizedSpread_{m,d} = \frac{2BuySell_m \times (Price_m - Midpoint_{t+\tau})}{Midpoint_{t+\tau}}; \quad (2)$$

where τ is the time horizon in which we measure the future quote midpoint. In this paper, we set τ equal to 60 seconds. $RealizedSpread_d$ is the share-weighted mean of all $RealizedSpread_{m,d}$ on day d for each stock.

We calculate the price impact of trade m on day d as follows:

$$PriceImpact_{m,d} = \frac{2BuySell_m \times (Midpoint_{t+\tau} - Midpoint_t)}{Midpoint_t}. \quad (3)$$

Again, we set τ equal to 60 seconds. We then use the share-weighted $PriceImpact_d$ for day d for each stock in the regressions.

Market centers in the U.S. are required to file a monthly report on execution quality under SEC Rule 605. We downloaded the Rule 605 reports from 11 lit venues (excluding IEX and NSX) from the exchange websites and the Transaction Auditing Group.¹⁵ Rule 605 data include the number of price-improved shares, the number of at-the-quote shares, and the number of outside-the-quote shares. We compute the percentage of price-improved shares and call it *PriceImprovement*.

3.2.3. Informed trading

We use various metrics as our empirical proxies for the extent of informed trading. The first proxy is the probability of informed trading (*PIN*) based on the Easley et al. (1996) model. We estimate the model parameters monthly using the daily number of buyer- and seller-initiated

¹⁵ Reports from the Investors Exchange (IEX) are dropped due to the venue's transition from an ATS to a national securities exchange in September 2016. Reports from the NYSE National (NSX) are dropped due to the venue's acquisition and subsequent cessation of operation in February 2017.

trades obtained from the Wharton Data Research Services (WRDS).¹⁶ We then obtain PIN_m for month m for each stock using the following formula:

$$PIN_m = \frac{\alpha\mu}{\alpha\mu + 2\varepsilon}, \quad (4)$$

where α is the probability that an information event has occurred, μ is the arrival rate of informed traders given that an event has occurred, and ε is the arrival rate of uninformed traders.

Our second measure of informed trading is PSI_{FF3} , motivated by Chen et al. (2007), Morck et al. (2013), and Chung et al. (2020). To compute this measure, we regress daily excess stock returns obtained from the CRSP database on the Fama and French (1992) three factors, grouped by quarter.¹⁷ PSI_{FF3} is the difference between one and the R^2 value from the regression. We calculate the third measure (PSI_{BSW}) similarly. However, instead of using the Fama and French (1992) factors, we regress daily stock return on the value-weighted industry (based on three-digit SIC) and market returns following Bennett et al. (2020), using data obtained from the CRSP. Larger values of PSI_{FF3} and PSI_{BSW} indicate more firm-specific information impounded into prices, indicating a greater extent of informed trading. Following Bennett et al. (2020) and Chung et al. (2020), we logistically transform PSI_{FF3} and PSI_{BSW} in the regression analysis, where the logistically transformed value of X is equal to the natural logarithm of $X / (1 - X)$.

We also use the following three variables as additional empirical proxies for informed trading: variance ratios (VR), return predictability of order imbalance ($OIB_{Predictability}$), and return autocorrelation ($Autocorrelation$). Each of these variables captures non-randomness in price

¹⁶ The PIN data we use are computed by applying the trade classification algorithm proposed by Lee and Ready (1991).

¹⁷ The Fama and French (1992) three factors include the equity premium, the difference between small- and large-cap stocks, and the difference between the high and low book-to-market ratio stocks. We obtain the factor data from Kenneth French's website: <http://mba.tuck.dartmouth.edu/pages/faculty/ken.french>.

changes and returns. In an efficient market, price changes follow a random walk because prices move in response to the arrival of new information, which is, by definition, random. An implicit assumption in this line of thought is that informed traders do not trade strategically. However, prior research shows that informed traders submit orders strategically. For instance, they submit many small child orders to minimize the price impact of their trades. High-frequency traders spend their time and resources to detect a large underlying parent order based on observed small child orders and use the information in their trading. Hence, price changes and returns are unlikely to follow a random walk at the time of informed trading and in a stock with a high degree of informed trading.¹⁸ We expect a high degree of informed trading in stocks with large *VR*, *OIB_Predictability*, and *Autocorrelation* based on these considerations.

Variance ratio (*VR*) is the absolute value of one minus the ratio of midpoint variance at the one-minute interval to four times that of the 15-second interval. We expect the ratio of the midpoint variances at different time intervals to be close to one (i.e., *VR* close to 0) for a stock that follows a random walk (see Comerton-Forde et al., 2019). We measure the return predictability of order imbalance (*OIB_Predictability*) by the R^2 value of the regression of one-minute stock return at interval t on the order imbalance at interval $t - 1$. Order imbalance is the difference in dollar trading volume between buyer- and seller-initiated trades (see Chordia et al., 2005; Röscher et al., 2017; Chung et al., 2020). A higher value of *OIB_Predictability* is interpreted as a high degree of informed trading. We measure return autocorrelation (*Autocorrelation*) by the R^2 value of the regression of quote-midpoint return during minute t on quote-midpoint return during minute $t - 1$. We use the logistically transformed values of *OIB_Predictability* and *Autocorrelation*.

¹⁸ This prediction would hold even if some uninformed traders also trade strategically, as long as informed traders do more strategic trading (e.g., order splitting) than uninformed traders.

3.2.4. Trading activity and order submission strategies

We calculate the daily dollar trading volume of each stock in million U.S. dollars (*DollarVolume_d*). We also calculate the daily average trade size in dollars (*TradeSize_d*) and the daily number of trades (*Num_Trades_d*). Rule 605 data include stock-month data on the number of shares executed by order types. We calculate the proportion of shares executed using market orders (*OT_MO*), marketable limit orders (*OT_MLO*), inside-the-quote limit orders (*OT_IQL*), at-the-quote limit orders (*OT_AQL*), and near-the-quote limit orders (*OT_NQL*), respectively, where near-the-quote limit orders are limit orders placed within 10 cents outside the NBBO. For example, when the national best bid price is \$3.25 and the best ask price is \$3.30, a limit buy order submitted at \$3.26 (\$3.20) is classified as *OT_IQL* (*OT_NQL*).

3.2.5. Strategies of algorithmic traders

As hidden liquidity increases, we expect increases in pinging activities and message traffic. Our empirical proxies for pinging activities are the cancellation ratio (*Cancellations*) and the odd-lot volume ratio (*OddLot*). We obtain the number of cancellation messages (full or partial), the number of trades executed against displayed orders, the sum of odd-lot volumes (in shares), and the sum of trade volume from exchanges that report odd-lot volumes from SEC MIDAS. Then, *Cancellations* is the ratio of the number of cancellation messages to the number of trades executed against displayed orders. *OddLot* is the ratio of the odd-lot trading volume to the total trade volume. To examine the change in message traffic, we also obtain the order volume and the number of shares executed against displayed orders from SEC MIDAS and scale the order volume data by the number of executed shares to create the order-to-trade ratio variable (*OTR*).

The cancellation data from SEC MIDAS do not account for message traffics generated by

immediate-or-cancel orders widely used by algorithmic traders. Unless changes in hidden liquidity mechanically affect non-algorithmic traders' cancellation activities or algorithmic traders' motivation to use immediate-or-cancel orders instead of other order types, it is reasonable to assume that changes in *Cancellations* and *OTR* capture changes in algorithmic traders' pinging and trading activities even with the absence of immediate-or-cancel order data. Moreover, results based on *OddLot* are not affected by this consideration.

3.2.6. Control variables

Our control variables include the daily CBOE Volatility Index (*VIX*) from the CBOE website, the inverse of daily closing price ($1 / Price$) from the CRSP, log of the market capitalization of each stock as of the previous year-end (*MarketCap*) from the CRSP, log of the daily trading volume in million U.S. dollars (*DollarVolume*) from the Daily TAQ database, and daily return volatility (*Volatility*) measured by the square root of the sum of squared one-minute quote midpoint returns from the Daily TAQ database.¹⁹ Foley and Putniņš (2016) show that a type of dark pool dubbed “dark limit order markets” improves liquidity and pricing efficiency. Comerton-Forde et al. (2019) show that the trade-at rule in the Tick Size Pilot Program has affected the market shares of inverted venues and dark pools.²⁰ We calculate the market share of inverted venues (*Inverted_Share*) and the market share of dark pools (*Dark_Share*) using the daily dollar trading volume for each stock.²¹

¹⁹ To reduce the impact of extreme values, we winsorize *QuotedSpread* at the 99th percentile values and *EffectiveSpread*, *RealizedSpread*, *Volatility*, *PriceImpact*, and *Autocorrelation* at the 1st and 99th percentile values.

²⁰ Inverted venues are exchanges that engage in the practice of taker-maker pricing, where the liquidity taker receives rebates and the liquidity supplier pays fees to the exchange in a transaction. This is in contrast to a more traditional maker-taker pricing model where the liquidity supplier receives rebates and the liquidity taker pays fees (see Comerton-Forde et al., 2019). Dark pools are off-exchange venues that do not publicly post their best quotes (see SEC, 2010).

²¹ Inverted venues in our sample include Bats BYX, Bats EDGA, and NASDAQ OMX BX.

3.2.7. Descriptive statistics

Table 1 provides descriptive statistics of the variables for each pilot program group. We obtain the stock-level mean for each variable before calculating the descriptive statistics. Columns (1) - (3) report the statistics for stocks in the control group, columns (4) - (6) report the statistics for stocks in Test Group 1, columns (7) - (9) report the statistics for stocks in Test Group 2, and columns (10) - (12) report the statistics for stocks in Test Group 3. We further separate each group by whether the date lies before or after the pilot program had started.

The results show that the mean values of *Hidden_PT* and *Hidden_PS* are around 21% before the pilot program started (see columns (1), (4), (7), and (10)). For the control group, *Hidden_PT* (*Hidden_PS*) changes from 20.94% (22.29%) to 21.10% (21.36%) after the program started. On the other hand, all three test groups exhibit significant reductions in hidden liquidity. *Hidden_PT* (*Hidden_PS*) for stocks in Test Group 1 decreased, on average, from 21.10% (22.44%) to 15.76% (16.41%), and the reduction is significant at the 1% level. While a formal test result is presented later, these figures suggest that the quote rule significantly reduced hidden liquidity. Although hidden liquidity decreased for stocks in Test Group 2, its magnitude in column (9) is comparable to that of Test Group 1 in column (6), suggesting that the trade rule may not have a material effect on hidden liquidity. On the other hand, hidden liquidity for Test Group 3 dropped to almost half (9.7 to 10.3 percentage points, column (12)) after the pilot, suggesting that the trade-at rule caused hidden liquidity to decrease even further.

Table 1 also shows that the pilot program is accompanied by significant changes in liquidity and information variables, trading activities, and order submission strategies for both the test and control groups. For instance, we find an increase in *Depth*, *PSI_FF3*, *PSI_BSW*, *OT_MLO*,

OddLot, and *Inverted_Share*, and a decrease in *VR*, *Cancellations*, *OTR*, and *Volatility* for both groups. Interestingly, we find opposite changes in *EffectiveSpread* between the control and test groups. For these reasons, we employ a difference-in-differences analysis to accurately measure the impact of the pilot program (the trade-at rule in particular) on hidden liquidity and examine the effect of hidden liquidity on liquidity, informed trading, and order submission strategies after controlling for market-wide changes in these variables.

4. Empirical findings

We employ the two-stage least squares (2SLS) method using the trade-at rule in the Tick Size Pilot Program as an instrument to explore causal relations between hidden liquidity and our variables of interest.²² Although the quote and trade rules may also affect hidden liquidity, we use them as explanatory variables (rather than instruments). These rules may affect our dependent variables directly, not just through hidden liquidity. Similar to Comerton-Forde et al. (2019), we also use the lagged value of hidden liquidity as an additional instrument. Based on these considerations, we estimate the following first-stage regression model:²³

$$\begin{aligned} Hidden_PT_{i,t} = & \beta_1 Q_i \times Event_t + \beta_2 T_i \times Event_t + \beta_3 TA_i \times Event_t \\ & + \beta_4 Hidden_PT_{i,t-1} + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t \\ & + Control_Variables + \varepsilon_{i,t}, \end{aligned} \quad (5)$$

where $Hidden_PT_{i,t}$ is the percentage of transactions against hidden orders for stock i on day t , $Event_t$ equals one for the post-event period and zero for the pre-event period, Q_i is the quote rule

²² Although Test Group 3 in the pilot program is the only group subject to the trade-at rule, it is also subject to the quote and trade rules during the pilot program. Therefore, we cannot simply assess the changes in Test Group 3 to examine the effect of the trade-at rule on hidden liquidity, as the quote and trade rules may also affect hidden liquidity.

²³ Since the results for $Hidden_PS$ and $Hidden_PT$ are qualitatively similar, we only report the results for $Hidden_PT$. The results for $Hidden_PS$ are available from the authors upon request.

dummy variable (which equals one for stocks quoted in \$0.05 increments and zero otherwise), T_i is the trade rule dummy variable (which equals one for stocks traded in \$0.05 increments and zero otherwise), and TA_i is the trade-at rule dummy variable (which equals one for stocks subject to the trade-at requirement and zero otherwise).²⁴

To compare the effect of hidden liquidity with that of dark pools, we also estimate the following first-stage regression model:

$$\begin{aligned} Dark_Share_{i,t} = & \beta_1 Q_i \times Event_t + \beta_2 T_i \times Event_t + \beta_3 TA_i \times Event_t + \\ & \beta_4 Dark_Share_{i,t-1} + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}, \end{aligned} \quad (6)$$

where $Dark_Share_{i,t}$ is the percentage market share of dark pools for stock i on day t , and all other variables are the same as defined in regression model (5).²⁵ As in Comerton-Forde et al. (2019), we include the lagged value of the market share of dark pools as an additional instrument.

We then estimate the following second-stage regression model using the predicted values of $Hidden_PT_{i,t}$ and $Dark_Share_{i,t}$ (denoted as $\widehat{Hidden_PT}_{i,t}$ and $\widehat{Dark_Share}_{i,t}$) based on the estimated coefficients in the first-stage regression models to examine the impact of hidden liquidity and dark pools on our variables of interest:

$$\begin{aligned} Var_{i,t} = & \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \\ & \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}, \end{aligned} \quad (7)$$

where Var is one of our dependent variables, and all other variables are the same as defined for

²⁴ Comerton-Forde et al. (2019) and Chung et al. (2020) also use the same dummy variables to show the impact of each rule in the pilot program. In an untabulated analysis, we estimate our regression models using only Test Group 2 and Test Group 3 stocks. Under this setting, the only difference between the new control group (Test Group 2) and the treatment group (Test Group 3) is that the treatment group is subject to the trade-at rule under the pilot program. The results are qualitatively similar to those using the three rule dummy variables.

²⁵ Dippel et al. (2022) show that the same instrumental variable can be used for two endogenous variables under certain conditions.

regression models (5) and (6). The coefficients on $\widehat{Hidden_PT}$ capture whether the trade-at rule influences liquidity, informed trading, and order placement strategies through its effect on hidden liquidity after controlling for its effects on *Dark_Share*. We include the market share of inverted venues (*Inverted_Share*) as a control variable in regression models (5), (6), and (7).²⁶ We run regressions with and without the stock and date fixed effects. Standard errors are clustered by firm and date unless otherwise noted.

4.1. First-stage regression results

We provide the first-stage regression results in Table 2.²⁷ Columns (1) and (2) show the results of regression model (5) when we do not include *QuotedSpread* as a control variable, while columns (3) and (4) show the results when we include it.²⁸ Columns (1) and (3) show the results when we do not include the stock and date fixed effects, while columns (2) and (4) show the results when we include them. Regardless of regression specification, the coefficient on *TA x Event* is negative and significant, indicating that the trade-at rule decreases hidden liquidity, consistent with Hypothesis 1. This result satisfies the relevance condition of the trade-at rule as an instrumental variable.²⁹ Table 2 shows that the level of hidden liquidity on day t is also significantly related to the level of hidden liquidity on day $t - 1$, satisfying the relevance condition of the lagged hidden

²⁶ Similar to Comerton-Forde et al. (2019), we also estimate regression model (6) with the market share of inverted venue as the dependent variable and include the predicted value of the market share of inverted venue as an independent variable in regression model (7). The results are qualitatively similar to our main results.

²⁷ We omit the results for the control variables in Table 2. We provide the full regression results in Table A1 in the [Online Appendix](#).

²⁸ We show the results with and without *QuotedSpread* as a control variable because we include *QuotedSpread* in the second-stage regression models for certain variables (i.e., *PriceImprovement*, *PIN*, *PSI_FF3*, *PSI_BSW*, *VR*, *OIB_Predictability*, *Autocorrelation*, *DollarVolume*, *TradeSize*, *Num_Trades*, *OT_MO*, *OT_MLO*, *OT_IQL*, *OT_AQL*, *OT_NQL*, *Cancellations*, *OddLot*, and *OTR*), and exclude it in the second-stage regression models for other variables (i.e., *QuotedSpread*, *EffectiveSpread*, and *RealizedSpread*).

²⁹ Roberts and Whited (2013) caution that small R^2 , sample size, or F -statistic in the first-stage regression may be a sign of weak instruments. Table 2 shows that our R^2 values range from 53% to 66%, sample sizes are over 800,000, and F -statistics are well over the critical values, suggesting that our instruments are not weak.

liquidity as an instrumental variable.³⁰

The quote rule applies to both displayed and hidden orders. Since informed traders are generally less likely to be front-run by other traders when the quote tick size is larger, we expect the quote rule to motivate informed traders to display their orders more often. The negative and significant coefficient on $Q \times Event$ indicates that the quote rule decreases hidden liquidity, which is consistent with the finding of Cox et al. (2019).

Columns (5) and (6) of Table 2 show the results of regression model (6) without and with the stock and date fixed effects, respectively. The results indicate that the quote rule increases the market share of dark pools, consistent with traders attempting to bypass the increased tick size by migrating to dark pools (Cox et al., 2019). On the other hand, the trade-at rule decreases the market share of dark pools because the rule reduces the incentive to use undisplayed orders. These results are consistent with the findings of Comerton-Forde et al. (2019).

As noted in Subsection 2.1, one way to continue trading using hidden orders under the trade-at rule is to offer a price improvement (e.g., submit a hidden limit buy order at one tick above the national best bid quote). If the required price improvement is larger because of a larger tick size, traders have less incentive to use such (aggressive) hidden orders. In the Tick Size Pilot Program, stocks subject to the trade-at rule are also subject to the quote and trade rules. Hence, the effect of the trade-at rule on hidden orders documented in Table 2 represents the effect of the trade-at rule on hidden orders for those stocks that require at least a five-cent price improvement to trade ahead of orders at the NBBO, which would be larger than the corresponding effect for those stocks that need only a one-cent price improvement.

To provide some insight on whether the validity of Hypothesis 1 depends on the difference

³⁰ Unlike the relevance condition, the exclusion condition cannot be tested because the regression error term is unobservable.

in price improvement requirements, we estimate for each stock the extent to which the five-cent tick size is a binding constraint (*TBC*) on spreads following Chung et al. (2020). Specifically, we calculate the proportion of one-second intervals during trading hours in the pre-pilot period that the quoted spread is less than or equal to five cents. A higher *TBC* indicates that the new tick size of five cents during the pilot program is more likely to be a binding constraint, therefore providing a price improvement of five cents would be more costly. Hence, traders are more likely to be discouraged from using hidden orders for stocks with a higher *TBC*. We then estimate regression model (5) with subsamples of stocks with a lower-than-median *TBC* and those with a higher-than-median *TBC* and present the results in Table A2 in the [Online Appendix](#). We find that the coefficient on $TA \times Event$ is negative and significant (i.e., traders have less incentive to use hidden orders) for both groups regardless of their *TBC* level, indicating that Hypothesis 1 still holds. Therefore, we expect Hypothesis 1 to hold even when the required price improvement is less than five cents.

Table A2 in the Online Appendix also shows that the coefficient on $Q \times Event$ is negative and significant for both *TBC* groups when *QuotedSpread* is included in the regression model (see columns (3), (4), (7), and (8)). However, the magnitude of the coefficient is much larger for stocks with a low *TBC*. For example, the coefficient on $Q \times Event$ is -0.0780 in column (3) for the low *TBC* group, compared to -0.0211 in column (7) for the high *TBC* group. These results suggest that stocks with a below-median *TBC* experience a further decline in hidden liquidity when the quote tick size increases.

4.2. Effects on liquidity

Table 3 shows the results of the second-stage regression model (7) for spreads and depths.³¹ In columns (1) and (2) of Panel A, the coefficients on $\widehat{Hidden_PT}$ are positive and significant, indicating that hidden liquidity increases the quoted spread, consistent with Hypothesis 2. The results are robust to including the stock and date fixed effects. By contrast, we find that the coefficients on $\widehat{Dark_Share}$ in columns (1) and (2) are negative and significant, indicating that the quoted spread decreases with the market share of dark pools, which is consistent with the findings of Buti et al. (2011), Foley and Putniņš (2016), and Comerton-Forde et al. (2019).

Possible explanations for the negative relation between the quoted spread and the market share of dark pools are as follows. Although dark pools divert liquidity away from lit venues, they could also create more trading opportunities by attracting new order flow to the market. Order migration from lit venues to dark pools decreases liquidity on lit venues. However, the new trading opportunities created by dark pools imply greater competition between lit venues and dark pools, forcing lit venues to provide better liquidity. These considerations suggest that the effect of dark pools on the liquidity of lit venues depends on the relative strength of the two opposing forces. Our results indicate that the effect of the competition is greater than the effect of order migration.

Furthermore, lit venues are used by both retail and institutional investors, while dark platforms are primarily used by institutional traders.³² To the extent that institutional investors are

³¹ We omit the results for the control variables in Tables 3 - 8. We provide the full regression results in Tables A3 - A8 in the [Online Appendix](#). We also report uninstrumented OLS results in Tables B1 - B6 in the [Online Appendix](#). Most of our 2SLS coefficient estimates in Tables 3 - 8 and their corresponding coefficients Tables B1 - B6 are in the same direction and of comparable magnitude. Therefore, our 2SLS results are not driven by a poor specification (Comerton-Forde et al., 2019).

³² Although the tick size pilot stocks are small-cap stocks, their institutional ownership is still significant. For instance, the average institutional ownership of each of the three test groups (Test Group 1, Test Group 2, and Test Group 3) during the study period is 70.2%, 71.4%, and 70.4%, respectively. When splitting each test group into two subgroups according to *TBC*, the average institutional ownership of each test group for stocks with low (high) *TBC* is 66.3% (74%), 64.7% (78%), and 63.7% (77%), respectively. Higher institutional ownerships for stocks with higher *TBC* may reflect that institutional investors prefer high liquidity stocks to minimize execution costs. Furthermore, institutional traders holding small-cap stocks are likely to be informed considering the trading cost and liquidity risk of holding small stocks. Hence, institutional investors likely play significant roles in price discovery for these stocks.

better informed than retail investors, the migration of informed institutional traders to dark pools would leave fewer informed traders on lit venues, increasing liquidity on lit venues.³³ O'Hara et al. (2014) suggest that odd-lots are frequently used in algorithmic and high-frequency trading, and odd-lot trades contribute 35% of price discovery. The authors interpret these results as evidence that informed traders split orders to avoid detection. We show later in Subsections 4.3 and 4.6 that *Dark_Share* is negatively related to the probability of informed trading (*PIN*), the price impact of trades (*PriceImpact*), and odd-lot volume ratio on lit venues (*OddLot*). These are additional reasons why liquidity on lit venues could increase with the market share of dark pools.³⁴

The effective and realized spreads increase with hidden liquidity, as evidenced by the positive and significant coefficients on $\widehat{Hidden_PT}$ in columns (3) - (6) of Panel A in Table 3. The quoted depth also increases with hidden liquidity, as indicated by the positive and significant coefficients on $\widehat{Hidden_PT}$ in columns (1) and (2) of Panel B in Table 3. The positive and significant coefficients on $Q \times Event$ throughout all columns in Panel A and columns (1) and (2) in Panel B suggest an increase in spreads and depths due to the quote rule, which is consistent with the finding of Chung et al. (2020).

We pointed out earlier that the effects of hidden liquidity on the quoted spread, price improvement, and trade size would determine the impact of hidden liquidity on the effective spread. To investigate whether price improvement contributes to the increase in the effective spread, we

³³ Prior research provides evidence that informed traders utilize dark pools because they could trade at low costs by hiding their orders (see, e.g., Balakrishnan and Taori, 2017; Reed et al., 2020; Ye and Zhu, 2020; Hu et al., 2021). Brogaard and Pan (2021) show that a higher level of dark pool trading is associated with more information acquisition and interpret the result as evidence that lower transaction costs increase information acquisition. Alfahoud et al. (2021) find that the market share of dark pools increases with insider trading, suggesting that investors are likely to trade in dark pools when they are endowed with private information. Hendershott et al. (2015) find evidence that institutions trade before the news announcement date to exploit their information advantages.

³⁴ The sub-penny tick size of dark pools is an important feature that attracts investors (Kwan et al., 2015), and changes in the market share of dark pools affect both the degree of market fragmentation and market quality (Comerton-Forde and Putniņš, 2015; Comerton-Forde et al., 2019). Therefore, the results from dark pool studies are likely driven, at least in part, by the sub-penny tick size and market fragmentation.

estimate regression model (7) with the percentage of shares that received price improvement (*PriceImprovement*) as the dependent variable. Panel B of Table 3 presents the results. Columns (3) and (4) show that the coefficients on $\widehat{Hidden_PT}$ are positive and significant, indicating that hidden liquidity increases the percentage of shares that received price improvement.³⁵ These results suggest that the positive effect of hidden liquidity on the quoted spread is greater than the positive effect of hidden liquidity on the price improvement.³⁶ We investigate the change in trade size in Subsection 4.4, which could be another source of changes in the effective spread. In contrast to the results for hidden liquidity, we find that price improvement decreases with the market share of dark pools, suggesting that hidden liquidity on lit exchanges and dark pools have different effects on liquidity providers.

4.3. Effects on informed trading and price impact

We present the results for informed trading and price impact in Table 4. In columns (1) and (2), the coefficients on $\widehat{Hidden_PT}$ are positive and significant, indicating that the probability of informed trading (*PIN*) increases with hidden liquidity. Similarly, columns (3) - (6) show that the coefficients on $\widehat{Hidden_PT}$ are all positive and significant in two regressions when the dependent variable is *PSI_FF3* or *PSI_BSW*. These results are consistent with Hypothesis 3 that hidden liquidity increases informed trading. As a result of increased informed trading that impounds more information into stock prices, we expect an increase in price impact (Chung et al., 2005). We find

³⁵ The proportion of trades that received a price improvement is not mechanically related to hidden liquidity. First, not all hidden orders may provide a price improvement, as hidden orders can be placed and executed at the NBBO. Second, not all price-improved trades may come from hidden liquidity, as orders from dark pools and odd-lot orders could provide price improvements as well.

³⁶ Bloomfield et al. (2015) show that the true spread, which includes quotes from both hidden and displayed orders, does not change across different opacity regimes. While we cannot empirically test whether this is also the case in our results since we only observe the displayed spread, it may be possible that the true spread also widens as a result of an increase in hidden liquidity, thus widening the effective spread.

this is indeed the case without (with) the stock and date fixed effects in column (7) (column (8)). In contrast, we find that *Dark_Share* is negatively and significantly related to *PIN* and *PriceImpact*, which is consistent with the idea that the migration of informed traders to dark pools leave fewer informed traders on lit venues. Columns (4) and (6) show that *Dark_Share* is also negatively related to both *PSI_FF3* and *PSI_BSW* once we include the stock and date fixed effects in the regression model. These results indicate that the effects of dark pools on informed trading and price impact are different from the effects of hidden liquidity on these variables.

We estimate regression model (7) using variance ratios (*VR*), return predictability of order imbalance (*OIB_Predictability*), and return autocorrelation (*Autocorrelation*), and show the results in Table 5. In columns (1), (3), and (5), the coefficients on $\widehat{Hidden_PT}$ are positive and significant, suggesting that hidden liquidity increases informed trading. These results are robust to including the stock and date fixed effects in the regression models (see columns (2), (4), and (6)). In contrast, the coefficients on *Dark_Share* are either significantly *negative* or insignificant, suggesting that the migration of informed traders to dark pools decreases informed trading on lit venues.

Prior research used the dependent variables we use in our regressions for Table 5 as inverse measures of pricing efficiency. For instance, large return autocorrelations and variance ratios are interpreted as an indication of inefficient pricing. Viewed from this perspective, the positive coefficients on $\widehat{Hidden_PT}$ in Table 5 imply that pricing efficiency decreases with hidden liquidity. Informed traders may employ certain trading strategies that reduce the randomness in price changes when they can use hidden orders, increasing the inverse measures of pricing efficiency in Table 5. For example, if patient informed traders submit only a fraction of their trading interest at a time using hidden orders to reduce the price impact, order submissions would be correlated. Thus, return autocorrelations and variance ratios would increase at the time of their

trading even though pricing efficiency would be higher after the completion of their trading.³⁷ Later in the paper, we show that traders' order submission strategies change, and algorithmic traders respond to the increase in hidden liquidity by engaging in more pinging activities, which likely contribute to the increase in our inverse measures of pricing efficiency.

Informed trading occurs when there are exploitable inefficiencies in prices. Pricing efficiency is achieved after informed traders' information has been impounded into prices through their trading. If a stock is already efficiently priced, informed traders do not have an incentive to trade. In other words, the fact that we find an increase in both informed trading and inverse measures of pricing efficiency indicates that more information is being impounded into stock prices, but this is possible only when the price was not efficient before such trades occurred. Therefore, hidden liquidity encourages more informed trading, but only when pricing efficiency is low enough.

4.4. Effects on trading activities

In this subsection, we analyze the effect of hidden liquidity on trading activities using regression model (7). Table 6 shows that the coefficients on $\widehat{Hidden_PT}$ in columns (1) - (4) are positive and significant, indicating that both dollar trading volume and trade size increase with hidden liquidity, consistent with Hypothesis 4.³⁸ Because both trading volume and trade size increase, it is unclear whether the number of trades increases or decreases with hidden liquidity.

³⁷ The Pearson correlations among the informed trading measures in Tables 4 and 5 range from 0.1688 to 0.8493.

³⁸ Foley et al. (2013) report that the introduction of dark orders led to no change in dollar trading volume. Buti and Rindi (2013) predict that trading volume decreases with iceberg orders because these orders increase gains from supplying liquidity, reducing incentives to take liquidity. In contrast, Moinas (2010) predicts that trading volume increases with hidden liquidity. Our result is not consistent with the prediction of Buti and Rindi (2013), perhaps because they do not have informed traders in their model. In our paper, the fact that informed and uninformed traders exist drives the difference in their aggressiveness (i.e., uninformed becomes more aggressive) and trading volume.

Columns (5) and (6) show that the number of transactions decreases with hidden liquidity.

The increase in trade size has an implication for the effective spread, as discussed in Section 2. As shown in Table 3, the effective spread is positively related to hidden liquidity. A possible explanation is that the effect of hidden liquidity on the quoted spread is larger than its effect on the price improvement. Another explanation is that trade size also increases with hidden liquidity. As trade size becomes larger, the probability that order size is larger than the quoted depth at the NBBO becomes higher, resulting in some shares beyond the NBBO being executed. When a transaction is executed beyond the NBBO, its effective spread becomes wider. Therefore, our finding that hidden liquidity increases trade size can be seen as a channel through which hidden liquidity widens the effective spread.

The results show that dollar volume, trade size, and the number of trades are positively related to dark pools' market share when we include the stock and day fixed effects. Hence, while hidden liquidity and dark pools have a similar impact on dollar volume and trade size, they have opposite effects on the number of trades.

4.5. Effects on order and trade aggressiveness

Table 7 provides the regression results for the effects of hidden liquidity on the proportion of shares executed using market orders (OT_MO), marketable limit orders (OT_MLO), inside-the-quote limit orders (OT_IQL), at-the-quote limit orders (OT_AQL), and near-the-quote limit orders (OT_NQL), respectively. The results without the fixed effects show that the coefficient on $\widehat{Hidden_PT}$ is positive and significant in the regression model of OT_IQL , but negative and significant in all other regression models. These results indicate that traders shift their order aggressiveness with hidden liquidity, leading to an increased proportion of shares executed using

inside-the-quote orders, supporting Hypothesis 5.³⁹ Although the results with the fixed effects show that the coefficients on $\widehat{Hidden_PT}$ are positive in the regression models of OT_MO and OT_NQL , their absolute magnitudes are much smaller than those in the regression models of OT_MLO , OT_IQL , and OT_AQL . Hence, the regression results with the fixed effects are also consistent with Hypothesis 5.

The results without the fixed effects show that the coefficients on $\widehat{Dark_Share}$ have opposite signs of the coefficients on $\widehat{Hidden_PT}$ in all regression models. The results with the fixed effects show that the coefficients on $\widehat{Dark_Share}$ have opposite signs of the coefficients on $\widehat{Hidden_PT}$ in the regression models of OT_IQL and OT_AQL , and the coefficient on $\widehat{Dark_Share}$ is not significant in the regression model of OT_MO . These results indicate that the effects of dark pools on order and trade aggressiveness are different from the corresponding effects of hidden liquidity on lit exchanges.

4.6. Effects on pinging activities and message traffic

Table 8 provides the regression results for the effects of hidden liquidity on the order cancellation ratio (*Cancellations*), odd-lot volume ratio (*OddLot*), and order-to-trade ratio (*OTR*). Columns (1), (3), and (5) show that the coefficients on $\widehat{Hidden_PT}$ are positive and significant when we do not include the stock and date fixed effects, suggesting that pinging activities and message traffic increase with hidden liquidity. These results are consistent with Hypothesis 6. When we include the stock and date fixed effects, however, the coefficient on $\widehat{Hidden_PT}$ is positive and significant when the dependent variable is *Cancellations* (column (2)). The coefficient

³⁹ Using Rule 605 data, we are also able to construct the dependent variables for Table 7 using the number of orders and the number of orders in shares. The results are qualitatively similar to those in Table 7.

is negative and significant when the dependent variable is *OddLot* (column (4)) and insignificant when the dependent variable is *OTR* (column (6)).

To reconcile the difference in results for *OddLot* depending on whether the stock and date fixed effects are included, we calculate its mean value for each stock during the pre-pilot period. We conjecture that the extent to which algorithmic traders respond to an increase in hidden liquidity depends on the level of algorithmic trading activities before the change in hidden liquidity. In an untabulated analysis that includes the stock and date fixed effects, we find that the coefficient on $\widehat{Hidden_PT}$ in the *OddLot* regression model is insignificant for stocks with higher than the median value of the pre-pilot period *OddLot* mean. In contrast, the coefficient continues to be negative and significant for stocks with lower than the median value of the pre-pilot period *OddLot* mean. While these results may indicate that hidden liquidity negatively affects ping activities for some stocks, our sample stocks are from the Tick Size Pilot Program, the goal of which was to examine the effects of changes in tick size and the trade-at rule on small and illiquid stocks. Therefore, our sample stocks are generally smaller and less liquid compared to an average publicly traded stock. On the other hand, algorithmic traders contributing the most to ping activities and message traffic are likely to be more active in larger and more liquid stocks.⁴⁰ Hence, we expect the effects of hidden liquidity on ping activities and message traffic to be positive for an average publicly traded stock. Table 8 also shows that the effects of dark pools are different from the effects of hidden liquidity, especially in regressions without the stock and date fixed effects.

4.7. Regression results for stocks with high and low TBC

⁴⁰ High-frequency traders, a subgroup within algorithmic traders that are commonly associated with activities such as ping and increased message traffic, account for a higher percentage of total trading volume in large-cap stocks than in middle-cap or small-cap stocks (see Brogaard et al. (2014) and Xu (2015)).

We explore whether the results in Tables 3 - 5 vary with the extent to which the five-cent tick size is a binding constraint (*TBC*) and provide the results in Tables 9 - 11, respectively.⁴¹ In Panels A and B of Table 9, columns (1) - (4) report the results for stocks with below-median *TBC* values, and columns (5) - 8 report the results for stocks with above-median *TBC* values. As in Table 3, Panel A of Table 9 shows that the quoted and effective spreads increase with hidden liquidity for both groups of stocks. By contrast, the quoted and effective spreads decrease (increase) with the market share of dark pools for stocks in the low (high) *TBC* group.

In Subsection 4.2, we interpret the negative relation between the spreads and the market share of dark pools as evidence that the effect of the competition is greater than the effect of order migration. The results in Table 9 suggest that the impact of the competition is greater than the effect of order migration only for stocks in the low *TBC* group (i.e., stocks with high equilibrium quoted and effective spreads). A possible interpretation of this finding is that competition matters more for these stocks because there is more room for reductions in spreads. Another possible interpretation is that the migration of informed institutional traders to dark pools occurs mainly in stocks with larger spreads, leaving fewer informed traders and increasing liquidity on lit venues for these stocks.⁴² For stocks with low equilibrium quoted and effective spreads (i.e., stocks in the high *TBC* group), the effect of competition appears smaller than the effect of order migration and/or there may be no significant migration of informed traders to dark pools.⁴³

Similarly, Panel B of Table 9 shows that for stocks with low equilibrium realized spreads,

⁴¹ We omit the results for the control variables. Tables C1 - C3 in the [Online Appendix](#) show the full regression results.

⁴² Institutional traders have larger incentives to migrate to dark pools for these stocks because of the high trading cost on lit venues before their migration.

⁴³ Institutional traders have smaller incentives to migrate to dark pools for these stocks because of the low trading cost on lit venues. The possibility of sub-tick trading should make dark pools more attractive when the spread is tick-constrained, and hence competition may matter more for high *TBC* stocks. Our results suggest that migration to dark pools may not be driven primarily by sub-tick trading. It could be driven considerably by traders' desire to hide their orders, as long as execution costs on lit venues are not excessively high.

the impact of competition appears smaller than the effect of order migration and/or there may be no significant migration of informed traders to dark pools. Similar to the results in Table 3, Panel B also shows that the effect of the dark pool market share on quoted depths is not statistically significant when we include the fixed effects for both groups of stocks. Panel C of Table 9 shows that price improvements decrease with the market share of dark pools for both groups of stocks, which is similar to the results in Panel B of Table 3.

Table 10 shows the effects of hidden liquidity and the dark pool market share on informed trading and price impact for each *TBC* group. Panel A shows the results for *PIN* and *PSI_FF3*, and Panel B shows the results for *PSI_BSW* and *PriceImpact*. Panel A shows that *PIN* is positively related to hidden liquidity for stocks in both *TBC* groups, which is identical to the full sample results in Table 4. We also find that *PIN* is negatively related to the market share of dark pools for stocks in both *TBC* groups, identical to the results in Table 4. These results indicate that the effects of hidden liquidity and dark pool trading on *PIN* do not vary with *TBC* values. The impacts of dark pool trading on *PSI_FF3* for stocks in both *TBC* groups are qualitatively similar to those for the whole sample in Table 4. We find that *PSI_FF3* is positively and significantly related to hidden liquidity for stocks in the low (high) *TBC* group when we include (exclude) the fixed effects. Panel B of Table 10 shows that the effects of hidden liquidity and dark pool trading on *PSI_BSW* and price impact are qualitatively identical to the full sample results in Table 4 for stocks in both *TBC* groups. Finally, Table 11 shows that the effects of hidden liquidity and dark pool trading on *VR*, *OIB_Predictability*, and *Autocorrelation* for stocks in both *TBC* groups are qualitatively similar to those for the whole sample.⁴⁴ On the whole, the results in Tables 10 and 11 suggest that the effects of hidden liquidity and dark pools on informed trading and price impact are similar between stocks

⁴⁴ We show the results with the fixed effects. We provide the results without the fixed effects in the [Online Appendix](#).

with high and low *TBC* values.

4.8. Regression results using an order-based hidden liquidity measure

Our hidden liquidity measure is based on hidden limit orders executed against market orders or marketable limit orders. In other words, the measure is based on hidden limit orders that are competitive enough (relative to displayed limit orders) to be executed against incoming market orders or marketable limit orders. Hence, it does not reflect hidden liquidity far behind the market (i.e., hidden buy limit orders with low limit prices and hidden sell limit orders with high limit prices). In this sense, our measure of hidden liquidity underestimates the level of hidden liquidity. The underestimation is unlikely to induce a systematic bias in our results as long as it is similar across stocks. To assess the robustness of our results to different measures of hidden liquidity, we replicate Tables 2 - 8 using an alternative measure of hidden liquidity based on all limit orders, including non-competitive hidden limit orders.

We use the Tick Size Pilot Program data to measure the hidden liquidity from nonmarketable limit orders submitted to exchanges. The Tick Size Pilot Program Appendix B.1 Data file provides data on three types of orders: fully displayed, partially displayed, and fully hidden. We report the results using the ratio of the number of fully hidden orders to the number of all orders. The results are identical when we use the ratio of the number of fully hidden and partially displayed orders to the number of all orders. We find that the Pearson correlation coefficient between the trade-based hidden liquidity and the order-based hidden liquidity ranges between 0.40 and 0.52, depending on whether we measure hidden liquidity by *Hidden_PT* or *Hidden_PS*.

We provide the regression results in Tables D1 - D7 in the Online Appendix. We find that

that the results in Tables D1 - D6 are all qualitatively identical to those in Tables 2 - 7, indicating that our results are robust to different measures of hidden liquidity.⁴⁵ As in Table 8, we find that the sign and magnitude of the coefficients in Table D7 are sensitive to whether or not we include the fixed effects in the regression. On the whole, our main inferences remain the same regardless of whether we measure hidden liquidity using only competitive (i.e., executed) hidden orders or all hidden orders.

5. Conclusion

Traders in most equity markets can instruct exchanges to hide or display their limit orders on the book. While hiding trading interests could be advantageous to liquidity providers, it increases uncertainties for liquidity demanders by reducing pre-trade transparency. While this feature of equity markets has long existed, how it affects market quality has not been well-understood, perhaps because it is challenging to measure hidden liquidity accurately. Another reason for the lack of research in hidden liquidity could be the absence of an exogenous shock in hidden liquidity that helps establish the causality of the relation between hidden liquidity and other variables of interest.

We provide evidence regarding causal relations between hidden liquidity and various market quality metrics using the data from the SEC's implementation of a new system that combines advanced technologies with empirical data (i.e., MIDAS) and an exogenous change in hidden liquidity that results from the implementation of the SEC's Tick Size Pilot Program. We show that an increase in hidden liquidity causes an increase in the quoted, effective, and realized

⁴⁵ Table D5 shows that the coefficient on $\widehat{Hidden_PT}$ in the dollar volume regression model (column (1)) is negative and significant. Once we include the stock and day fixed effects, however, the coefficient becomes positive and significant, which is consistent with the result in Table 6.

spreads, the probability of informed trading, the price impact of a trade, dollar trading volume, and trade size. We also show that an increase in hidden liquidity causes an increase in the proportion of shares executed using inside-the-quote limit orders, pinging activities, and message traffic.

Our results indicate that dark pools and hidden liquidity on lit exchanges play different roles in price discovery (e.g., informed trading decreases with the market share of dark pools, whereas it increases with hidden liquidity on lit exchanges). Our finding that the ability to hide trading intentions on lit exchanges helps informed traders to more actively participate in trading and impound their information into stock prices is in line with the conclusion of prior studies that transparency is not always a socially optimal feature (e.g., Diamond and Verrecchia, 1991; Goldstein and Yang, 2019).

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Table 1
Descriptive statistics

This table presents the mean, standard deviation, and difference between the pilot period mean and pre-pilot period mean and its *t*-statistics. *Hidden_PT* (*Hidden_PS*) is the percentage of transactions (shares) that are traded against hidden orders. *QuotedSpread*, *EffectiveSpread*, and *RealizedSpread* are the percentage quoted, effective, and realized spreads. *PriceImprovement* is the percentage of price-improved shares. *VR* is the variance ratio, which is the absolute value of one minus the ratio of midpoint variance at the one-minute interval to four times that of the 15-second interval. *OIB_Predictability* is the R^2 value of the regression of one-minute stock return at interval *t* on the order imbalance at interval *t*-1. *Autocorrelation* is the R^2 value of the regression of stock return at minute *t* on return at minute *t*-1. *PIN* is the probability of informed trading. *PSI_FF3* (*PSI_BSW*) is the difference between 1 and R^2 value from the regression of the stock excess return on Fama-French three factors (industry and market returns). *PriceImpact* is the price impact of trades. *DollarVolume* is the log of the sum of the trading volume. *TradeSize* is the daily average trade size in dollars. *Num_Trades* is the number of trades. *OT_MO*, *OT_MLO*, *OT_IQL*, *OT_AQL*, and *OT_NQL* are the proportions of shares executed using market, marketable limit, inside-the-quote limit orders, at-the-quote limit orders, and near-the-quote limit orders (*OT_NQL*), respectively. *Cancellations* is the ratio of the number of cancellation messages to the number of trades executed against displayed orders. *OddLot* is the ratio of the odd-lot trading volume to the total trade volume. *OTR* is the order-to-trade ratio, i.e., the ratio of order volumes to trade volumes that are executed against displayed orders. *Inverted_Share* (*Dark_Share*) is the market share of inverted venues (dark pools). *Price* is the daily closing price. *MarketCap* is the log of market capitalization. *Volatility* is the daily return volatility. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Control Group			Test Group 1			Test Group 2			Test Group 3		
	Pre-pilot period mean (S.D.)	Pilot period mean (S.D.)	Difference in means (<i>t</i> -stat)	Pre-pilot period mean (S.D.)	Pilot period mean (S.D.)	Difference in means (<i>t</i> -stat)	Pre-pilot period mean (S.D.)	Pilot period mean (S.D.)	Difference in means (<i>t</i> -stat)	Pre-pilot period mean (S.D.)	Pilot period mean (S.D.)	Difference in means (<i>t</i> -stat)
Panel A: Liquidity variables												
<i>Hidden_PT</i>	0.2094 (0.1151)	0.2110 (0.1079)	0.0016 (0.30)	0.2110 (0.1141)	0.1576 (0.0726)	-0.053*** (-6.76)	0.2071 (0.1127)	0.1561 (0.0769)	-0.051*** (-6.23)	0.2083 (0.1117)	0.1117 (0.0610)	-0.097*** (-12.45)
<i>Hidden_PS</i>	0.2229 (0.1017)	0.2136 (0.0921)	-0.0092** (-1.97)	0.2244 (0.1026)	0.1641 (0.0616)	-0.060*** (-8.62)	0.2205 (0.1011)	0.1613 (0.0636)	-0.059*** (-8.26)	0.2220 (0.0988)	0.1190 (0.0527)	-0.103*** (-15.08)
<i>QuotedSpread</i>	0.6898 (0.7761)	0.5924 (0.6490)	-0.097*** (-2.82)	0.6771 (0.7807)	0.7523 (0.6809)	0.0752 (1.24)	0.6767 (0.7469)	0.7434 (0.7129)	0.0667 (1.07)	0.6505 (0.7302)	0.7202 (0.7098)	0.0697 (1.12)
<i>EffectiveSpread</i>	0.2895 (0.2817)	0.2429 (0.2342)	-0.047*** (-3.73)	0.2828 (0.2785)	0.3842 (0.3046)	0.1014*** (4.20)	0.2853 (0.2756)	0.4044 (0.3499)	0.1191*** (4.45)	0.2728 (0.2579)	0.3689 (0.3300)	0.0961*** (3.76)
<i>RealizedSpread</i>	0.1605 (0.1967)	0.1357 (0.1680)	-0.025*** (-2.80)	0.1592 (0.2000)	0.2106 (0.2180)	0.0514*** (2.97)	0.1589 (0.1931)	0.2199 (0.2485)	0.0610*** (3.23)	0.1523 (0.1825)	0.1779 (0.2149)	0.0255 (1.48)
<i>Depth</i>	18.028 (19.358)	26.086 (88.289)	8.0576*** (2.61)	19.442 (14.636)	80.220 (119.02)	60.778*** (8.66)	18.510 (13.791)	98.958 (278.64)	80.449*** (4.80)	18.728 (15.052)	108.46 (197.44)	89.732*** (7.42)
<i>PriceImprovement</i>	0.1385 (0.0789)	0.1378 (0.0888)	-0.0007 (-0.16)	0.1438 (0.0841)	0.0716 (0.0476)	-0.072*** (-12.78)	0.1383 (0.0756)	0.0720 (0.0472)	-0.066*** (-12.39)	0.1412 (0.0839)	0.0683 (0.0446)	-0.073*** (-12.57)
Panel B: Information variables												
<i>VR</i>	0.1502 (0.0349)	0.1474 (0.0350)	-0.0028* (-1.66)	0.1504 (0.0349)	0.1460 (0.0285)	-0.0045* (-1.69)	0.1499 (0.0357)	0.1428 (0.0283)	-0.007*** (-2.59)	0.1515 (0.0351)	0.1419 (0.0251)	-0.010*** (-3.68)
<i>OIB_Predictability</i>	6.4209 (7.9297)	5.5222 (6.9461)	-0.8987** (-2.50)	6.3658 (7.8521)	5.7299 (7.0243)	-0.6359 (-1.03)	6.5063 (8.1399)	5.3590 (6.759)	-1.1473* (-1.80)	6.3763 (7.7672)	5.1836 (6.5683)	-1.1926* (-1.92)

<i>Autocorrelation</i>	3.1510 (3.6897)	3.0031 (3.6627)	-0.1479 (-0.83)	3.0771 (3.5008)	3.3865 (4.0573)	0.3094 (0.99)	3.1625 (3.7014)	3.1950 (3.9233)	0.0325 (0.10)	3.1428 (3.4144)	3.0435 (3.7827)	-0.0993 (-0.32)
<i>PIN</i>	0.1589 (0.0659)	0.1575 (0.0621)	-0.0013 (-0.43)	0.1571 (0.0640)	0.1646 (0.0637)	0.0075 (1.41)	0.1582 (0.0627)	0.1599 (0.0633)	0.0018 (0.33)	0.1556 (0.0625)	0.1552 (0.0590)	-0.0005 (-0.09)
<i>PSI_FF3</i>	72.825 (14.565)	75.389 (15.275)	2.5644*** (3.57)	71.935 (14.544)	76.144 (14.190)	4.2091*** (3.55)	72.988 (14.285)	75.647 (14.862)	2.6598** (2.15)	72.148 (14.576)	75.754 (15.209)	3.6063*** (2.81)
<i>PSI_BSW</i>	74.224 (16.698)	79.056 (15.859)	4.8325*** (6.11)	73.024 (17.334)	79.477 (15.099)	6.4523*** (4.76)	74.372 (16.142)	79.912 (14.675)	5.5393*** (4.20)	73.968 (16.367)	80.200 (15.114)	6.2323*** (4.54)
<i>PriceImpact</i>	0.1291 (0.0973)	0.1071 (0.0768)	-0.022*** (-5.18)	0.1236 (0.0925)	0.1734 (0.1057)	0.0498*** (6.06)	0.1264 (0.0948)	0.1845 (0.1219)	0.0581*** (6.26)	0.1204 (0.0868)	0.1908 (0.1331)	0.0704*** (7.25)
Panel C: Trading activities												
<i>DollarVolume</i>	4.8101 (7.2811)	6.3510 (10.877)	1.5409*** (3.45)	4.9219 (6.2396)	5.9702 (7.6440)	1.0482* (1.82)	5.3232 (8.9669)	6.3609 (9.3323)	1.0377 (1.34)	5.1632 (7.7968)	6.3373 (10.178)	1.1741 (1.50)
<i>TradeSize</i>	2287.8 (1617.7)	2481.3 (1774.4)	193.43** (2.36)	2427.7 (1753.7)	2772.6 (1871.6)	344.88** (2.30)	2333.8 (1671.2)	2728.9 (1742.7)	395.15*** (2.72)	2472.0 (2207.8)	2769.6 (2284.8)	297.55 (1.53)
<i>Num_Trades</i>	1697.7 (1718.5)	1970.0 (1996.9)	272.36*** (3.03)	1773.0 (1825.5)	1741.9 (1762.7)	-31.112 (-0.21)	1811.1 (1913.5)	1792.9 (1778.4)	-18.160 (-0.12)	1748.8 (1812.1)	1803.3 (1783.9)	54.480 (0.35)
Panel D: Order submission strategies												
<i>OT_MO</i>	0.0021 (0.0018)	0.0018 (0.0013)	-0.000*** (-4.08)	0.0021 (0.0021)	0.0019 (0.0015)	-0.0002 (-1.06)	0.0019 (0.0016)	0.0017 (0.0012)	-0.0002 (-1.62)	0.0019 (0.0015)	0.0015 (0.0013)	-0.000*** (-3.20)
<i>OT_MLO</i>	0.4162 (0.0678)	0.4305 (0.0602)	0.0143*** (4.62)	0.4123 (0.0668)	0.4721 (0.0616)	0.0599*** (11.26)	0.4126 (0.0724)	0.4759 (0.0611)	0.0633*** (11.12)	0.4159 (0.0672)	0.4999 (0.0642)	0.0839*** (14.78)
<i>OT_IQL</i>	0.2197 (0.1022)	0.2209 (0.1075)	0.0012 (0.24)	0.2204 (0.1061)	0.1255 (0.0999)	-0.095*** (-11.12)	0.2196 (0.1054)	0.1217 (0.0978)	-0.098*** (-11.34)	0.2189 (0.1017)	0.1222 (0.0999)	-0.097*** (-11.11)
<i>OT_AQL</i>	0.2505 (0.0611)	0.2493 (0.0638)	-0.0012 (-0.40)	0.2531 (0.0669)	0.3263 (0.0484)	0.0732*** (15.15)	0.2531 (0.0655)	0.3268 (0.0458)	0.0738*** (15.35)	0.2480 (0.0628)	0.3077 (0.043)	0.0597*** (12.85)
<i>OT_NQL</i>	0.1115 (0.0462)	0.0974 (0.0357)	-0.014*** (-7.04)	0.1122 (0.0482)	0.0741 (0.0369)	-0.038*** (-10.72)	0.1128 (0.0476)	0.0739 (0.0366)	-0.039*** (-10.78)	0.1152 (0.0538)	0.0687 (0.0320)	-0.047*** (-12.16)
<i>Cancellations</i>	53.267 (53.382)	35.686 (26.278)	-17.58*** (-8.67)	56.582 (70.831)	27.793 (27.272)	-28.79*** (-6.49)	53.484 (44.344)	25.134 (18.438)	-28.35*** (-9.84)	55.741 (58.180)	26.763 (96.412)	-28.98*** (-4.22)
<i>OddLot</i>	0.1670 (0.0746)	0.2116 (0.1000)	0.0445*** (10.47)	0.1692 (0.0770)	0.1962 (0.0987)	0.027*** (3.69)	0.1675 (0.0691)	0.1932 (0.0943)	0.0257*** (3.66)	0.1680 (0.0746)	0.1871 (0.0936)	0.0191*** (2.62)
<i>OTR</i>	157.14 (569.13)	76.744 (160.92)	-80.39*** (-3.99)	231.12 (1599.7)	76.209 (238.84)	-154.91 (-1.64)	179.22 (621.36)	56.958 (77.099)	-122.3*** (-3.26)	172.36 (789.60)	49.353 (138.69)	-123.01** (-2.52)
Panel E: Control variables												
<i>Inverted_Share</i>	0.0901 (0.0328)	0.0955 (0.0331)	0.0054*** (3.42)	0.0919 (0.0347)	0.1743 (0.0525)	0.0824*** (22.42)	0.0896 (0.0332)	0.1780 (0.0525)	0.0884*** (23.75)	0.0893 (0.0315)	0.2536 (0.0696)	0.1642*** (35.26)
<i>Dark_Share</i>	0.3539 (0.0811)	0.3476 (0.0919)	-0.0064 (-1.53)	0.3519 (0.0829)	0.3830 (0.0965)	0.0311*** (4.19)	0.3550 (0.0824)	0.3796 (0.0914)	0.0246*** (3.34)	0.3575 (0.0788)	0.2514 (0.0732)	-0.106*** (-16.18)
<i>1 / Price</i>	0.0993 (0.1067)	0.1072 (0.1717)	0.0079 (1.15)	0.0960 (0.1064)	0.0900 (0.1070)	-0.0060 (-0.68)	0.1017 (0.1409)	0.0885 (0.1123)	-0.0132 (-1.22)	0.0976 (0.1295)	0.0895 (0.1064)	-0.0081 (-0.80)
<i>MarketCap</i>	693.74 (701.09)	783.91 (789.64)	90.163** (2.51)	744.97 (730.92)	826.93 (815.46)	81.956 (1.28)	727.93 (706.75)	806.95 (764.22)	79.023 (1.27)	703.65 (662.44)	825.52 (797.59)	121.86* (1.93)

<i>Volatility</i>	0.0242 (0.0110)	0.0205 (0.0089)	-0.004*** (-7.68)	0.0229 (0.0101)	0.0195 (0.0072)	-0.003*** (-4.70)	0.0233 (0.0100)	0.0200 (0.0079)	-0.003*** (-4.30)	0.0231 (0.0103)	0.0204 (0.0085)	-0.003*** (-3.36)
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Table 2

Effects of the Tick Size Pilot Program on hidden liquidity and the market share of dark pools

This table presents the results of regression models (5) and (6): $Hidden_PT_{i,t} = \beta_1 Q_i \times Event_t + \beta_2 T_i \times Event_t + \beta_3 TA_i \times Event_t + \beta_4 Hidden_PT_{i,t-1} + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$ and $Dark_Share_{i,t} = \beta_1 Q_i \times Event_t + \beta_2 T_i \times Event_t + \beta_3 TA_i \times Event_t + \beta_4 Dark_Share_{i,t-1} + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where $Hidden_PT_{i,t}$ is the percentage of transactions against hidden orders for stock i on day t , $Dark_Share_{i,t}$ is the percentage market share of dark pools for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. Control variables include the CBOE Volatility Index (VIX), the inverse of closing price ($1 / Price$), the log of market capitalization ($MarketCap$), the log of trading volume in million U.S. dollar ($DollarVolume$), daily return volatility ($Volatility$), and the market share of inverted venues ($Inverted_Share$). The regressions for columns (3) and (4) also include quoted spread ($QuotedSpread$) as a control variable. Standard errors are clustered by firm and date. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. t -statistics are in parentheses. See Table A1 for full results.

	(1) <i>Hidden_PT</i>	(2) <i>Hidden_PT</i>	(3) <i>Hidden_PT</i>	(4) <i>Hidden_PT</i>	(5) <i>Dark_Share</i>	(6) <i>Dark_Share</i>
<i>Q x Event</i>	-0.0233*** (-11.43)	-0.0479*** (-14.66)	-0.0488*** (-22.80)	-0.0655*** (-23.24)	0.0787*** (25.55)	0.0847*** (25.83)
<i>T x Event</i>	0.0014 (0.59)	0.0013 (0.31)	0.0002 (0.06)	-0.0001 (-0.03)	-0.0003 (-0.10)	-0.0013 (-0.35)
<i>TA x Event</i>	-0.0188*** (-7.26)	-0.0413*** (-9.82)	-0.0291*** (-10.90)	-0.0438*** (-11.88)	-0.0464*** (-11.70)	-0.0563*** (-13.31)
<i>Hidden_PT (t - 1)</i>	0.6524*** (83.87)	0.3783*** (65.15)	0.5164*** (56.64)	0.2969*** (58.15)		
<i>Dark_Share (t - 1)</i>					0.2786*** (47.15)	0.1889*** (67.28)
<i>Q</i>	0.0018 (0.85)		0.0008 (0.37)		0.0015 (0.69)	
<i>T</i>	-0.0017 (-0.66)		-0.0006 (-0.25)		-0.0011 (-0.42)	
<i>TA</i>	0.0012 (0.43)		0.0013 (0.49)		0.0021 (0.76)	
<i>Event</i>	0.0047*** (4.66)		0.0027*** (2.69)		-0.0112*** (-5.16)	
<i>N</i>	803,462	803,462	803,251	803,251	803,938	803,938
<i>Stock and date F.E.</i>	No	Yes	No	Yes	No	Yes
<i>Adjusted R²</i>	0.5314	0.6144	0.5917	0.6589	0.4093	0.4753
<i>F-statistic</i>	1,961.16	966.28	1,542.79	1,318.76	1,041.33	619.97

Table 3
Effects of hidden liquidity on spreads and depth

This table presents the spreads and depth results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is one of quoted spread (*QuotedSpread*), effective spread (*EffectiveSpread*), realized spread (*RealizedSpread*), quoted depth (*Depth*), or the proportion of executed shares that received price improvement (*PriceImprovement*), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders (*Hidden_PT*) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), and the market share of inverted venues (*Inverted_Share*). Standard errors are clustered by firm and date. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. t -statistics are reported in the parentheses. See Table A3 for full results.

Panel A: Regression results for spread measures						
	(1) <i>QuotedSpread</i>	(2) <i>QuotedSpread</i>	(3) <i>EffectiveSpread</i>	(4) <i>EffectiveSpread</i>	(5) <i>RealizedSpread</i>	(6) <i>RealizedSpread</i>
<i>Hidden_PT</i>	2.8312*** (34.21)	2.6970*** (32.03)	0.8798*** (26.34)	0.9373*** (24.47)	0.7180*** (30.24)	0.7211*** (26.28)
<i>Dark_Share</i>	-0.1553** (-1.98)	-0.1371* (-1.70)	0.0517 (1.19)	-0.0137 (-0.32)	0.1007*** (3.31)	0.0963*** (3.01)
<i>Q x Event</i>	0.4238*** (21.09)	0.3637*** (16.07)	0.2199*** (20.76)	0.2117*** (18.98)	0.1381*** (18.76)	0.1229*** (15.48)
<i>T x Event</i>	0.1001*** (4.85)	0.0765*** (3.29)	0.0443*** (3.83)	0.0349*** (2.81)	0.0311*** (3.98)	0.0236*** (2.75)
<i>Q</i>	0.0103 (0.54)		0.0039 (0.59)		0.0009 (0.18)	
<i>T</i>	-0.0547** (-2.55)		-0.0128* (-1.71)		-0.0099* (-1.79)	
<i>TA</i>	0.0850*** (4.44)		0.0143* (1.92)		0.0125** (2.30)	
<i>Event</i>	0.0117* (1.66)		-0.0061** (-2.04)		-0.0024 (-0.94)	
<i>N</i>	803,251	803,251	803,342	803,342	801,653	801,653
Stock and date F.E.	No	Yes	No	Yes	No	Yes
Adj. R ²	0.6488	0.7632	0.5236	0.6050	0.3250	0.3851

Panel B: Regression results for depth and proportion of shares that received price improvement				
	(1) <i>Depth</i>	(2) <i>Depth</i>	(3) <i>PriceImprovement</i>	(4) <i>PriceImprovement</i>
<i>Hidden_PT</i>	226.19*** (4.20)	460.27*** (3.28)	0.8078*** (18.03)	0.5546*** (8.30)
<i>Dark_Share</i>	156.80*** (2.74)	11.6755 (0.19)	-0.1886*** (-7.75)	-0.0543 (-1.66)
<i>Q x Event</i>	40.276*** (4.67)	74.350*** (7.53)	-0.0167** (-2.52)	-0.0431*** (-5.13)
<i>T x Event</i>	26.200** (2.02)	31.170** (2.00)	0.0037 (1.30)	0.0076** (2.09)
<i>Q</i>	-1.8571 (-1.26)		0.0024 (0.93)	
<i>T</i>	-3.5679 (-0.76)		-0.0030 (-1.03)	
<i>TA</i>	8.2877 (0.93)		0.0036 (1.39)	
<i>Event</i>	-1.2614 (-0.41)		-0.0030 (-1.13)	
<i>N</i>	803,251	803,251	38,394	38,394
Stock and date F.E.	No	Yes	No	Yes
Adjusted R ²	0.0330	0.1316	0.6241	0.7830

Table 4
Effects of hidden liquidity on informed trading and price impact

This table presents the informed trading and price impact results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is the probability of informed trading (PIN), the price nonsynchronicity measure based on the Fama-French 3 factors (PSI_FF3), the price nonsynchronicity measure based on the market and industry returns (PSI_BSW), or the price impact ($PriceImpact$), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders ($Hidden_PT$) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools ($Dark_Share$) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. Control variables include the CBOE Volatility Index (VIX), the inverse of the closing price ($1 / Price$), the log of market capitalization ($MarketCap$), the log of trading volume in million U.S. dollar ($DollarVolume$), daily return volatility ($Volatility$), quoted spread ($QuotedSpread$), and the market share of inverted venues ($Inverted_Share$). Standard errors are clustered by firm and date in columns (1), (2), (7), and (8), and by firm in columns (3) - (6). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. t-statistics are reported in the parentheses. See Table A4 for full results.

	(1) <i>PIN</i>	(2) <i>PIN</i>	(3) <i>PSI_FF3</i>	(4) <i>PSI_FF3</i>	(5) <i>PSI_BSW</i>	(6) <i>PSI_BSW</i>	(7) <i>PriceImpact</i>	(8) <i>PriceImpact</i>
<i>Hidden_PT</i>	0.0919*** (6.10)	0.0855** (2.21)	0.1568 (0.65)	2.0773*** (3.49)	0.4452 (1.57)	2.2843*** (2.85)	0.1615*** (12.61)	0.2149*** (12.79)
<i>Dark_Share</i>	-0.0558*** (-3.14)	-0.0659** (-2.17)	3.2805*** (10.51)	-1.4185** (-2.30)	3.9275*** (10.30)	-1.7377** (-2.22)	-0.0487*** (-2.73)	-0.1091*** (-5.45)
<i>Q x Event</i>	0.0226*** (5.39)	0.0230*** (3.24)	0.0475 (0.77)	0.4837*** (4.57)	-0.0042 (-0.06)	0.5080*** (3.70)	0.0816*** (17.73)	0.0885*** (18.67)
<i>T x Event</i>	-0.0004 (-0.10)	-0.0022 (-0.49)	0.2162*** (4.13)	-0.0201 (-0.36)	0.2923*** (4.56)	-0.0079 (-0.12)	0.0132*** (2.67)	0.0113** (2.25)
<i>Q</i>	-0.0007 (-0.24)		-0.0325 (-0.76)		-0.0288 (-0.55)		0.0030 (1.13)	
<i>T</i>	-0.0011 (-0.33)		-0.0905* (-1.68)		-0.0814 (-1.25)		-0.0029 (-0.91)	
<i>TA</i>	0.0013 (0.45)		0.2718*** (5.26)		0.2975*** (4.81)		0.0017 (0.59)	
<i>Event</i>	0.0044 (1.52)		-0.0627** (-2.28)		0.2445*** (7.47)		-0.0036*** (-2.64)	
<i>N</i>	38,288	38,288	11,849	11,849	11,614	11,614	801,653	801,653
Stock and time F.E.	No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R ²	0.2139	0.2752	0.3844	0.5717	0.3977	0.5723	0.2555	0.2878

Table 5

Effects of hidden liquidity on other measures of informed trading

This table presents the pricing efficiency results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is one of variance ratios (VR), return predictability of order imbalance ($OIB_Predictability$), or return autocorrelation ($Autocorrelation$), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders ($Hidden_PT$) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools ($Dark_Share$) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. Control variables include the CBOE Volatility Index (VIX), the inverse of the closing price ($1 / Price$), the log of market capitalization ($MarketCap$), the log of trading volume in million U.S. dollar ($DollarVolume$), daily return volatility ($Volatility$), quoted spread ($QuotedSpread$), and the market share of inverted venues ($Inverted_Share$). Standard errors are clustered by firm and date. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. t -statistics are reported in the parentheses. See Table A5 for full results.

	(1) VR	(2) VR	(3) <i>OIB_Predictability</i>	(4) <i>OIB_Predictability</i>	(5) <i>Autocorrelation</i>	(6) <i>Autocorrelation</i>
<i>Hidden_PT</i>	0.1007*** (14.57)	0.1024*** (10.93)	3.4354*** (25.85)	0.9708*** (7.25)	2.3651*** (16.47)	0.8703*** (4.53)
<i>Dark_Share</i>	0.0033 (0.44)	0.0032 (0.34)	-1.7128*** (-11.06)	-0.4121*** (-2.95)	-0.5266*** (-3.00)	0.0779 (0.41)
<i>Q x Event</i>	0.0059*** (2.59)	0.0068*** (2.73)	0.2037*** (6.17)	-0.1558*** (-5.06)	0.1904*** (3.99)	-0.0206 (-0.40)
<i>T x Event</i>	0.0001 (0.04)	-0.0003 (-0.12)	0.0564** (2.05)	-0.0018 (-0.07)	0.1059** (2.08)	0.0386 (0.77)
<i>Q</i>	0.0009 (0.48)		0.0113 (0.50)		0.0053 (0.22)	
<i>T</i>	-0.0015 (-0.72)		-0.0419 (-1.46)		-0.0612* (-1.95)	
<i>TA</i>	0.0027* (1.65)		0.0712** (2.33)		0.1641*** (4.88)	
<i>Event</i>	-0.0011 (-0.86)		-0.0262* (-1.67)		-0.0590*** (-3.52)	
<i>N</i>	801,697	801,697	713,923	713,923	785,473	785,473
Stock and date F.E.	No	Yes	No	Yes	No	Yes
Adjusted R ²	0.0230	0.0459	0.1234	0.1478	0.0442	0.0737

Table 6
Effects of hidden liquidity on trading activity

This table presents the trading activity results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is one of log of trading volume in million U.S. dollar (*DollarVolume*), log of trade size in U.S. dollars (*TradeSize*), or log of the number of trades (*Num_Trades*), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders (*Hidden_PT*) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). Standard errors are clustered by firm and date. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. t -statistics are reported in the parentheses. See Table A6 for full results.

	(1) <i>DollarVolume</i>	(2) <i>DollarVolume</i>	(3) <i>TradeSize</i>	(4) <i>TradeSize</i>	(5) <i>Num_Trades</i>	(6) <i>Num_Trades</i>
<i>Hidden_PT</i>	0.3209* (1.95)	0.6614*** (5.26)	2.9139*** (19.63)	0.8774*** (12.38)	-2.5930*** (-21.09)	-0.2160** (-2.43)
<i>Dark_Share</i>	1.8514*** (10.40)	1.2258*** (10.56)	-0.2026 (-1.58)	0.4909*** (8.16)	2.0539*** (15.01)	0.7349*** (8.69)
<i>Q x Event</i>	0.0186 (0.43)	0.0731** (2.01)	0.4948*** (17.78)	0.2363*** (12.50)	-0.4762*** (-14.10)	-0.1632*** (-6.17)
<i>T x Event</i>	0.1520*** (3.83)	0.1762*** (4.80)	0.1396*** (8.19)	0.1020*** (5.88)	0.0124 (0.39)	0.0742*** (2.66)
<i>Q</i>	0.0259 (0.62)		0.0125 (0.55)		0.0134 (0.42)	
<i>T</i>	-0.0719 (-1.36)		-0.0691** (-2.43)		-0.0028 (-0.07)	
<i>TA</i>	0.1226** (2.38)		0.1433*** (5.24)		-0.0207 (-0.50)	
<i>Event</i>	0.1748*** (8.83)		0.0056 (0.55)		0.1693*** (11.01)	
<i>N</i>	803,251	803,251	803,251	803,251	803,251	803,251
Stock and day F.E.	No	Yes	No	Yes	No	Yes
Adjusted R ²	0.7596	0.8763	0.5998	0.8151	0.7407	0.8788

Table 7
Effects of hidden liquidity on order aggressiveness

This table presents the order aggressiveness results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is the proportion of shares executed using market (OT_MO), marketable limit (OT_MLO), inside-the-quote limit orders (OT_IQL), at-the-quote limit orders (OT_AQL), or near-the-quote limit orders (OT_NQL), respectively. $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders ($Hidden_PT$) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools ($Dark_Share$) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. Control variables include the CBOE Volatility Index (VIX), the inverse of the closing price ($1 / Price$), the log of market capitalization ($MarketCap$), the log of trading volume in million U.S. dollar ($DollarVolume$), daily return volatility ($Volatility$), quoted spread ($QuotedSpread$), and the market share of inverted venues ($Inverted_Share$). Standard errors are clustered by firm and date. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. t -statistics are reported in the parentheses. See Table A7 for full results.

	(1) <i>OT_MO</i>	(2) <i>OT_MO</i>	(3) <i>OT_MLO</i>	(4) <i>OT_MLO</i>	(5) <i>OT_IQL</i>	(6) <i>OT_IQL</i>	(7) <i>OT_AQL</i>	(8) <i>OT_AQL</i>	(9) <i>OT_NQL</i>	(10) <i>OT_NQL</i>
<i>Hidden_PT</i>	-0.0013*** (-3.21)	0.0024** (2.21)	-0.2385*** (-11.25)	-0.0951*** (-3.33)	0.8170*** (17.73)	0.3253*** (8.66)	-0.4794*** (-17.79)	-0.2754*** (-7.50)	-0.0977*** (-7.64)	0.0428* (2.02)
<i>Dark_Share</i>	0.0047*** (6.52)	-0.0003 (-0.30)	0.0961*** (3.46)	-0.0689** (-2.10)	-0.4238*** (-6.57)	-0.1830*** (-4.39)	0.1181*** (3.76)	0.2017*** (5.03)	0.2049*** (8.50)	0.0505** (2.52)
<i>Q x Event</i>	0.0002 (1.07)	0.0007** (2.79)	0.0161** (2.38)	0.0486*** (7.70)	0.0101 (0.77)	-0.0594*** (-7.79)	0.0135** (2.35)	0.0288*** (4.17)	-0.0400*** (-9.36)	-0.0187*** (-4.85)
<i>T x Event</i>	0.0003** (2.26)	0.0000 (0.34)	0.0041 (1.12)	0.0047 (1.25)	0.0085** (2.21)	-0.0003 (-0.08)	-0.0201*** (-6.09)	-0.0077* (-2.04)	0.0072*** (2.88)	0.0033 (1.49)
<i>Q</i>	0.0001 (0.47)		-0.0017 (-0.74)		0.0003 (0.09)		0.0011 (0.35)		0.0003 (0.14)	
<i>T</i>	-0.0004*** (-2.93)		-0.0022 (-0.79)		-0.0051 (-1.09)		0.0102** (2.37)		-0.0025 (-1.05)	
<i>TA</i>	0.0003*** (3.21)		0.0045* (1.66)		0.0124** (2.18)		-0.0256*** (-4.88)		0.0084*** (3.84)	
<i>Event</i>	-0.0007*** (-2.64)		0.0120* (1.68)		0.0050 (1.04)		-0.0077* (-1.94)		-0.0086** (-2.51)	
<i>N</i>	38,394	38,394	38,394	38,394	38,394	38,394	38,394	38,394	38,394	38,394
Stock and day F.E.	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R ²	0.0745	0.1443	0.5057	0.6532	0.6319	0.8461	0.4725	0.6947	0.3955	0.6001

Table 8

Effects of hidden liquidity on cancellations, odd-lot volumes, and order-to-trade ratio

This table presents the algorithmic trading strategies results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is the ratio of the number of cancellation messages to the number of trades executed against displayed orders (*Cancellations*), the ratio of the odd-lot trading volume to the total trade volume (*OddLot*), or the ratio of order volumes to trade volumes that are executed against displayed orders (*OTR*), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders (*Hidden_PT*) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). Standard errors are clustered by firm and date. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. t -statistics are reported in the parentheses. See Table A8 for full results.

	(1) <i>Cancellations</i>	(2) <i>Cancellations</i>	(3) <i>OddLot</i>	(4) <i>OddLot</i>	(5) <i>OTR</i>	(6) <i>OTR</i>
<i>Hidden_PT</i>	160.93*** (8.55)	148.32*** (4.78)	0.4135*** (15.62)	-0.0207* (-1.80)	818.58** (2.40)	659.23 (0.97)
<i>Dark_Share</i>	-10.912 (-0.51)	63.881** (2.13)	-0.3956*** (-15.64)	-0.0463*** (-4.10)	49.729 (0.22)	633.65* (1.86)
<i>Q x Event</i>	-4.1993 (-1.21)	-11.733*** (-2.61)	0.0451*** (8.17)	-0.0188*** (-5.55)	-34.113 (-0.82)	-107.15* (-1.73)
<i>T x Event</i>	1.2142 (0.37)	5.5605 (1.55)	-0.0119*** (-3.15)	-0.0055* (-1.71)	-22.664 (-0.63)	5.8822 (0.15)
<i>Q</i>	1.8364 (0.82)		-0.0021 (-0.74)		3.8320 (0.14)	
<i>T</i>	-1.4589 (-0.54)		0.0062* (1.68)		13.521 (0.38)	
<i>TA</i>	3.8436 (1.27)		-0.0075* (-1.70)		1.1903 (0.03)	
<i>Event</i>	-6.7889*** (-4.78)		0.0361*** (17.13)		-54.584*** (-3.09)	
<i>N</i>	802,895	802,895	803,222	803,222	802,996	802,996
Stock and date F.E.	No	Yes	No	Yes	No	Yes
Adjusted R ²	0.0213	0.0302	0.3748	0.6114	0.0020	0.0037

Table 9

Effects of hidden liquidity on spreads and depth, separated by *TBC* groups

This table presents the spreads and depth results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where *Var* is one of quoted spread (*QuotedSpread*), effective spread (*EffectiveSpread*), realized spread (*RealizedSpread*), quoted depth (*Depth*), or the proportion of executed shares that received price improvement (*PriceImprovement*), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders (*Hidden_PT*) for stock *i* on day *t*, $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock *i* on day *t*, Q_i equals 1 if stock *i* is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock *i* is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock *i* is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date *t* is in the pilot period and 0 otherwise. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), and the market share of inverted venues (*Inverted_Share*). *TBC* is the mean percentage of one-second intervals with quoted spreads less than or equal to 5 cents during the day, measured during the pre-pilot period. For Panel A and Panel B, Columns (1) - (4) report the results for stocks with below median *TBC* values, whereas columns (5) - (8) report the results for stocks with above median *TBC* values. For Panel C, Columns (1) and (2) report the results for stocks with below median *TBC* values, whereas columns (3) and (4) report the results for stocks with above median *TBC* values. Standard errors are clustered by firm and date. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. *t*-statistics are reported in the parentheses. See Table C1 for full results.

Panel A: Regression results for the quoted spread and the effective spread

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low <i>TBC</i> group				High <i>TBC</i> group			
	<i>QuotedSpread</i>	<i>QuotedSpread</i>	<i>EffectiveSpread</i>	<i>EffectiveSpread</i>	<i>QuotedSpread</i>	<i>QuotedSpread</i>	<i>EffectiveSpread</i>	<i>EffectiveSpread</i>
$\widehat{Hidden_PT}$	3.2402*** (25.75)	2.9807*** (31.50)	0.9604*** (16.54)	0.8768*** (22.84)	2.1165*** (15.70)	1.8326*** (12.41)	0.8863*** (14.46)	0.8434*** (10.40)
$\widehat{Dark_Share}$	-0.3361** (-2.20)	-0.4590*** (-4.14)	-0.1068 (-1.23)	-0.1687*** (-2.97)	0.2814*** (3.60)	0.1196 (1.22)	0.1623*** (3.71)	0.0675 (1.23)
$Q \times Event$	0.5020*** (15.08)	0.4328*** (13.51)	0.2194*** (14.04)	0.1955*** (14.87)	0.3674*** (13.33)	0.3442*** (12.33)	0.2358*** (15.19)	0.2301*** (14.72)
$T \times Event$	0.1235*** (4.61)	0.0954*** (3.36)	0.0612*** (5.02)	0.0470*** (3.85)	0.0704** (2.27)	0.0430 (1.30)	0.0270 (1.44)	0.0176 (0.90)
Q	0.0263 (0.83)		0.0052 (0.49)		0.0035 (0.27)		0.0052 (0.87)	
T	-0.1058*** (-3.04)		-0.0310** (-2.57)		-0.0229 (-1.37)		-0.0003 (-0.03)	
TA	0.0934*** (3.15)		0.0191* (1.76)		0.0718*** (3.93)		0.0109 (1.12)	
$Event$	-0.0031 (-0.22)		-0.0187*** (-3.03)		0.0150** (2.36)		0.0004 (0.14)	
<i>N</i>	390,605	390,605	390,696	390,696	412,646	412,646	412,646	412,646

Stock and day F.E.	No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R ²	0.6436	0.7459	0.4854	0.5616	0.6916	0.7870	0.6022	0.6780
Panel B: Regression results for the realized spread and depth								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low TBC group				High TBC group			
	<i>RealizedSpread</i>	<i>RealizedSpread</i>	<i>Depth</i>	<i>Depth</i>	<i>RealizedSpread</i>	<i>RealizedSpread</i>	<i>Depth</i>	<i>Depth</i>
<i>Hidden_PT</i>	0.7947*** (19.04)	0.7318*** (23.41)	88.4379*** (5.70)	65.3952** (2.47)	0.6576*** (16.17)	0.6095*** (11.48)	1,021.6356*** (2.85)	1,241.4472*** (2.64)
<i>Dark_Share</i>	0.0094 (0.16)	-0.0262 (-0.63)	5.3905 (0.57)	-9.3425 (-0.57)	0.2205*** (6.79)	0.1921*** (4.60)	150.0797*** (2.61)	-44.8262 (-0.44)
<i>Q x Event</i>	0.1469*** (12.64)	0.1265*** (12.07)	36.0677*** (11.08)	34.6768*** (7.83)	0.1320*** (12.52)	0.1195*** (11.13)	35.3357* (1.82)	74.1036*** (4.03)
<i>T x Event</i>	0.0413*** (4.44)	0.0318*** (3.40)	3.8323 (0.97)	2.3435 (0.62)	0.0210* (1.68)	0.0122 (0.91)	54.3534** (2.07)	63.7186** (1.99)
<i>Q</i>	0.0026 (0.31)		-1.9526 (-1.29)		0.0012 (0.27)		1.2947 (0.53)	
<i>T</i>	-0.0203** (-2.23)		-2.5977 (-1.48)		-0.0030 (-0.53)		-9.5131 (-1.11)	
<i>TA</i>	0.0118 (1.42)		7.9354** (2.35)		0.0124* (1.86)		13.3941 (0.83)	
<i>Event</i>	-0.0081* (-1.66)		-0.5774 (-0.70)		-0.0000 (-0.01)		-12.4034* (-1.76)	
<i>N</i>	389,854	389,854	390,605	390,605	411,799	411,799	412,646	412,646
Stock and day F.E.	No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R ²	0.2951	0.3523	0.0706	0.1412	0.3588	0.4152	0.0444	0.1396

Panel C: Regression results for the proportion of shares that received price improvement

	(1)	(2)	(3)	(4)
	Low TBC group		High TBC group	
	<i>PriceImprovement</i>	<i>PriceImprovement</i>	<i>PriceImprovement</i>	<i>PriceImprovement</i>
<i>Hidden_PT</i>	0.7389*** (12.14)	0.4479*** (6.35)	0.5936*** (10.07)	0.5341*** (6.88)
<i>Dark_Share</i>	-0.2125*** (-5.75)	-0.0854* (-1.87)	-0.0892*** (-4.07)	-0.0983** (-2.67)
<i>Q x Event</i>	-0.0396*** (-3.85)	-0.0712*** (-7.01)	-0.0178*** (-3.30)	-0.0179** (-2.21)
<i>T x Event</i>	0.0049 (1.01)	0.0096* (1.74)	0.0015 (0.76)	0.0012 (0.52)
<i>Q</i>	0.0036 (0.88)		-0.0014 (-1.08)	
<i>T</i>	-0.0033 (-0.72)		0.0004 (0.23)	
<i>TA</i>	0.0031 (0.73)		0.0025* (1.72)	
<i>Event</i>	-0.0015 (-0.40)		-0.0027 (-1.03)	
<i>N</i>	19,133	19,133	19,261	19,261
Stock and month F.E.	No	Yes	No	Yes
Adjusted R ²	0.5895	0.7438	0.4058	0.5798

Table 10

Effects of hidden liquidity on informed trading and price impact, separated by *TBC* groups

This table presents the informed trading and price impact results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where *Var* is the probability of informed trading (*PIN*), the price nonsynchronicity measure based on the Fama-French 3 factors (*PSI_FF3*), the price nonsynchronicity measure based on the market and industry returns (*PSI_BSW*), or the price impact (*PriceImpact*), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders (*Hidden_PT*) for stock *i* on day *t*, $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock *i* on day *t*, Q_i equals 1 if stock *i* is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock *i* is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock *i* is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date *t* is in the pilot period and 0 otherwise. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). *TBC* is the mean percentage of one-second intervals with quoted spreads less than or equal to 5 cents during the day, measured during the pre-pilot period. Columns (1) - (4) report the results for stocks with below median *TBC* values, whereas columns (5) - (8) report the results for stocks with above median *TBC* values. Standard errors are clustered by firm and date in columns (1), (2), (7), and (8), and by firm in columns (3) - (6). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. *t*-statistics are reported in the parentheses. See Table C2 for full results.

Panel A: Regression results for *PIN* and *PSI_FF3*

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low <i>TBC</i> group				High <i>TBC</i> group			
	<i>PIN</i>	<i>PIN</i>	<i>PSI_FF3</i>	<i>PSI_FF3</i>	<i>PIN</i>	<i>PIN</i>	<i>PSI_FF3</i>	<i>PSI_FF3</i>
$\widehat{Hidden_PT}$	0.0487** (2.27)	0.0816 (1.31)	-0.5214 (-1.30)	2.6437* (1.95)	0.1344*** (3.56)	0.1045** (2.67)	1.2541** (2.08)	0.6609 (0.67)
$\widehat{Dark_Share}$	-0.0569* (-1.95)	-0.0792* (-1.73)	2.2479*** (4.80)	-1.6249* (-1.73)	-0.0577** (-2.27)	-0.0591 (-1.13)	4.5899*** (9.05)	-1.3426 (-1.06)
$Q \times Event$	0.0220*** (3.63)	0.0263** (2.50)	0.2435*** (2.71)	0.7352*** (3.45)	0.0144** (2.25)	0.0133 (1.29)	-0.4725*** (-4.29)	0.1927 (0.94)
$T \times Event$	0.0020 (0.42)	-0.0001 (-0.01)	0.1106 (1.52)	-0.0442 (-0.56)	-0.0053 (-0.95)	-0.0059 (-0.98)	0.2633*** (3.46)	-0.0162 (-0.19)
<i>Q</i>	-0.0026 (-0.71)		-0.1087* (-1.75)		0.0010 (0.27)		0.0483 (0.83)	
<i>T</i>	0.0003 (0.06)		0.0128 (0.16)		-0.0004 (-0.08)		-0.1559** (-2.15)	
<i>TA</i>	0.0010 (0.24)		0.2938*** (3.97)		0.0002 (0.07)		0.1595** (2.19)	
<i>Event</i>	0.0022 (0.58)		-0.1694*** (-4.06)		0.0058* (1.94)		0.0141 (0.40)	
<i>N</i>	18,998	18,998	5,928	5,928	19,290	19,290	5,921	5,921
Stock and date F.E.	No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R ²	0.2117	0.2747	0.4153	0.5856	0.1645	0.2230	0.3465	0.5507

Panel B: Regression results for *PSI_BSW* and *PriceImpact*

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low TBC group				High TBC group			
	<i>PSI_BSW</i>	<i>PSI_BSW</i>	<i>PriceImpact</i>	<i>PriceImpact</i>	<i>PSI_BSW</i>	<i>PSI_BSW</i>	<i>PriceImpact</i>	<i>PriceImpact</i>
<i>Hidden_PT</i>	-0.2210 (-0.47)	2.4230 (1.35)	0.1651*** (8.27)	0.1430*** (7.87)	1.7390** (2.39)	1.2922 (1.02)	0.2281*** (8.38)	0.2335*** (6.63)
<i>Dark_Share</i>	3.1509*** (5.30)	-1.6971 (-1.31)	-0.1155*** (-3.44)	-0.1414*** (-4.72)	5.3672*** (8.76)	-2.0164 (-1.36)	-0.0577*** (-3.45)	-0.1238*** (-5.44)
<i>Q x Event</i>	0.2322** (2.10)	0.7283** (2.55)	0.0723*** (11.48)	0.0687*** (11.69)	-0.5637*** (-4.20)	0.2664 (1.10)	0.1036*** (17.21)	0.1103*** (17.73)
<i>T x Event</i>	0.2236** (2.55)	0.0018 (0.02)	0.0200*** (3.70)	0.0153*** (2.81)	0.2979*** (3.15)	-0.0482 (-0.50)	0.0060 (0.80)	0.0055 (0.73)
<i>Q</i>	-0.1288* (-1.79)		0.0026 (0.62)		0.0724 (0.97)		0.0040 (1.56)	
<i>T</i>	0.0509 (0.55)		-0.0106** (-2.16)		-0.1669* (-1.82)		0.0028 (0.78)	
<i>TA</i>	0.3270*** (3.79)		0.0073* (1.85)		0.1732* (1.94)		-0.0016 (-0.40)	
<i>Event</i>	0.1125** (2.30)		-0.0105*** (-4.15)		0.3522*** (8.26)		0.0005 (0.37)	
<i>N</i>	5,771	5,771	389,854	389,854	5,843	5,843	411,799	411,799
Stock and date F.E.	No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R ²	0.4240	0.5622	0.2167	0.2429	0.3585	0.5743	0.3443	0.3801

Table 11

Effects of hidden liquidity on other measures of informed trading, separated by *TBC* groups

This table presents the pricing efficiency results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where *Var* is one of the variance ratio (*VR*), the return predictability of order imbalance (*OIB_Predictability*), or the return autocorrelation (*Autocorrelation*), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders (*Hidden_PT*) for stock *i* on day *t*, $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock *i* on day *t*, Q_i equals 1 if stock *i* is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock *i* is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock *i* is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date *t* is in the pilot period and 0 otherwise. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). *TBC* is the mean percentage of one-second intervals with quoted spreads less than or equal to 5 cents during the day, measured during the pre-pilot period. We include the fixed effects. Columns (1) - (3) report the results for stocks with below median *TBC* values, whereas columns (4) - (6) report the results for stocks with above median *TBC* values. Standard errors are clustered by firm and date. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. *t*-statistics are reported in the parentheses. See Table C3 for full results.

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>VR</i>	Low <i>TBC</i> group <i>OIB_Predictability</i>	<i>Autocorrelation</i>	<i>VR</i>	High <i>TBC</i> group <i>OIB_Predictability</i>	<i>Autocorrelation</i>
$\widehat{Hidden_PT}$	0.0645*** (5.21)	1.2103*** (6.21)	1.1403*** (5.64)	0.1187*** (6.94)	0.8551*** (3.56)	0.3195 (0.79)
$\widehat{Dark_Share}$	0.0082 (0.63)	-0.4241** (-2.07)	0.3056 (1.30)	0.0002 (0.02)	-0.4175** (-2.09)	0.1714 (0.65)
$Q \times Event$	-0.0046 (-1.38)	0.0046 (0.10)	0.1082** (2.00)	0.0119*** (3.28)	-0.2694*** (-6.07)	-0.0949 (-1.17)
$T \times Event$	-0.0046 (-1.50)	0.0150 (0.40)	-0.0524 (-1.15)	0.0035 (1.01)	-0.0220 (-0.62)	0.1148 (1.38)
<i>N</i>	390,157	331,849	378,814	411,540	382,074	406,659
Stock and date F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.0449	0.1469	0.0798	0.0381	0.0892	0.0443

**Online Appendix for
“Hidden Liquidity, Market Quality, and
Order Submission Strategies”**

Table A1

Effects of the Tick Size Pilot Program on hidden liquidity and the market share of dark pools

This table presents the results of regression models (5) and (6): $Hidden_PT_{i,t} = \beta_1 Q_i \times Event_t + \beta_2 T_i \times Event_t + \beta_3 TA_i \times Event_t + \beta_4 Hidden_PT_{i,t-1} + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$ and $Dark_Share_{i,t} = \beta_1 Q_i \times Event_t + \beta_2 T_i \times Event_t + \beta_3 TA_i \times Event_t + \beta_4 Dark_Share_{i,t-1} + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where $Hidden_PT_{i,t}$ is the percentage of transactions against hidden orders for stock i on day t , $Dark_Share_{i,t}$ is the percentage market share of dark pools for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (VIX), the inverse of closing price ($1 / Price$), the log of market capitalization ($MarketCap$), the log of trading volume in million U.S. dollar ($DollarVolume$), daily return volatility ($Volatility$), quoted spread ($QuotedSpread$), and the market share of inverted venues ($Inverted_Share$). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1) <i>Hidden_PT</i>	(2) <i>Hidden_PT</i>	(3) <i>Hidden_PT</i>	(4) <i>Hidden_PT</i>	(5) <i>Dark_Share</i>	(6) <i>Dark_Share</i>
<i>Q x Event</i>	-0.0233*** (-11.43)	-0.0479*** (-14.66)	-0.0488*** (-22.80)	-0.0655*** (-23.24)	0.0787*** (25.55)	0.0847*** (25.83)
<i>T x Event</i>	0.0014 (0.59)	0.0013 (0.31)	0.0002 (0.06)	-0.0001 (-0.03)	-0.0003 (-0.10)	-0.0013 (-0.35)
<i>TA x Event</i>	-0.0188*** (-7.26)	-0.0413*** (-9.82)	-0.0291*** (-10.90)	-0.0438*** (-11.88)	-0.0464*** (-11.70)	-0.0563*** (-13.31)
<i>Hidden_PT (t - 1)</i>	0.6524*** (83.87)	0.3783*** (65.15)	0.5164*** (56.64)	0.2969*** (58.15)		
<i>Dark_Share (t - 1)</i>					0.2786*** (47.15)	0.1889*** (67.28)
<i>Q</i>	0.0018 (0.85)		0.0008 (0.37)		0.0015 (0.69)	
<i>T</i>	-0.0017 (-0.66)		-0.0006 (-0.25)		-0.0011 (-0.42)	
<i>TA</i>	0.0012 (0.43)		0.0013 (0.49)		0.0021 (0.76)	
<i>Event</i>	0.0047*** (4.66)		0.0027*** (2.69)		-0.0112*** (-5.16)	
<i>VIX</i>	-0.0003*** (-2.60)		-0.0004*** (-3.37)		-0.0016*** (-6.35)	
<i>1 / Price</i>	-0.0681*** (-5.05)	-0.0175*** (-3.58)	-0.0963*** (-5.25)	-0.0302*** (-3.64)	0.1089*** (4.83)	0.0676*** (4.09)
<i>MarketCap</i>	-0.0025***	-0.0093***	-0.0039***	-0.0046***	-0.0444***	-0.0396***

	(-2.61)	(-5.29)	(-3.58)	(-3.15)	(-25.91)	(-17.80)
<i>DollarVolume</i>	-0.0091***	-0.0031***	0.0074***	0.0130***	0.0198***	0.0308***
	(-12.38)	(-4.63)	(12.21)	(24.20)	(21.87)	(33.04)
<i>Volatility</i>	0.4503***	0.8613***	-0.4871***	-0.1400***	-0.1852***	-0.6563***
	(10.38)	(22.70)	(-11.46)	(-4.53)	(-3.22)	(-16.79)
<i>QuotedSpread</i>			0.0727***	0.0764***		
			(44.48)	(58.94)		
<i>Inverted_Share</i>	0.0303***	0.1688***	0.1133***	0.1829***	-0.6212***	-0.6424***
	(3.87)	(21.03)	(14.27)	(26.68)	(-40.45)	(-62.49)
Constant	0.1216***		0.1440***		1.2051***	
	(6.09)		(6.21)		(34.67)	
<i>N</i>	803,462	803,462	803,251	803,251	803,938	803,938
Stock and date F.E.	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.5314	0.6144	0.5917	0.6589	0.4093	0.4753
<i>F</i> -statistic	1,961.16	966.28	1,542.79	1,318.76	1,041.33	619.97

Table A2

Effects of the Tick Size Pilot Program on hidden liquidity measures, separated by *TBC* groups

This table presents the results of regression model (5): $Hidden_PT_{i,t} = \beta_1 Q_i \times Event_t + \beta_2 T_i \times Event_t + \beta_3 TA_i \times Event_t + \beta_4 Hidden_PT_{i,t-1} + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where $Hidden_PT_{i,t}$ is the percentage of transactions against hidden orders for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). *TBC* is the mean percentage of one-second intervals with quoted spreads less than or equal to 5 cents during the day, measured during the pre-pilot period. Columns (1) through (4) report the results for stocks with below median *TBC* values, whereas columns (5) through (8) report the results for stocks with above median *TBC* values. N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low <i>TBC</i> group				High <i>TBC</i> group			
	<i>Hidden_PT</i>	<i>Hidden_PT</i>	<i>Hidden_PT</i>	<i>Hidden_PT</i>	<i>Hidden_PT</i>	<i>Hidden_PT</i>	<i>Hidden_PT</i>	<i>Hidden_PT</i>
<i>Q x Event</i>	-0.0543*** (-18.46)	-0.0888*** (-24.80)	-0.0780*** (-23.60)	-0.0996*** (-30.57)	0.0033 (1.31)	-0.0027 (-0.84)	-0.0211*** (-8.15)	-0.0261*** (-8.74)
<i>T x Event</i>	0.0015 (0.48)	0.0014 (0.34)	-0.0019 (-0.60)	-0.0013 (-0.33)	-0.0008 (-0.25)	-0.0012 (-0.30)	-0.0002 (-0.07)	-0.0008 (-0.19)
<i>TA x Event</i>	-0.0241*** (-6.99)	-0.0421*** (-8.74)	-0.0334*** (-9.51)	-0.0454*** (-10.79)	-0.0192*** (-5.78)	-0.0311*** (-7.64)	-0.0259*** (-7.86)	-0.0341*** (-9.04)
<i>Hidden_PT (t - 1)</i>	0.5572*** (50.12)	0.3197*** (59.16)	0.4165*** (28.51)	0.2456*** (49.83)	0.5044*** (61.46)	0.3798*** (47.60)	0.4388*** (55.58)	0.3306*** (46.65)
<i>Q</i>	-0.0004 (-0.10)		-0.0022 (-0.72)		-0.0005 (-0.29)		-0.0007 (-0.41)	
<i>T</i>	-0.0021 (-0.47)		0.0033 (0.85)		0.0000 (0.01)		-0.0009 (-0.44)	
<i>TA</i>	0.0005 (0.11)		-0.0001 (-0.02)		-0.0006 (-0.26)		-0.0000 (-0.00)	
<i>Event</i>	0.0033** (2.00)		0.0021 (1.26)		0.0045*** (4.76)		0.0031*** (3.43)	
<i>VIX</i>	-0.0001 (-0.79)		-0.0004** (-2.14)		-0.0005*** (-5.65)		-0.0005*** (-6.20)	
$1 / Price$	-0.1258*** (-2.75)	-0.0707*** (-2.60)	-0.1978*** (-2.83)	-0.0805** (-2.12)	0.0063 (1.54)	0.0076 (1.31)	-0.0236*** (-3.53)	-0.0095 (-1.48)
<i>MarketCap</i>	-0.0009 (-0.45)	-0.0017 (-0.56)	-0.0028 (-1.08)	0.0010 (0.40)	-0.0088*** (-10.81)	-0.0140*** (-8.22)	-0.0085*** (-10.03)	-0.0078*** (-5.56)

<i>DollarVolume</i>	-0.0140*** (-11.59)	-0.0113*** (-10.90)	0.0053*** (5.92)	0.0079*** (9.04)	0.0012** (2.16)	0.0055*** (9.34)	0.0103*** (17.77)	0.0154*** (27.48)
<i>Volatility</i>	0.5484*** (6.69)	1.0538*** (21.88)	-0.7243*** (-9.30)	-0.2074*** (-5.52)	0.2491*** (6.48)	0.3877*** (8.01)	-0.1563*** (-3.64)	-0.0970** (-2.18)
<i>QuotedSpread</i>			0.0768*** (32.87)	0.0753*** (56.44)			0.0577*** (17.50)	0.0645*** (21.50)
<i>Inverted_Share</i>	0.0643*** (4.83)	0.1993*** (17.76)	0.1469*** (11.78)	0.2075*** (21.25)	-0.0180*** (-3.34)	0.0517*** (8.97)	0.0471*** (7.83)	0.0875*** (15.80)
Constant	0.1365*** (3.22)		0.1749*** (3.08)		0.2421*** (14.57)		0.2252*** (12.88)	
<i>N</i>	390,816	390,816	390,605	390,605	412,646	412,646	412,646	412,646
Stock and date F.E.	No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.4885	0.5736	0.5642	0.6248	0.3371	0.3979	0.3828	0.4374
<i>F</i> -statistic	1,447.73	870.99	1,395.90	1,214.66	538.49	544.98	601.32	596.50

Table A3
Effects of hidden liquidity on spreads and depth

This table presents the spreads and depth results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is one of quoted spread (*QuotedSpread*), effective spread (*EffectiveSpread*), realized spread (*RealizedSpread*), quoted depth (*Depth*), or the proportion of executed shares that received price improvement (*PriceImprovement*), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders (*Hidden_PT*) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), and the market share of inverted venues (*Inverted_Share*). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

Panel A: Regression results for spread measures						
	(1) <i>QuotedSpread</i>	(2) <i>QuotedSpread</i>	(3) <i>EffectiveSpread</i>	(4) <i>EffectiveSpread</i>	(5) <i>RealizedSpread</i>	(6) <i>RealizedSpread</i>
$\widehat{Hidden_PT}$	2.8312*** (34.21)	2.6970*** (32.03)	0.8798*** (26.34)	0.9373*** (24.47)	0.7180*** (30.24)	0.7211*** (26.28)
$\widehat{Dark_Share}$	-0.1553** (-1.98)	-0.1371* (-1.70)	0.0517 (1.19)	-0.0137 (-0.32)	0.1007*** (3.31)	0.0963*** (3.01)
$Q \times Event$	0.4238*** (21.09)	0.3637*** (16.07)	0.2199*** (20.76)	0.2117*** (18.98)	0.1381*** (18.76)	0.1229*** (15.48)
$T \times Event$	0.1001*** (4.85)	0.0765*** (3.29)	0.0443*** (3.83)	0.0349*** (2.81)	0.0311*** (3.98)	0.0236*** (2.75)
Q	0.0103 (0.54)		0.0039 (0.59)		0.0009 (0.18)	
T	-0.0547** (-2.55)		-0.0128* (-1.71)		-0.0099* (-1.79)	
TA	0.0850*** (4.44)		0.0143* (1.92)		0.0125** (2.30)	
$Event$	0.0117* (1.66)		-0.0061** (-2.04)		-0.0024 (-0.94)	
VIX	0.0014** (2.14)		0.0010*** (3.47)		0.0020*** (8.08)	
$1 / Price$	0.6036*** (6.78)	0.2264*** (2.84)	0.4801*** (7.05)	0.2786*** (5.12)	0.3400*** (8.26)	0.2122*** (6.50)
$MarketCap$	0.0149	-0.0448***	-0.0068	-0.0474***	0.0026	-0.0167***

	(1.59)	(-2.76)	(-1.63)	(-5.86)	(0.92)	(-3.14)
<i>DollarVolume</i>	-0.1971***	-0.1973***	-0.0649***	-0.0587***	-0.0500***	-0.0464***
	(-33.18)	(-33.23)	(-22.08)	(-21.65)	(-24.25)	(-22.58)
<i>Volatility</i>	11.642***	10.730***	4.3698***	3.8419***	-0.4097***	-1.0294***
	(27.48)	(32.30)	(23.44)	(30.11)	(-3.37)	(-9.51)
<i>Inverted_Share</i>	-1.2609***	-0.6589***	-0.3181***	-0.1715***	-0.3188***	-0.1437***
	(-16.97)	(-10.76)	(-8.95)	(-6.38)	(-11.56)	(-6.29)
Constant	-0.3556*		0.0926		-0.0949	
	(-1.79)		(1.02)		(-1.53)	
<i>N</i>	803,251	803,251	803,342	803,342	801,653	801,653
Stock and date F.E.	No	Yes	No	Yes	No	Yes
Adjusted <i>R</i> ²	0.6488	0.7632	0.5236	0.6050	0.3250	0.3851

Panel B: Regression results for depth and proportion of shares that received price improvement

	(1)	(2)	(3)	(4)
	<i>Depth</i>	<i>Depth</i>	<i>PriceImprovement</i>	<i>PriceImprovement</i>
<i>Hidden_PT</i>	226.19***	460.27***	0.8078***	0.5546***
	(4.20)	(3.28)	(18.03)	(8.30)
<i>Dark_Share</i>	156.80***	11.6755	-0.1886***	-0.0543
	(2.74)	(0.19)	(-7.75)	(-1.66)
<i>Q x Event</i>	40.276***	74.350***	-0.0167**	-0.0431***
	(4.67)	(7.53)	(-2.52)	(-5.13)
<i>T x Event</i>	26.200**	31.170**	0.0037	0.0076**
	(2.02)	(2.00)	(1.30)	(2.09)
<i>Q</i>	-1.8571		0.0024	
	(-1.26)		(0.93)	
<i>T</i>	-3.5679		-0.0030	
	(-0.76)		(-1.03)	
<i>TA</i>	8.2877		0.0036	
	(0.93)		(1.39)	
<i>Event</i>	-1.2614		-0.0030	
	(-0.41)		(-1.13)	
<i>VIX</i>	0.1699		0.0009***	
	(0.76)		(2.71)	
<i>1 / Price</i>	73.211***	21.818***	0.0255	-0.0099
	(3.60)	(3.02)	(1.47)	(-1.01)
<i>MarketCap</i>	1.4072	-18.754***	0.0032**	0.0070***
	(0.35)	(-2.94)	(2.10)	(3.06)
<i>DollarVolume</i>	18.094***	23.662***	0.0020*	0.0020

	(5.79)	(4.50)	(1.65)	(1.53)
<i>Volatility</i>	-1,075.8***	-1,647.8***	-0.0201	-0.2109**
	(-3.93)	(-3.57)	(-0.20)	(-2.15)
<i>QuotedSpread</i>			-0.0332***	-0.0035
			(-5.17)	(-0.44)
<i>Inverted_Share</i>	296.14***	87.108**	0.0356	0.0560
	(3.94)	(2.43)	(0.73)	(1.49)
Constant	-128.70		-0.0254	
	(-1.29)		(-0.71)	
<i>N</i>	803,251	803,251	38,394	38,394
Stock and date F.E.	No	Yes	No	Yes
Adjusted R^2	0.0330	0.1316	0.6241	0.7830

Table A4

Effects of hidden liquidity on informed trading and price impact

This table presents the informed trading and price impact results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is the probability of informed trading (PIN), the price nonsynchronicity measure based on the Fama-French 3 factors (PSI_FF3), the price nonsynchronicity measure based on the market and industry returns (PSI_BSW), or the price impact ($PriceImpact$), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders ($Hidden_PT$) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools ($Dark_Share$) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (VIX), the inverse of the closing price ($1 / Price$), the log of market capitalization ($MarketCap$), the log of trading volume in million U.S. dollar ($DollarVolume$), daily return volatility ($Volatility$), quoted spread ($QuotedSpread$), and the market share of inverted venues ($Inverted_Share$). N indicates the sample size. Standard errors are clustered by firm and date in columns 1, 2, 7, and 8, and by firm in columns 3, 4, 5, and 6. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1) <i>PIN</i>	(2) <i>PIN</i>	(3) <i>PSI_FF3</i>	(4) <i>PSI_FF3</i>	(5) <i>PSI_BSW</i>	(6) <i>PSI_BSW</i>	(7) <i>PriceImpact</i>	(8) <i>PriceImpact</i>
<i>Hidden_PT</i>	0.0919*** (6.10)	0.0855** (2.21)	0.1568 (0.65)	2.0773*** (3.49)	0.4452 (1.57)	2.2843*** (2.85)	0.1615*** (12.61)	0.2149*** (12.79)
<i>Dark_Share</i>	-0.0558*** (-3.14)	-0.0659** (-2.17)	3.2805*** (10.51)	-1.4185** (-2.30)	3.9275*** (10.30)	-1.7377** (-2.22)	-0.0487*** (-2.73)	-0.1091*** (-5.45)
<i>Q x Event</i>	0.0226*** (5.39)	0.0230*** (3.24)	0.0475 (0.77)	0.4837*** (4.57)	-0.0042 (-0.06)	0.5080*** (3.70)	0.0816*** (17.73)	0.0885*** (18.67)
<i>T x Event</i>	-0.0004 (-0.10)	-0.0022 (-0.49)	0.2162*** (4.13)	-0.0201 (-0.36)	0.2923*** (4.56)	-0.0079 (-0.12)	0.0132*** (2.67)	0.0113** (2.25)
<i>Q</i>	-0.0007 (-0.24)		-0.0325 (-0.76)		-0.0288 (-0.55)		0.0030 (1.13)	
<i>T</i>	-0.0011 (-0.33)		-0.0905* (-1.68)		-0.0814 (-1.25)		-0.0029 (-0.91)	
<i>TA</i>	0.0013 (0.45)		0.2718*** (5.26)		0.2975*** (4.81)		0.0017 (0.59)	
<i>Event</i>	0.0044 (1.52)		-0.0627** (-2.28)		0.2445*** (7.47)		-0.0036*** (-2.64)	
<i>VIX</i>	0.0006** (2.11)		-0.0744*** (-21.78)		-0.0517*** (-12.53)		-0.0010*** (-6.77)	
<i>1 / Price</i>	0.0184*** (2.81)	0.0217*** (3.44)	-0.2113 (-1.21)	0.7078** (2.51)	-0.3925** (-2.19)	0.7544*** (3.38)	0.1399*** (5.01)	0.0662*** (2.89)
<i>MarketCap</i>	-0.0122*** (-7.45)	-0.0156*** (-4.64)	-0.4267*** (-14.88)	-0.5468*** (-9.71)	-0.5008*** (-15.09)	-0.5491*** (-8.65)	-0.0094*** (-4.93)	-0.0309*** (-7.86)

<i>DollarVolume</i>	-0.0121*** (-11.05)	-0.0022 (-1.40)	0.2123*** (11.52)	0.2651*** (11.11)	0.1731*** (7.93)	0.2541*** (8.90)	-0.0150*** (-12.28)	-0.0123*** (-9.75)
<i>Volatility</i>	-1.2401*** (-10.34)	-1.1184*** (-6.62)					4.7774*** (41.92)	4.8717*** (51.26)
<i>QuotedSpread</i>	0.0153*** (5.10)	0.0169*** (3.64)	0.3739*** (8.79)	0.1357* (1.80)	0.3652*** (7.09)	0.1278 (1.35)		
<i>Inverted_Share</i>	-0.1394*** (-5.70)	-0.1333*** (-4.01)	-1.2771*** (-3.26)	-2.6106*** (-4.65)	-0.9397** (-1.98)	-2.6785*** (-3.82)	0.0012 (0.08)	-0.0267* (-1.90)
Constant	0.4245*** (12.17)		9.4456*** (14.80)		10.435*** (13.90)		0.1878*** (4.48)	
<i>N</i>	38,288	38,288	11,849	11,849	11,614	11,614	801,653	801,653
Stock and date F.E.	No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.2139	0.2752	0.3844	0.5717	0.3977	0.5723	0.2555	0.2878

Table A5

Effects of hidden liquidity on other measures of informed trading

This table presents the pricing efficiency results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is one of the variance ratio (VR), the return predictability of order imbalance ($OIB_Predictability$), or the return autocorrelation ($Autocorrelation$), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders ($Hidden_PT$) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools ($Dark_Share$) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (VIX), the inverse of the closing price ($1 / Price$), the log of market capitalization ($MarketCap$), the log of trading volume in million U.S. dollar ($DollarVolume$), daily return volatility ($Volatility$), quoted spread ($QuotedSpread$), and the market share of inverted venues ($Inverted_Share$). Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1) VR	(2) VR	(3) <i>OIB_Predictability</i>	(4) <i>OIB_Predictability</i>	(5) <i>Autocorrelation</i>	(6) <i>Autocorrelation</i>
<i>Hidden_PT</i>	0.1007*** (14.57)	0.1024*** (10.93)	3.4354*** (25.85)	0.9708*** (7.25)	2.3651*** (16.47)	0.8703*** (4.53)
<i>Dark_Share</i>	0.0033 (0.44)	0.0032 (0.34)	-1.7128*** (-11.06)	-0.4121*** (-2.95)	-0.5266*** (-3.00)	0.0779 (0.41)
<i>Q x Event</i>	0.0059*** (2.59)	0.0068*** (2.73)	0.2037*** (6.17)	-0.1558*** (-5.06)	0.1904*** (3.99)	-0.0206 (-0.40)
<i>T x Event</i>	0.0001 (0.04)	-0.0003 (-0.12)	0.0564** (2.05)	-0.0018 (-0.07)	0.1059** (2.08)	0.0386 (0.77)
<i>Q</i>	0.0009 (0.48)		0.0113 (0.50)		0.0053 (0.22)	
<i>T</i>	-0.0015 (-0.72)		-0.0419 (-1.46)		-0.0612* (-1.95)	
<i>TA</i>	0.0027* (1.65)		0.0712** (2.33)		0.1641*** (4.88)	
<i>Event</i>	-0.0011 (-0.86)		-0.0262* (-1.67)		-0.0590*** (-3.52)	
<i>VIX</i>	-0.0000 (-0.33)		0.0015 (0.86)		-0.0141*** (-7.58)	
<i>1 / Price</i>	-0.0076* (-1.65)	-0.0065 (-1.61)	-0.6579*** (-2.97)	-0.5795*** (-5.95)	-0.2834* (-1.66)	-0.2433*** (-3.36)
<i>MarketCap</i>	0.0026*** (3.18)	-0.0014 (-1.00)	-0.0505*** (-3.68)	0.0191 (0.93)	-0.0680*** (-4.34)	0.0228 (0.80)
<i>DollarVolume</i>	-0.0127***	-0.0090***	-0.4085***	-0.4635***	-0.1761***	-0.2021***

	(-25.56)	(-15.56)	(-43.26)	(-48.30)	(-18.55)	(-17.67)
<i>Volatility</i>			-4.6073***	6.5274***	-1.3002*	9.9372***
			(-7.55)	(12.87)	(-1.92)	(12.24)
<i>QuotedSpread</i>	-0.0089***	-0.0102***	0.0802***	0.0277	-0.0595**	-0.2101***
	(-8.89)	(-9.33)	(3.68)	(1.46)	(-2.48)	(-9.21)
<i>Inverted_Share</i>	-0.0192***	-0.0163**	-2.6210***	-0.4008***	-2.7516***	-1.1186***
	(-2.72)	(-2.07)	(-17.25)	(-3.17)	(-15.37)	(-7.64)
Constant	0.0904***		-3.2421***		-3.6032***	
	(5.12)		(-10.71)		(-10.13)	
<i>N</i>	801,697	801,697	713,923	713,923	785,473	785,473
Stock and date F.E.	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.0230	0.0459	0.1234	0.1478	0.0442	0.0737

Table A6
Effects of hidden liquidity on trading activity

This table presents the trading activity results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is one of log of trading volume in million U.S. dollar ($DollarVolume$), log of trade size in U.S. dollars ($TradeSize$), or log of the number of trades (Num_Trades), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders ($Hidden_PT$) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools ($Dark_Share$) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (VIX), the inverse of the closing price ($1 / Price$), the log of market capitalization ($MarketCap$), daily return volatility ($Volatility$), quoted spread ($QuotedSpread$), and the market share of inverted venues ($Inverted_Share$). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1) <i>DollarVolume</i>	(2) <i>DollarVolume</i>	(3) <i>TradeSize</i>	(4) <i>TradeSize</i>	(5) <i>Num_Trades</i>	(6) <i>Num_Trades</i>
<i>Hidden_PT</i>	0.3209* (1.95)	0.6614*** (5.26)	2.9139*** (19.63)	0.8774*** (12.38)	-2.5930*** (-21.09)	-0.2160** (-2.43)
<i>Dark_Share</i>	1.8514*** (10.40)	1.2258*** (10.56)	-0.2026 (-1.58)	0.4909*** (8.16)	2.0539*** (15.01)	0.7349*** (8.69)
<i>Q x Event</i>	0.0186 (0.43)	0.0731** (2.01)	0.4948*** (17.78)	0.2363*** (12.50)	-0.4762*** (-14.10)	-0.1632*** (-6.17)
<i>T x Event</i>	0.1520*** (3.83)	0.1762*** (4.80)	0.1396*** (8.19)	0.1020*** (5.88)	0.0124 (0.39)	0.0742*** (2.66)
<i>Q</i>	0.0259 (0.62)		0.0125 (0.55)		0.0134 (0.42)	
<i>T</i>	-0.0719 (-1.36)		-0.0691** (-2.43)		-0.0028 (-0.07)	
<i>TA</i>	0.1226** (2.38)		0.1433*** (5.24)		-0.0207 (-0.50)	
<i>Event</i>	0.1748*** (8.83)		0.0056 (0.55)		0.1693*** (11.01)	
<i>VIX</i>	-0.0070*** (-3.32)		-0.0008 (-0.95)		-0.0062*** (-3.41)	
<i>1 / Price</i>	-0.7651*** (-4.62)	-0.8087*** (-3.99)	-0.8962*** (-3.66)	-0.7538*** (-5.31)	0.1311 (1.15)	-0.0549 (-0.71)
<i>MarketCap</i>	1.0638*** (50.04)	0.4108*** (13.65)	0.3360*** (29.29)	0.1533*** (8.60)	0.7278*** (48.19)	0.2575*** (13.22)

<i>Volatility</i>	30.7689*** (39.53)	27.3551*** (52.20)	-3.6192*** (-9.92)	2.1909*** (11.38)	34.3880*** (53.27)	25.1641*** (60.64)
<i>QuotedSpread</i>	-0.8730*** (-26.77)	-0.6494*** (-28.98)	-0.3327*** (-16.61)	-0.1937*** (-20.52)	-0.5403*** (-23.24)	-0.4557*** (-28.73)
<i>Inverted_Share</i>	-0.2070 (-1.29)	-0.9179*** (-9.84)	-2.5890*** (-21.68)	-1.3381*** (-25.06)	2.3820*** (19.08)	0.4202*** (6.64)
Constant	-21.4270*** (-46.04)		0.9234*** (3.82)		-8.5348*** (-25.83)	
<i>N</i>	803,251	803,251	803,251	803,251	803,251	803,251
Stock and day F.E.	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.7596	0.8763	0.5998	0.8151	0.7407	0.8788

Table A7

Effects of hidden liquidity on order aggressiveness

This table presents the order aggressiveness results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is the proportion of shares executed using market (OT_MO), marketable limit (OT_MLO), inside-the-quote limit orders (OT_IQL), at-the-quote limit orders (OT_AQL), or near-the-quote limit orders (OT_NQL), respectively. $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders (*Hidden_PT*) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 2015 to September 2016, and the pilot period is from November 2016 to October 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

Panel A: Regression results without stock and month fixed effects					
	(1) <i>OT_MO</i>	(2) <i>OT_MLO</i>	(3) <i>OT_IQL</i>	(4) <i>OT_AQL</i>	(5) <i>OT_NQL</i>
$\widehat{Hidden_PT}$	-0.0013*** (-3.21)	-0.2385*** (-11.25)	0.8170*** (17.73)	-0.4794*** (-17.79)	-0.0977*** (-7.64)
$\widehat{Dark_Share}$	0.0047*** (6.52)	0.0961*** (3.46)	-0.4238*** (-6.57)	0.1181*** (3.76)	0.2049*** (8.50)
$Q \times Event$	0.0002 (1.07)	0.0161** (2.38)	0.0101 (0.77)	0.0135** (2.35)	-0.0400*** (-9.36)
$T \times Event$	0.0003** (2.26)	0.0041 (1.12)	0.0085** (2.21)	-0.0201*** (-6.09)	0.0072*** (2.88)
Q	0.0001 (0.47)	-0.0017 (-0.74)	0.0003 (0.09)	0.0011 (0.35)	0.0003 (0.14)
T	-0.0004*** (-2.93)	-0.0022 (-0.79)	-0.0051 (-1.09)	0.0102** (2.37)	-0.0025 (-1.05)
TA	0.0003*** (3.21)	0.0045* (1.66)	0.0124** (2.18)	-0.0256*** (-4.88)	0.0084*** (3.84)
$Event$	-0.0007*** (-2.64)	0.0120* (1.68)	0.0050 (1.04)	-0.0077* (-1.94)	-0.0086** (-2.51)
VIX	-0.0001*** (-2.68)	-0.0006 (-0.62)	0.0005 (0.88)	-0.0007* (-1.70)	0.0009* (1.69)
$1 / Price$	-0.0002 (-0.55)	0.0329** (2.23)	-0.0830** (-2.06)	-0.0333** (-2.45)	0.0836*** (4.71)
$MarketCap$	-0.0002** (-2.08)	-0.0074*** (-4.36)	-0.0080*** (-3.20)	0.0142*** (8.35)	0.0015 (0.97)

<i>DollarVolume</i>	0.0008*** (8.58)	0.0087*** (6.04)	-0.0088*** (-5.07)	0.0046*** (4.07)	-0.0053*** (-4.89)
<i>Volatility</i>	0.0251*** (5.36)	1.4708*** (14.12)	0.5122*** (3.32)	-1.0983*** (-8.51)	-0.9097*** (-10.39)
<i>QuotedSpread</i>	0.0004*** (3.41)	-0.0289*** (-6.85)	-0.0256*** (-4.14)	0.0429*** (10.20)	0.0112*** (3.89)
<i>Inverted_Share</i>	-0.0046*** (-3.11)	0.2421*** (4.58)	-0.5565*** (-4.51)	0.2880*** (5.63)	0.0309 (0.90)
Constant	0.0055** (2.21)	0.5442*** (13.86)	0.4169*** (5.71)	0.0138 (0.30)	0.0195 (0.50)
<i>N</i>	38,394	38,394	38,394	38,394	38,394
Adjusted <i>R</i> ²	0.0745	0.5057	0.6319	0.4725	0.3955

Panel B: Regression results with stock and month fixed effects

	(1) <i>OT_MO</i>	(2) <i>OT_MLO</i>	(3) <i>OT_IQL</i>	(4) <i>OT_AQL</i>	(5) <i>OT_NQL</i>
<i>Hidden_PT</i>	0.0024** (2.21)	-0.0951*** (-3.33)	0.3253*** (8.66)	-0.2754*** (-7.50)	0.0428* (2.02)
<i>Dark_Share</i>	-0.0003 (-0.30)	-0.0689** (-2.10)	-0.1830*** (-4.39)	0.2017*** (5.03)	0.0505** (2.52)
<i>Q x Event</i>	0.0007** (2.79)	0.0486*** (7.70)	-0.0594*** (-7.79)	0.0288*** (4.17)	-0.0187*** (-4.85)
<i>T x Event</i>	0.0000 (0.34)	0.0047 (1.25)	-0.0003 (-0.08)	-0.0077* (-2.04)	0.0033 (1.49)
<i>1 / Price</i>	0.0009** (2.44)	0.0401*** (3.29)	-0.0702*** (-3.56)	-0.0371*** (-4.68)	0.0663*** (6.29)
<i>MarketCap</i>	-0.0003** (-2.71)	-0.0195*** (-5.29)	0.0136** (2.75)	0.0086*** (3.01)	-0.0024 (-1.49)
<i>DollarVolume</i>	0.0010*** (6.54)	0.0260*** (14.66)	-0.0097*** (-6.44)	-0.0047*** (-3.13)	-0.0127*** (-11.45)
<i>Volatility</i>	0.0255*** (4.84)	1.1508*** (9.78)	1.0012*** (9.13)	-1.0858*** (-11.35)	-1.0917*** (-11.33)
<i>QuotedSpread</i>	0.0004** (2.59)	-0.0240*** (-6.32)	0.0070 (1.50)	0.0205*** (3.90)	-0.0039 (-1.30)
<i>Inverted_Share</i>	-0.0051*** (-3.44)	0.0137 (0.38)	-0.2006*** (-3.64)	0.2403*** (6.63)	-0.0484** (-2.23)
<i>N</i>	38,394	38,394	38,394	38,394	38,394
Adjusted <i>R</i> ²	0.1443	0.6532	0.8461	0.6947	0.6001

Table A8

Effects of hidden liquidity on cancellations, odd-lot volumes, and order-to-trade ratio

This table presents the algorithmic trading strategies results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is the ratio of the number of cancellation messages to the number of trades executed against displayed orders (*Cancellations*), the ratio of the odd-lot trading volume to the total trade volume (*OddLot*), or the ratio of order volumes to trade volumes that are executed against displayed orders (*OTR*), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders (*Hidden_PT*) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1) <i>Cancellations</i>	(2) <i>Cancellations</i>	(3) <i>OddLot</i>	(4) <i>OddLot</i>	(5) <i>OTR</i>	(6) <i>OTR</i>
$\widehat{Hidden_PT}$	160.93*** (8.55)	148.32*** (4.78)	0.4135*** (15.62)	-0.0207* (-1.80)	818.58** (2.40)	659.23 (0.97)
$\widehat{Dark_Share}$	-10.912 (-0.51)	63.881** (2.13)	-0.3956*** (-15.64)	-0.0463*** (-4.10)	49.729 (0.22)	633.65* (1.86)
$Q \times Event$	-4.1993 (-1.21)	-11.733*** (-2.61)	0.0451*** (8.17)	-0.0188*** (-5.55)	-34.113 (-0.82)	-107.15* (-1.73)
$T \times Event$	1.2142 (0.37)	5.5605 (1.55)	-0.0119*** (-3.15)	-0.0055* (-1.71)	-22.664 (-0.63)	5.8822 (0.15)
Q	1.8364 (0.82)		-0.0021 (-0.74)		3.8320 (0.14)	
T	-1.4589 (-0.54)		0.0062* (1.68)		13.521 (0.38)	
TA	3.8436 (1.27)		-0.0075* (-1.70)		1.1903 (0.03)	
$Event$	-6.7889*** (-4.78)		0.0361*** (17.13)		-54.584*** (-3.09)	
VIX	1.3555*** (6.48)		-0.0011*** (-5.66)		1.3735 (0.54)	
$1 / Price$	-44.797*** (-3.47)	-43.845*** (-4.33)	-0.0911** (-2.22)	-0.0796*** (-3.95)	-216.57** (-2.49)	-220.12*** (-3.05)
$MarketCap$	16.472*** (8.59)	11.535*** (4.88)	0.0353*** (13.95)	0.0509*** (18.23)	99.729*** (2.95)	73.648*** (2.97)

<i>DollarVolume</i>	-14.049*** (-13.47)	-24.027*** (-14.23)	-0.0285*** (-18.45)	-0.0420*** (-34.28)	-49.279*** (-2.80)	-113.68*** (-3.39)
<i>Volatility</i>	128.99 (1.41)	537.88*** (5.29)	-0.7607*** (-9.62)	-0.0331 (-0.86)	-1,013.1 (-0.77)	1,380.1 (0.81)
<i>QuotedSpread</i>	15.959*** (5.09)	15.335*** (4.09)	-0.0301*** (-9.10)	0.0016 (1.21)	172.04*** (3.69)	185.98*** (2.75)
<i>Inverted_Share</i>	-18.703 (-1.39)	21.692 (1.19)	-0.2954*** (-12.84)	-0.0501*** (-5.43)	363.30 (1.56)	742.93** (2.36)
Constant	-328.48*** (-7.14)		-0.3763*** (-6.86)		-2,120.5*** (-2.65)	
<i>N</i>	802,895	802,895	803,222	803,222	802,996	802,996
Stock and date F.E.	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.0213	0.0302	0.3748	0.6114	0.0020	0.0037

Table B1
OLS results for spreads and depth

This table presents the spreads and depth results from the following regression model: $Var_{i,t} = \beta_1 Hidden_PT_{i,t} + \beta_2 Dark_Share_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is one of quoted spread (*QuotedSpread*), effective spread (*EffectiveSpread*), realized spread (*RealizedSpread*), quoted depth (*Depth*), or the proportion of executed shares that received price improvement (*PriceImprovement*), $Hidden_PT_{i,t}$ is the percentage of transactions against hidden orders for stock i on day t , $Dark_Share_{i,t}$ is the percentage market share of dark pools for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), and the market share of inverted venues (*Inverted_Share*). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

Panel A: Regression results for spread measures						
	(1) <i>QuotedSpread</i>	(2) <i>QuotedSpread</i>	(3) <i>EffectiveSpread</i>	(4) <i>EffectiveSpread</i>	(5) <i>RealizedSpread</i>	(6) <i>RealizedSpread</i>
<i>Hidden_PT</i>	2.2230*** (39.27)	1.6681*** (47.48)	0.7051*** (31.28)	0.5843*** (39.68)	0.5954*** (36.81)	0.4828*** (40.34)
<i>Dark_Share</i>	0.1024*** (4.06)	0.2293*** (17.00)	0.0109 (0.73)	0.0081 (0.88)	0.1694*** (15.85)	0.1962*** (26.16)
<i>Q x Event</i>	0.3662*** (20.62)	0.2566*** (13.74)	0.2149*** (22.78)	0.1854*** (19.37)	0.1250*** (19.57)	0.0968*** (14.95)
<i>T x Event</i>	0.1016*** (5.11)	0.0643*** (3.10)	0.0404*** (3.65)	0.0266** (2.35)	0.0319*** (4.30)	0.0212*** (2.78)
<i>Q</i>	0.0130 (0.71)		0.0048 (0.76)		0.0013 (0.27)	
<i>T</i>	-0.0564*** (-2.73)		-0.0113 (-1.54)		-0.0105* (-1.95)	
<i>TA</i>	0.0833*** (4.49)		0.0099 (1.30)		0.0128** (2.36)	
<i>Event</i>	0.0239*** (3.54)		-0.0042 (-1.52)		0.0003 (0.10)	
<i>VIX</i>	0.0016*** (2.76)		0.0008*** (3.15)		0.0021*** (9.16)	
<i>1 / Price</i>	0.4514*** (4.85)	0.1690** (2.23)	0.4531*** (6.54)	0.2674*** (4.97)	0.3066*** (7.77)	0.1978*** (6.72)
<i>MarketCap</i>	0.0261*** (2.91)	-0.0419*** (-2.92)	-0.0102** (-2.53)	-0.0511*** (-6.86)	0.0059** (2.21)	-0.0153*** (-3.19)

<i>DollarVolume</i>	-0.2185*** (-38.41)	-0.2152*** (-41.00)	-0.0685*** (-26.67)	-0.0615*** (-27.45)	-0.0547*** (-30.12)	-0.0511*** (-31.24)
<i>Volatility</i>	12.124*** (29.60)	12.129*** (39.85)	4.4886*** (24.40)	4.2474*** (36.80)	-0.3084*** (-2.68)	-0.6943*** (-7.34)
<i>Inverted_Share</i>	-1.1298*** (-20.28)	-0.2239*** (-6.78)	-0.3664*** (-13.94)	-0.0979*** (-5.84)	-0.2803*** (-14.08)	-0.0324** (-2.44)
Constant	-0.5515*** (-3.01)		0.2174*** (2.62)		-0.1617*** (-2.94)	
<i>N</i>	804,884	804,884	804,975	804,975	803,286	803,286
Stock and date F.E.	No	Yes	No	Yes	No	Yes
Adjusted <i>R</i> ²	0.6880	0.7896	0.5445	0.6195	0.3548	0.4083

Panel B: Regression results for depth and proportion of shares that received price improvement

	(1) <i>Depth</i>	(2) <i>Depth</i>	(3) <i>PriceImprovement</i>	(4) <i>PriceImprovement</i>
<i>Hidden_PT</i>	149.92*** (4.22)	193.03*** (3.38)	0.7676*** (50.17)	0.5551*** (37.55)
<i>Dark_Share</i>	52.244*** (3.08)	4.7091 (0.46)	-0.0721*** (-5.27)	0.0256** (2.16)
<i>Q x Event</i>	46.74*** (6.20)	56.667*** (7.38)	-0.0324*** (-9.22)	-0.0509*** (-14.25)
<i>T x Event</i>	21.295* (1.80)	24.026* (1.91)	0.0070*** (2.75)	0.0105*** (3.86)
<i>Q</i>	-1.2590 (-0.89)		0.0025 (1.20)	
<i>T</i>	-1.4335 (-0.27)		-0.0045* (-1.82)	
<i>TA</i>	3.3975 (0.34)		0.0062*** (2.60)	
<i>Event</i>	-1.6983 (-0.55)		-0.0015 (-0.55)	
<i>VIX</i>	-0.0963 (-0.59)		0.0011*** (3.21)	
<i>1 / Price</i>	74.278*** (3.66)	14.926** (2.44)	-0.0019 (-0.34)	-0.0159*** (-3.45)
<i>MarketCap</i>	-5.0707 (-1.55)	-22.666*** (-3.11)	0.0085*** (6.96)	0.0098*** (6.10)
<i>DollarVolume</i>	18.508*** (5.87)	22.320*** (4.72)	0.0007 (0.78)	0.0007 (0.71)
<i>Volatility</i>	-1,032.5***	-1,351.7***	-0.1009	-0.1235

<i>QuotedSpread</i>	(-3.91)	(-3.51)	(-1.29)	(-1.46)
			-0.0283***	-0.0056
<i>Inverted_Share</i>	210.30***	125.87***	(-9.92)	(-1.64)
	(4.44)	(5.01)	0.1329***	0.1165***
Constant	62.880		(3.79)	(4.85)
	(1.02)		-0.1738***	
			(-5.78)	
<i>N</i>	804,884	804,884	40,038	40,038
Stock and date F.E.	No	Yes	No	Yes
Adjusted R^2	0.0333	0.1320	0.7414	0.8415

Table B2

OLS results for informed trading and price impact

This table presents the informed trading and price impact results from the following regression model: $Var_{i,t} = \beta_1 Hidden_PT_{i,t} + \beta_2 Dark_Share_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is the probability of informed trading (PIN), the price nonsynchronicity measure based on the Fama-French 3 factors (PSI_FF3), the price nonsynchronicity measure based on the market and industry returns (PSI_BSW), or the price impact ($PriceImpact$), $Hidden_PT_{i,t}$ is the percentage of transactions against hidden orders for stock i on day t , $Dark_Share_{i,t}$ is the percentage market share of dark pools for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (VIX), the inverse of the closing price ($1 / Price$), the log of market capitalization ($MarketCap$), the log of trading volume in million U.S. dollar ($DollarVolume$), daily return volatility ($Volatility$), quoted spread ($QuotedSpread$), and the market share of inverted venues ($Inverted_Share$). N indicates the sample size. Standard errors are clustered by firm and date in columns 1, 2, 7, and 8, and by firm in columns 3, 4, 5, and 6. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1) <i>PIN</i>	(2) <i>PIN</i>	(3) <i>PSI_FF3</i>	(4) <i>PSI_FF3</i>	(5) <i>PSI_BSW</i>	(6) <i>PSI_BSW</i>	(7) <i>PriceImpact</i>	(8) <i>PriceImpact</i>
<i>Hidden_PT</i>	0.0679*** (5.65)	0.0352** (2.39)	0.4216** (2.25)	0.1566 (0.63)	0.4274* (1.86)	0.1628 (0.53)	0.1098*** (11.97)	0.1017*** (14.38)
<i>Dark_Share</i>	-0.0440*** (-3.97)	-0.0403*** (-2.86)	3.1602*** (16.05)	2.5586*** (11.57)	3.6993*** (14.79)	2.1913*** (8.17)	-0.1582*** (-22.25)	-0.1878*** (-32.61)
<i>Q x Event</i>	0.0192*** (5.33)	0.0160*** (3.95)	0.0705 (1.28)	-0.0194 (-0.32)	0.0225 (0.33)	0.0168 (0.23)	0.0897*** (19.86)	0.0884*** (19.43)
<i>T x Event</i>	-0.0007 (-0.18)	-0.0027 (-0.67)	0.2369*** (4.86)	0.1250** (2.52)	0.3122*** (5.28)	0.1451** (2.53)	0.0086* (1.68)	0.0054 (1.06)
<i>Q</i>	-0.0005 (-0.17)		-0.0184 (-0.47)		-0.0190 (-0.39)		0.0035 (1.36)	
<i>T</i>	-0.0007 (-0.22)		-0.0865* (-1.75)		-0.0811 (-1.34)		-0.0008 (-0.26)	
<i>TA</i>	0.0010 (0.37)		0.2301*** (4.82)		0.2538*** (4.37)		-0.0029 (-0.91)	
<i>Event</i>	0.0043 (1.55)		-0.1045*** (-3.97)		0.1963*** (6.26)		-0.0044*** (-3.21)	
<i>VIX</i>	0.0007** (2.19)		-0.0716*** (-21.89)		-0.0503*** (-12.75)		-0.0013*** (-7.93)	
<i>1 / Price</i>	0.0097 (1.48)	0.0152** (2.38)	-0.0295 (-0.26)	0.3839** (2.36)	-0.2405* (-1.66)	0.4347*** (2.72)	0.1463*** (4.76)	0.0694*** (2.72)
<i>MarketCap</i>	-0.0115*** (-6.83)	-0.0144*** (-5.44)	-0.4201*** (-17.06)	-0.3271*** (-8.99)	-0.5008*** (-17.71)	-0.3137*** (-7.59)	-0.0161*** (-8.53)	-0.0360*** (-9.48)

<i>DollarVolume</i>	-0.0120*** (-10.57)	-0.0013 (-0.84)	0.2079*** (12.35)	0.2460*** (11.58)	0.1733*** (8.66)	0.2335*** (9.37)	-0.0138*** (-12.23)	-0.0105*** (-9.13)
<i>Volatility</i>	-1.2818*** (-11.45)	-1.1737*** (-7.13)					4.7946*** (40.91)	4.9403*** (52.59)
<i>QuotedSpread</i>	0.0192*** (6.53)	0.0238*** (6.27)	0.3571*** (9.36)	0.3145*** (6.68)	0.3819*** (8.16)	0.3483*** (6.43)		
<i>Inverted_Share</i>	-0.1309*** (-6.29)	-0.1082*** (-5.07)	-1.4464*** (-4.07)	0.0523 (0.14)	-1.3148*** (-3.05)	-0.2995 (-0.69)	-0.0857*** (-6.84)	-0.0651*** (-6.40)
Constant	0.4098*** (11.91)		9.3203*** (17.79)		10.558*** (17.39)		0.3794*** (9.71)	
<i>N</i>	39,917	39,917	13,436	13,436	13,171	13,170	803,286	803,286
Stock and date F.E.	No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.2160	0.2763	0.3996	0.5747	0.4119	0.5804	0.2636	0.2973

Table B3

OLS results for other measures of informed trading

This table presents the pricing efficiency results from the following regression model: $Var_{i,t} = \beta_1 Hidden_PT_{i,t} + \beta_2 Dark_Share_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is one of the variance ratio (VR), the return predictability of order imbalance ($OIB_Predictability$), or the return autocorrelation ($Autocorrelation$), $Hidden_PT_{i,t}$ is the percentage of transactions against hidden orders for stock i on day t , $Dark_Share_{i,t}$ is the percentage market share of dark pools for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (VIX), the inverse of the closing price ($1 / Price$), the log of market capitalization ($MarketCap$), the log of trading volume in million U.S. dollar ($DollarVolume$), daily return volatility ($Volatility$), quoted spread ($QuotedSpread$), and the market share of inverted venues ($Inverted_Share$). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1) VR	(2) VR	(3) <i>OIB_Predictability</i>	(4) <i>OIB_Predictability</i>	(5) <i>Autocorrelation</i>	(6) <i>Autocorrelation</i>
<i>Hidden_PT</i>	0.0630*** (15.48)	0.0438*** (13.04)	2.0453*** (25.68)	0.3396*** (5.94)	1.3770*** (16.84)	0.4248*** (7.15)
<i>Dark_Share</i>	0.0033 (1.34)	0.0004 (0.19)	-1.2698*** (-21.20)	-0.7221*** (-18.27)	-0.1869*** (-3.27)	0.1630*** (4.74)
<i>Q x Event</i>	0.0027 (1.26)	0.0020 (0.91)	0.0430 (1.64)	-0.1805*** (-7.20)	0.0743* (1.70)	-0.0658 (-1.50)
<i>T x Event</i>	-0.0007 (-0.29)	-0.0019 (-0.82)	0.0402 (1.53)	-0.0292 (-1.13)	0.0962* (1.91)	0.0296 (0.60)
<i>Q</i>	0.0010 (0.59)		0.0113 (0.50)		0.0069 (0.29)	
<i>T</i>	-0.0012 (-0.58)		-0.0341 (-1.21)		-0.0564* (-1.80)	
<i>TA</i>	0.0020 (1.21)		0.0541* (1.87)		0.1527*** (4.50)	
<i>Event</i>	-0.0008 (-0.64)		-0.0121 (-0.78)		-0.0477*** (-2.90)	
<i>VIX</i>	-0.0001 (-0.73)		0.0020 (1.10)		-0.0141*** (-7.74)	
<i>1 / Price</i>	-0.0155*** (-4.03)	-0.0087** (-2.29)	-0.9544*** (-5.30)	-0.5788*** (-6.52)	-0.5101*** (-3.29)	-0.2687*** (-3.74)
<i>MarketCap</i>	0.0026*** (3.42)	-0.0020 (-1.42)	-0.0369*** (-3.09)	0.0020 (0.10)	-0.0562*** (-4.01)	0.0232 (0.87)
<i>DollarVolume</i>	-0.0126*** (-25.61)	-0.0081*** (-16.09)	-0.3962*** (-46.39)	-0.4354*** (-51.89)	-0.1731*** (-19.25)	-0.1994*** (-21.33)

<i>Volatility</i>			-6.5738*** (-11.05)	5.8447*** (11.90)	-2.4006*** (-3.54)	9.9674*** (12.29)
<i>QuotedSpread</i>	-0.0044*** (-5.48)	-0.0047*** (-5.88)	0.2961*** (16.86)	0.1226*** (7.88)	0.0719*** (3.29)	-0.1730*** (-8.18)
<i>Inverted_Share</i>	-0.0168*** (-2.87)	-0.0079 (-1.55)	-2.1599*** (-19.79)	-0.5268*** (-6.67)	-2.4145*** (-18.50)	-0.9800*** (-10.27)
Constant	0.0964*** (6.36)		-3.4985*** (-14.31)		-3.8235*** (-13.30)	
<i>N</i>	803,331	803,331	713,943	713,943	787,042	787,042
Stock and date F.E.	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.0234	0.0460	0.1261	0.1486	0.0446	0.0739

Table B4

OLS results for trading activity

This table presents the trading activity results the following regression model: $Var_{i,t} = \beta_1 Hidden_PT_{i,t} + \beta_2 Dark_Share_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is one of log of trading volume in million U.S. dollar (*DollarVolume*), the log of trade size in U.S. dollars (*TradeSize*), or the log of the number of trades (*Num_Trades*), *Hidden_PT_{i,t}* is the percentage of transactions against hidden orders for stock i on day t , *Dark_Share_{i,t}* is the percentage market share of dark pools for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1) <i>DollarVolume</i>	(2) <i>DollarVolume</i>	(3) <i>TradeSize</i>	(4) <i>TradeSize</i>	(5) <i>Num_Trades</i>	(6) <i>Num_Trades</i>
<i>Hidden_PT</i>	0.5593*** (5.70)	0.5983*** (11.87)	1.6980*** (19.42)	0.2610*** (8.83)	-1.1387*** (-15.34)	0.3372*** (10.59)
<i>Dark_Share</i>	1.4178*** (23.92)	1.2409*** (46.27)	0.5593*** (12.13)	0.9533*** (59.53)	0.8585*** (18.94)	0.2875*** (17.03)
<i>Q x Event</i>	0.0786** (2.19)	0.0683** (2.08)	0.3169*** (16.77)	0.1411*** (8.74)	-0.2383*** (-8.22)	-0.0728*** (-2.92)
<i>T x Event</i>	0.1408*** (3.76)	0.1748*** (4.96)	0.1405*** (8.89)	0.1020*** (6.28)	0.0003 (0.01)	0.0727*** (2.64)
<i>Q</i>	0.0266 (0.64)		0.0133 (0.60)		0.0133 (0.41)	
<i>T</i>	-0.0664 (-1.27)		-0.0696** (-2.51)		0.0031 (0.08)	
<i>TA</i>	0.1127** (2.25)		0.1418*** (5.30)		-0.0292 (-0.71)	
<i>Event</i>	0.1681*** (8.49)		0.0244** (2.51)		0.1437*** (9.37)	
<i>VIX</i>	-0.0079*** (-3.79)		0.0004 (0.48)		-0.0083*** (-4.54)	
$1 / Price$	-0.6646*** (-4.01)	-0.8072*** (-3.96)	-1.2367*** (-5.22)	-0.8032*** (-5.48)	0.5721*** (5.42)	-0.0039 (-0.05)

<i>MarketCap</i>	1.0534*** (52.17)	0.4088*** (13.69)	0.3618*** (29.62)	0.1647*** (9.08)	0.6916*** (47.09)	0.2441*** (12.60)
<i>Volatility</i>	31.073*** (39.73)	27.492*** (52.62)	-4.8585*** (-12.20)	2.4153*** (12.32)	35.931*** (55.38)	25.076*** (60.72)
<i>QuotedSpread</i>	-0.8936*** (-31.52)	-0.6590*** (-32.92)	-0.1837*** (-12.66)	-0.1534*** (-20.53)	-0.7099*** (-34.37)	-0.5056*** (-33.06)
<i>Inverted_Share</i>	-0.5386*** (-4.07)	-0.8900*** (-12.44)	-1.9237*** (-23.58)	-0.8767*** (-22.76)	1.3850*** (13.39)	-0.0133 (-0.25)
Constant	-21.076*** (-50.50)		0.2789 (1.09)		-7.5395*** (-24.81)	
<i>N</i>	804,884	804,884	804,884	804,884	804,884	804,884
Stock and day F.E.	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.7685	0.8831	0.6171	0.8359	0.7394	0.8797

Table B5
OLS results for order aggressiveness

This table presents the order aggressiveness results from the following regression model: $Var_{i,t} = \beta_1 Hidden_PT_{i,t} + \beta_2 Dark_Share_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is the proportion of shares of trades executed as one of market (OT_MO), marketable limit (OT_MLO), inside-the-quote limit orders (OT_IQL), at-the-quote limit orders (OT_AQL), or near-the-quote limit orders (OT_NQL). $Hidden_PT_{i,t}$ is the percentage of transactions against hidden orders for stock i on day t , $Dark_Share_{i,t}$ is the percentage market share of dark pools for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 2015 to September 2016, and the pilot period is from November 2016 to October 2017. Control variables include the CBOE Volatility Index (VIX), the inverse of the closing price ($1 / Price$), the log of market capitalization ($MarketCap$), the log of trading volume in million U.S. dollar ($DollarVolume$), daily return volatility ($Volatility$), quoted spread ($QuotedSpread$), and the market share of inverted venues ($Inverted_Share$). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

Panel A: Regression results without stock and month fixed effects					
	(1) OT_MO	(2) OT_MLO	(3) OT_IQL	(4) OT_AQL	(5) OT_NQL
<i>Hidden_PT</i>	-0.0011*** (-2.59)	-0.2108*** (-17.17)	0.7340*** (31.23)	-0.4325*** (-26.08)	-0.0895*** (-9.31)
<i>Dark_Share</i>	0.0041*** (8.81)	0.0452*** (2.61)	-0.2656*** (-6.95)	0.0609*** (2.72)	0.1554*** (13.51)
<i>Q x Event</i>	0.0003* (1.66)	0.0239*** (4.59)	-0.0136 (-1.55)	0.0236*** (5.18)	-0.0342*** (-9.96)
<i>T x Event</i>	0.0003** (2.46)	0.0030 (0.84)	0.0120*** (3.14)	-0.0208*** (-6.64)	0.0055** (2.34)
<i>Q</i>	0.0001 (0.52)	-0.0018 (-0.80)	0.0002 (0.06)	0.0010 (0.34)	0.0006 (0.29)
<i>T</i>	-0.0004*** (-2.98)	-0.0018 (-0.63)	-0.0063 (-1.39)	0.0101** (2.42)	-0.0016 (-0.67)
<i>TA</i>	0.0003*** (3.10)	0.0035 (1.34)	0.0146** (2.45)	-0.0253*** (-4.81)	0.0070*** (3.34)
<i>Event</i>	-0.0006*** (-2.68)	0.0104 (1.43)	0.0085 (1.59)	-0.0078** (-2.02)	-0.0104*** (-2.88)
<i>VIX</i>	-0.0001*** (-2.74)	-0.0007 (-0.73)	0.0008 (1.43)	-0.0008* (-1.83)	0.0008 (1.42)
<i>1 / Price</i>	0.0000 (0.19)	0.0480*** (4.28)	-0.1238*** (-4.55)	-0.0148* (-1.90)	0.0906*** (6.32)
<i>MarketCap</i>	-0.0003*** (-2.58)	-0.0091*** (-5.39)	-0.0016 (-0.82)	0.0117*** (7.79)	-0.0008 (-0.63)

<i>DollarVolume</i>	0.0008*** (8.89)	0.0091*** (6.23)	-0.0099*** (-5.86)	0.0051*** (4.99)	-0.0051*** (-4.59)
<i>Volatility</i>	0.0236*** (4.89)	1.5312*** (15.46)	0.3551** (2.56)	-1.0353*** (-8.45)	-0.8746*** (-9.97)
<i>QuotedSpread</i>	0.0004*** (3.14)	-0.0331*** (-9.52)	-0.0148*** (-4.73)	0.0374*** (12.30)	0.0101*** (3.78)
<i>Inverted_Share</i>	-0.0051*** (-4.14)	0.1995*** (4.62)	-0.4188*** (-4.36)	0.2367*** (5.75)	-0.0122 (-0.39)
Constant	0.0066*** (2.84)	0.5960*** (17.22)	0.2327*** (4.59)	0.0781** (2.10)	0.0865*** (2.89)
<i>N</i>	40,038	40,038	40,038	40,038	40,038
Adjusted <i>R</i> ²	0.0744	0.5160	0.6783	0.5101	0.4080

Panel B: Regression results with stock and month fixed effects

	(1) <i>OT_MO</i>	(2) <i>OT_MLO</i>	(3) <i>OT_IQL</i>	(4) <i>OT_AQL</i>	(5) <i>OT_NQL</i>
<i>Hidden_PT</i>	0.0005 (0.90)	-0.0798*** (-6.32)	0.3410*** (22.41)	-0.2541*** (-17.25)	-0.0076 (-0.67)
<i>Dark_Share</i>	0.0022*** (3.37)	-0.0551*** (-3.95)	-0.0868*** (-5.30)	0.0617*** (5.21)	0.0780*** (8.28)
<i>Q x Event</i>	0.0003 (1.61)	0.0488*** (10.65)	-0.0668*** (-13.67)	0.0430*** (11.05)	-0.0253*** (-8.45)
<i>T x Event</i>	0.0001 (1.02)	0.0060 (1.67)	0.0037 (1.07)	-0.0128*** (-3.82)	0.0030 (1.34)
<i>1 / Price</i>	0.0006** (2.10)	0.0425*** (3.95)	-0.0775*** (-5.13)	-0.0250*** (-4.42)	0.0595*** (6.44)
<i>MarketCap</i>	-0.0003** (-2.25)	-0.0183*** (-5.63)	0.0169*** (4.23)	0.0038 (1.61)	-0.0022 (-1.50)
<i>DollarVolume</i>	0.0010*** (6.48)	0.0261*** (15.16)	-0.0105*** (-7.73)	-0.0033** (-2.46)	-0.0132*** (-11.12)
<i>Volatility</i>	0.0240*** (4.30)	1.1314*** (10.06)	1.0642*** (9.23)	-1.1237*** (-11.72)	-1.0958*** (-11.11)
<i>QuotedSpread</i>	0.0006*** (3.73)	-0.0252*** (-10.28)	0.0063** (2.07)	0.0186*** (5.36)	-0.0002 (-0.07)
<i>Inverted_Share</i>	-0.0031** (-2.56)	0.0180 (0.60)	-0.1371*** (-3.53)	0.1463*** (6.94)	-0.0241 (-1.37)
<i>N</i>	40,038	40,038	40,038	40,038	40,038
Adjusted <i>R</i> ²	0.1437	0.6555	0.8547	0.7079	0.6034

Table B6

OLS results for cancellations, odd-lot volumes, and order-to-trade ratio

This table presents the algorithmic trading strategies results from the following regression model: $Var_{i,t} = \beta_1 Hidden_PT_{i,t} + \beta_2 Dark_Share_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is the ratio of the number of cancellation messages to the number of trades executed against displayed orders (*Cancellations*), the ratio of the odd-lot trading volume to the total trade volume (*OddLot*), or the ratio of order volumes to trade volumes that are executed against displayed orders (*OTR*), $Hidden_PT_{i,t}$ is the percentage of transactions against hidden orders for stock i on day t , $Dark_Share_{i,t}$ is the percentage market share of dark pools for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1) <i>Cancellations</i>	(2) <i>Cancellations</i>	(3) <i>OddLot</i>	(4) <i>OddLot</i>	(5) <i>OTR</i>	(6) <i>OTR</i>
<i>Hidden_PT</i>	168.32*** (12.57)	167.61*** (12.28)	0.2396*** (15.23)	-0.0013 (-0.23)	1,331.2*** (5.98)	1,541.7*** (6.00)
<i>Dark_Share</i>	27.830*** (5.89)	57.172*** (10.86)	-0.0607*** (-6.64)	0.0732*** (21.36)	226.13*** (4.14)	390.21*** (6.61)
<i>Q x Event</i>	-7.3238*** (-2.95)	-9.1372*** (-3.43)	-0.0030 (-0.74)	-0.0284*** (-8.59)	-7.2873 (-0.21)	-7.0517 (-0.19)
<i>T x Event</i>	2.8116 (0.92)	5.8166* (1.78)	-0.0035 (-0.94)	-0.0003 (-0.10)	-4.5275 (-0.12)	20.393 (0.51)
<i>Q</i>	1.7043 (0.78)		-0.0025 (-0.86)		2.7235 (0.10)	
<i>T</i>	-2.2018 (-0.86)		0.0023 (0.63)		4.9639 (0.14)	
<i>TA</i>	5.3581** (2.17)		0.0001 (0.01)		19.2869 (0.74)	
<i>Event</i>	-6.2558*** (-4.63)		0.0421*** (20.27)		-55.349*** (-3.11)	
<i>VIX</i>	1.4552*** (7.06)		-0.0004** (-2.29)		2.0493 (0.75)	
$1 / Price$	-49.302*** (-5.31)	-42.438*** (-4.56)	-0.1739*** (-4.55)	-0.0878*** (-4.15)	-144.40*** (-2.95)	-163.85*** (-2.77)
<i>MarketCap</i>	18.829*** (12.78)	11.284*** (7.26)	0.0536*** (22.26)	0.0563*** (19.51)	113.89*** (5.01)	67.846*** (4.32)
<i>DollarVolume</i>	-15.156***	-24.597***	-0.0344***	-0.0467***	-59.152***	-122.09***

	(-15.66)	(-18.91)	(-23.91)	(-38.71)	(-4.85)	(-5.89)
<i>Volatility</i>	145.94*	562.95***	-0.8909***	0.0934**	-433.52	1,550.3
	(1.66)	(5.69)	(-10.39)	(2.41)	(-0.39)	(1.05)
<i>QuotedSpread</i>	14.745***	11.728***	-0.0077***	-0.0031***	104.27***	90.914***
	(5.13)	(4.20)	(-3.18)	(-2.88)	(3.76)	(2.64)
<i>Inverted_Share</i>	9.2928	13.496	-0.0312**	0.0326***	447.05***	409.42**
	(1.12)	(1.50)	(-2.19)	(4.40)	(3.21)	(2.24)
Constant	-393.27***		-0.8556***		-2,560.4***	
	(-12.45)		(-17.24)		(-5.30)	
<i>N</i>	804,520	804,520	804,854	804,854	804,623	804,623
Stock and date F.E.	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.0253	0.0341	0.3698	0.6156	0.0029	0.0047

Table C1

Effects of hidden liquidity on spreads and depth, separated by *TBC* groups

This table presents the spreads and depth results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where *Var* is one of quoted spread (*QuotedSpread*), effective spread (*EffectiveSpread*), realized spread (*RealizedSpread*), quoted depth (*Depth*), or the proportion of executed shares that received price improvement (*PriceImprovement*), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders (*Hidden_PT*) for stock *i* on day *t*, $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock *i* on day *t*, Q_i equals 1 if stock *i* is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock *i* is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock *i* is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date *t* is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), and the market share of inverted venues (*Inverted_Share*). *TBC* is the mean percentage of one-second intervals with quoted spreads less than or equal to 5 cents during the day, measured during the pre-pilot period. For Panel A and Panel B, Columns (1) through (4) report the results for stocks with below median *TBC* values, whereas columns (5) through (8) report the results for stocks with above median *TBC* values. For Panel C, Columns (1) and (2) report the results for stocks with below median *TBC* values, whereas columns (3) through (4) report the results for stocks with above median *TBC* values. *N* indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

Panel A: Regression results for the quoted spread and the effective spread

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low <i>TBC</i> group				High <i>TBC</i> group			
	<i>QuotedSpread</i>	<i>QuotedSpread</i>	<i>EffectiveSpread</i>	<i>EffectiveSpread</i>	<i>QuotedSpread</i>	<i>QuotedSpread</i>	<i>EffectiveSpread</i>	<i>EffectiveSpread</i>
$\widehat{Hidden_PT}$	3.2402*** (25.75)	2.9807*** (31.50)	0.9604*** (16.54)	0.8768*** (22.84)	2.1165*** (15.70)	1.8326*** (12.41)	0.8863*** (14.46)	0.8434*** (10.40)
$\widehat{Dark_Share}$	-0.3361** (-2.20)	-0.4590*** (-4.14)	-0.1068 (-1.23)	-0.1687*** (-2.97)	0.2814*** (3.60)	0.1196 (1.22)	0.1623*** (3.71)	0.0675 (1.23)
$Q \times Event$	0.5020*** (15.08)	0.4328*** (13.51)	0.2194*** (14.04)	0.1955*** (14.87)	0.3674*** (13.33)	0.3442*** (12.33)	0.2358*** (15.19)	0.2301*** (14.72)
$T \times Event$	0.1235*** (4.61)	0.0954*** (3.36)	0.0612*** (5.02)	0.0470*** (3.85)	0.0704** (2.27)	0.0430 (1.30)	0.0270 (1.44)	0.0176 (0.90)
<i>Q</i>	0.0263 (0.83)		0.0052 (0.49)		0.0035 (0.27)		0.0052 (0.87)	
<i>T</i>	-0.1058*** (-3.04)		-0.0310** (-2.57)		-0.0229 (-1.37)		-0.0003 (-0.03)	
<i>TA</i>	0.0934*** (3.15)		0.0191* (1.76)		0.0718*** (3.93)		0.0109 (1.12)	

<i>Event</i>	-0.0031 (-0.22)		-0.0187*** (-3.03)		0.0150** (2.36)		0.0004 (0.14)	
<i>VIX</i>	0.0030*** (2.97)		0.0022*** (5.30)		0.0022*** (4.17)		0.0003 (1.09)	
<i>1 / Price</i>	1.4136*** (4.09)	0.3901** (2.16)	0.7935*** (3.55)	0.3448** (2.25)	0.4770*** (4.00)	0.2430** (2.33)	0.3811*** (5.31)	0.2616*** (4.58)
<i>MarketCap</i>	0.0044 (0.30)	-0.0527* (-1.76)	-0.0176*** (-3.16)	-0.0504*** (-3.47)	0.0266*** (2.80)	-0.0660*** (-4.26)	0.0058 (0.94)	-0.0451*** (-5.14)
<i>DollarVolume</i>	-0.1966*** (-19.24)	-0.2042*** (-26.82)	-0.0591*** (-11.09)	-0.0545*** (-14.70)	-0.1653*** (-21.44)	-0.1674*** (-19.83)	-0.0656*** (-14.87)	-0.0601*** (-14.74)
<i>Volatility</i>	14.7985*** (21.27)	13.3386*** (33.12)	4.7387*** (13.49)	4.1052*** (25.81)	6.6181*** (14.43)	6.9577*** (16.61)	3.5312*** (13.76)	3.5139*** (18.77)
<i>Inverted_Share</i>	-1.4454*** (-13.21)	-0.9360*** (-11.02)	-0.4555*** (-8.43)	-0.2623*** (-7.18)	-0.7266*** (-8.72)	-0.4596*** (-6.75)	-0.2324*** (-5.72)	-0.1575*** (-4.43)
<i>Constant</i>	-0.3407 (-1.12)		0.3054*** (2.63)		-0.5954*** (-2.91)		-0.1674 (-1.26)	
<i>N</i>	390,605	390,605	390,696	390,696	412,646	412,646	412,646	412,646
<i>Stock and day F.E.</i>	No	Yes	No	Yes	No	Yes	No	Yes
<i>Adjusted R²</i>	0.6436	0.7459	0.4854	0.5616	0.6916	0.7870	0.6022	0.6780

Panel B: Regression results for the realized spread and depth

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low <i>TBC</i> group				High <i>TBC</i> group			
	<i>RealizedSpread</i>	<i>RealizedSpread</i>	<i>Depth</i>	<i>Depth</i>	<i>RealizedSpread</i>	<i>RealizedSpread</i>	<i>Depth</i>	<i>Depth</i>
<i>Hidden_PT</i>	0.7947*** (19.04)	0.7318*** (23.41)	88.4379*** (5.70)	65.3952** (2.47)	0.6576*** (16.17)	0.6095*** (11.48)	1,021.6356*** (2.85)	1,241.4472*** (2.64)
<i>Dark_Share</i>	0.0094 (0.16)	-0.0262 (-0.63)	5.3905 (0.57)	-9.3425 (-0.57)	0.2205*** (6.79)	0.1921*** (4.60)	150.0797*** (2.61)	-44.8262 (-0.44)
<i>Q x Event</i>	0.1469*** (12.64)	0.1265*** (12.07)	36.0677*** (11.08)	34.6768*** (7.83)	0.1320*** (12.52)	0.1195*** (11.13)	35.3357* (1.82)	74.1036*** (4.03)
<i>T x Event</i>	0.0413*** (4.44)	0.0318*** (3.40)	3.8323 (0.97)	2.3435 (0.62)	0.0210* (1.68)	0.0122 (0.91)	54.3534** (2.07)	63.7186** (1.99)
<i>Q</i>	0.0026		-1.9526		0.0012		1.2947	

	(0.31)		(-1.29)		(0.27)		(0.53)	
<i>T</i>	-0.0203**		-2.5977		-0.0030		-9.5131	
	(-2.23)		(-1.48)		(-0.53)		(-1.11)	
<i>TA</i>	0.0118		7.9354**		0.0124*		13.3941	
	(1.42)		(2.35)		(1.86)		(0.83)	
<i>Event</i>	-0.0081*		-0.5774		-0.0000		-12.4034*	
	(-1.66)		(-0.70)		(-0.01)		(-1.76)	
<i>VIX</i>	0.0024***		-0.2167***		0.0021***		1.4262**	
	(6.34)		(-3.78)		(9.41)		(2.11)	
<i>1 / Price</i>	0.5660***	0.2507***	37.7812***	7.7436	0.2761***	0.2056***	63.5067***	26.1493*
	(3.86)	(2.87)	(3.16)	(0.93)	(6.10)	(5.72)	(2.74)	(1.75)
<i>MarketCap</i>	0.0018	-0.0143	0.5648	-6.4230	0.0058	-0.0229***	2.7136	-19.9887**
	(0.43)	(-1.55)	(0.36)	(-1.23)	(1.59)	(-3.87)	(0.37)	(-1.97)
<i>DollarVolume</i>	-0.0504***	-0.0493***	9.2044***	9.7147***	-0.0445***	-0.0401***	26.1296***	33.5060***
	(-13.09)	(-17.66)	(6.82)	(4.12)	(-15.71)	(-13.49)	(4.43)	(3.42)
<i>Volatility</i>	-0.1340	-0.8776***	-457.2998***	-446.0059***	-1.0347***	-1.2670***	-2,206.4927***	-3,118.8799***
	(-0.58)	(-6.62)	(-5.72)	(-2.82)	(-6.58)	(-7.61)	(-3.26)	(-3.17)
<i>Inverted_Share</i>	-0.3638***	-0.2065***	51.0191**	47.5065***	-0.2219***	-0.0834***	458.2206***	127.7912
	(-8.81)	(-6.55)	(2.10)	(2.82)	(-6.93)	(-2.87)	(3.54)	(1.43)
<i>Constant</i>	-0.0929		-11.3332		-0.1849**		-260.0696	
	(-1.04)		(-0.40)		(-2.31)		(-1.36)	
<i>N</i>	389,854	389,854	390,605	390,605	411,799	411,799	412,646	412,646
<i>Stock and day F.E.</i>	No	Yes	No	Yes	No	Yes	No	Yes
<i>Adjusted R²</i>	0.2951	0.3523	0.0706	0.1412	0.3588	0.4152	0.0444	0.1396

Panel C: Regression results for the proportion of shares that received price improvement

	(1)	(2)	(3)	(4)
	Low <i>TBC</i> group		High <i>TBC</i> group	
	<i>PriceImprovement</i>	<i>PriceImprovement</i>	<i>PriceImprovement</i>	<i>PriceImprovement</i>
<i>Hidden_PT</i>	0.7389***	0.4479***	0.5936***	0.5341***
	(12.14)	(6.35)	(10.07)	(6.88)
<i>Dark_Share</i>	-0.2125***	-0.0854*	-0.0892***	-0.0983**
	(-5.75)	(-1.87)	(-4.07)	(-2.67)

<i>Q x Event</i>	-0.0396*** (-3.85)	-0.0712*** (-7.01)	-0.0178*** (-3.30)	-0.0179** (-2.21)
<i>T x Event</i>	0.0049 (1.01)	0.0096* (1.74)	0.0015 (0.76)	0.0012 (0.52)
<i>Q</i>	0.0036 (0.88)		-0.0014 (-1.08)	
<i>T</i>	-0.0033 (-0.72)		0.0004 (0.23)	
<i>TA</i>	0.0031 (0.73)		0.0025* (1.72)	
<i>Event</i>	-0.0015 (-0.40)		-0.0027 (-1.03)	
<i>VIX</i>	0.0007* (1.70)		0.0009*** (3.87)	
<i>1 / Price</i>	0.0389 (0.75)	-0.0262 (-0.70)	-0.0193** (-2.16)	-0.0067 (-1.04)
<i>MarketCap</i>	0.0066** (2.53)	0.0129** (2.78)	0.0005 (0.65)	0.0014 (0.98)
<i>DollarVolume</i>	0.0057*** (3.41)	0.0024 (1.18)	0.0014 (1.58)	0.0032*** (3.10)
<i>Volatility</i>	-0.2741* (-1.76)	-0.4703*** (-3.05)	-0.0555 (-0.95)	-0.1537* (-1.97)
<i>QuotedSpread</i>	-0.0235*** (-2.99)	0.0031 (0.38)	-0.0091* (-1.86)	0.0011 (0.16)
<i>Inverted_Share</i>	0.0881 (1.39)	0.0083 (0.21)	0.0322 (1.25)	0.0074 (0.19)
Constant	-0.0639 (-1.13)		0.0137 (0.72)	
<i>N</i>	19,133	19,133	19,261	19,261
Stock and month F.E.	No	Yes	No	Yes
Adjusted R^2	0.5895	0.7438	0.4058	0.5798

Table C2

Effects of hidden liquidity on informed trading and price impact, separated by *TBC* groups

This table presents the informed trading and price impact results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is the probability of informed trading (*PIN*), the price nonsynchronicity measure based on the Fama-French 3 factors (*PSI_FF3*), the price nonsynchronicity measure based on the market and industry returns (*PSI_BSW*), or the price impact (*PriceImpact*), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders (*Hidden_PT*) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). *TBC* is the mean percentage of one-second intervals with quoted spreads less than or equal to 5 cents during the day, measured during the pre-pilot period. Columns (1) through (4) report the results for stocks with below median *TBC* values, whereas columns (5) through (8) report the results for stocks with above median *TBC* values. N indicates the sample size. Standard errors are clustered by firm and date in columns 1, 2, 7, and 8, and by firm in columns 3, 4, 5, and 6. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

Panel A: Regression results for *PIN* and *PSI_FF3*

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Low <i>TBC</i> group				High <i>TBC</i> group			
	<i>PIN</i>	<i>PIN</i>	<i>PSI_FF3</i>	<i>PSI_FF3</i>	<i>PIN</i>	<i>PIN</i>	<i>PSI_FF3</i>	<i>PSI_FF3</i>
$\widehat{Hidden_PT}$	0.0487** (2.27)	0.0816 (1.31)	-0.5214 (-1.30)	2.6437* (1.95)	0.1344*** (3.56)	0.1045** (2.67)	1.2541** (2.08)	0.6609 (0.67)
$\widehat{Dark_Share}$	-0.0569* (-1.95)	-0.0792* (-1.73)	2.2479*** (4.80)	-1.6249* (-1.73)	-0.0577** (-2.27)	-0.0591 (-1.13)	4.5899*** (9.05)	-1.3426 (-1.06)
$Q \times Event$	0.0220*** (3.63)	0.0263** (2.50)	0.2435*** (2.71)	0.7352*** (3.45)	0.0144** (2.25)	0.0133 (1.29)	-0.4725*** (-4.29)	0.1927 (0.94)
$T \times Event$	0.0020 (0.42)	-0.0001 (-0.01)	0.1106 (1.52)	-0.0442 (-0.56)	-0.0053 (-0.95)	-0.0059 (-0.98)	0.2633*** (3.46)	-0.0162 (-0.19)
Q	-0.0026 (-0.71)		-0.1087* (-1.75)		0.0010 (0.27)		0.0483 (0.83)	
T	0.0003 (0.06)		0.0128 (0.16)		-0.0004 (-0.08)		-0.1559** (-2.15)	
TA	0.0010 (0.24)		0.2938*** (3.97)		0.0002 (0.07)		0.1595** (2.19)	

<i>Event</i>	0.0022 (0.58)		-0.1694*** (-4.06)		0.0058* (1.94)		0.0141 (0.40)	
<i>VIX</i>	0.0005 (1.30)		-0.0791*** (-16.17)		0.0007* (1.96)		-0.0612*** (-12.08)	
<i>1 / Price</i>	-0.0032 (-0.19)	0.0002 (0.02)	0.1714 (0.46)	1.7002** (2.29)	0.0098 (1.60)	0.0174*** (3.00)	-0.3767** (-2.05)	0.4239 (1.35)
<i>MarketCap</i>	-0.0151*** (-6.93)	-0.0198*** (-3.70)	-0.4620*** (-10.82)	-0.5415*** (-7.32)	-0.0082*** (-2.91)	-0.0106** (-2.81)	-0.3860*** (-10.28)	-0.5112*** (-6.04)
<i>DollarVolume</i>	-0.0061*** (-3.91)	0.0056** (2.54)	0.2195*** (8.04)	0.2199*** (6.35)	-0.0183*** (-12.48)	-0.0085*** (-4.15)	0.2211*** (8.75)	0.3468*** (10.16)
<i>Volatility</i>	-1.5112*** (-9.77)	-1.4610*** (-6.79)			-0.9426*** (-6.70)	-0.7118*** (-3.49)		
<i>QuotedSpread</i>	0.0235*** (5.60)	0.0203*** (3.63)	0.3432*** (6.07)	-0.0253 (-0.20)	0.0149*** (3.31)	0.0232*** (3.89)	0.4233*** (4.57)	0.4873*** (3.85)
<i>Inverted_Share</i>	-0.2137*** (-6.15)	-0.1934*** (-4.48)	-3.3980*** (-6.34)	-4.0396*** (-5.15)	-0.0503 (-1.43)	-0.0602 (-0.99)	2.4392*** (3.36)	-1.5678 (-1.24)
Constant	0.5040*** (10.42)		11.0261*** (11.57)		0.3275*** (5.14)		7.4034*** (8.53)	
<i>N</i>	18,998	18,998	5,928	5,928	19,290	19,290	5,921	5,921
Stock and date F.E.	No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.2117	0.2747	0.4153	0.5856	0.1645	0.2230	0.3465	0.5507

Panel B: Regression results for *PSI_BSW* and *PriceImpact*

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		Low <i>TBC</i> group				High <i>TBC</i> group		
	<i>PSI_BSW</i>	<i>PSI_BSW</i>	<i>PriceImpact</i>	<i>PriceImpact</i>	<i>PSI_BSW</i>	<i>PSI_BSW</i>	<i>PriceImpact</i>	<i>PriceImpact</i>
<i>Hidden_PT</i>	-0.2210 (-0.47)	2.4230 (1.35)	0.1651*** (8.27)	0.1430*** (7.87)	1.7390** (2.39)	1.2922 (1.02)	0.2281*** (8.38)	0.2335*** (6.63)
<i>Dark_Share</i>	3.1509*** (5.30)	-1.6971 (-1.31)	-0.1155*** (-3.44)	-0.1414*** (-4.72)	5.3672*** (8.76)	-2.0164 (-1.36)	-0.0577*** (-3.45)	-0.1238*** (-5.44)
<i>Q x Event</i>	0.2322** (2.10)	0.7283** (2.55)	0.0723*** (11.48)	0.0687*** (11.69)	-0.5637*** (-4.20)	0.2664 (1.10)	0.1036*** (17.21)	0.1103*** (17.73)
<i>T x Event</i>	0.2236**	0.0018	0.0200***	0.0153***	0.2979***	-0.0482	0.0060	0.0055

	(2.55)	(0.02)	(3.70)	(2.81)	(3.15)	(-0.50)	(0.80)	(0.73)
<i>Q</i>	-0.1288*		0.0026		0.0724		0.0040	
	(-1.79)		(0.62)		(0.97)		(1.56)	
<i>T</i>	0.0509		-0.0106**		-0.1669*		0.0028	
	(0.55)		(-2.16)		(-1.82)		(0.78)	
<i>TA</i>	0.3270***		0.0073*		0.1732*		-0.0016	
	(3.79)		(1.85)		(1.94)		(-0.40)	
<i>Event</i>	0.1125**		-0.0105***		0.3522***		0.0005	
	(2.30)		(-4.15)		(8.26)		(0.37)	
<i>VIX</i>	-0.0568***		-0.0002		-0.0361***		-0.0018***	
	(-9.54)		(-0.94)		(-5.95)		(-8.48)	
<i>1 / Price</i>	0.1892	1.3034	0.2265***	0.0937	-0.5166***	0.5971**	0.1051***	0.0558**
	(0.41)	(1.38)	(2.90)	(1.39)	(-3.07)	(2.39)	(3.80)	(2.46)
<i>MarketCap</i>	-0.5138***	-0.5715***	-0.0194***	-0.0362***	-0.4730***	-0.5090***	-0.0001	-0.0224***
	(-10.57)	(-6.16)	(-7.95)	(-5.11)	(-10.44)	(-5.81)	(-0.02)	(-5.11)
<i>DollarVolume</i>	0.2016***	0.2146***	-0.0088***	-0.0052***	0.1497***	0.3217***	-0.0211***	-0.0201***
	(6.27)	(5.34)	(-4.64)	(-2.73)	(4.89)	(7.56)	(-11.20)	(-11.66)
<i>Volatility</i>			4.8727***	4.9870***			4.5619***	4.7769***
			(31.06)	(45.60)			(25.31)	(29.73)
<i>QuotedSpread</i>	0.3816***	0.0404			0.2511**	0.3661**		
	(5.48)	(0.24)			(2.43)	(2.47)		
<i>Inverted_Share</i>	-2.7640***	-3.8758***	-0.0914***	-0.0547***	3.1087***	-1.9482	-0.0096	-0.0731***
	(-4.21)	(-3.88)	(-4.11)	(-2.72)	(3.55)	(-1.26)	(-0.58)	(-4.00)
<i>Constant</i>	11.4048***		0.3990***		8.5873***		0.0173	
	(10.40)		(7.56)		(8.07)		(0.28)	
<i>N</i>	5,771	5,771	389,854	389,854	5,843	5,843	411,799	411,799
<i>Stock and date F.E.</i>	No	Yes	No	Yes	No	Yes	No	Yes
<i>Adjusted R²</i>	0.4240	0.5622	0.2167	0.2429	0.3585	0.5743	0.3443	0.3801

Table C3

Effects of hidden liquidity on other measures of informed trading, separated by *TBC* groups

This table presents the pricing efficiency results from regression model (7): $Var_{i,t} = \beta_1 \widehat{Hidden_PT}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where *Var* is one of the variance ratio (*VR*), the return predictability of order imbalance (*OIB_Predictability*), or the return autocorrelation (*Autocorrelation*), $\widehat{Hidden_PT}_{i,t}$ is the predicted value of the percentage of transactions against hidden orders (*Hidden_PT*) for stock *i* on day *t*, $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock *i* on day *t*, Q_i equals 1 if stock *i* is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock *i* is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock *i* is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date *t* is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). *TBC* is the mean percentage of one-second intervals with quoted spreads less than or equal to 5 cents during the day, measured during the pre-pilot period. Columns (1) through (3) report the results for stocks with below median *TBC* values, whereas columns (4) through (6) report the results for stocks with above median *TBC* values. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

Panel A: Regression results without stock and day fixed effects

	(1)	(2)	(3)	(4)	(5)	(6)
		Low <i>TBC</i> group			High <i>TBC</i> group	
	<i>VR</i>	<i>OIB_Predictability</i>	<i>Autocorrelation</i>	<i>VR</i>	<i>OIB_Predictability</i>	<i>Autocorrelation</i>
$\widehat{Hidden_PT}$	0.0406*** (4.53)	2.6045*** (14.57)	1.5704*** (7.86)	0.2015*** (14.36)	1.9673*** (8.31)	2.4196*** (7.23)
$\widehat{Dark_Share}$	0.0065 (0.64)	-0.9876*** (-4.76)	0.4371** (2.11)	-0.0040 (-0.40)	-0.9545*** (-5.74)	-0.4089* (-1.95)
$Q \times Event$	-0.0091*** (-2.91)	0.2476*** (5.41)	0.1806*** (3.52)	0.0134*** (3.94)	-0.1334*** (-3.12)	0.0595 (0.76)
$T \times Event$	-0.0051* (-1.70)	0.0342 (0.99)	-0.0058 (-0.13)	0.0061* (1.79)	0.0550 (1.53)	0.2136** (2.45)
Q	0.0035 (1.29)	0.0177 (0.60)	0.0087 (0.26)	-0.0017 (-0.85)	-0.0194 (-0.68)	0.0013 (0.04)
T	0.0014 (0.43)	-0.0846** (-2.22)	-0.0715 (-1.64)	-0.0051** (-2.13)	0.0175 (0.49)	-0.0558 (-1.26)
TA	-0.0025 (-1.03)	0.1076*** (2.86)	0.1249*** (2.91)	0.0088*** (4.19)	0.0048 (0.13)	0.1944*** (4.04)

<i>Event</i>	-0.0001 (-0.03)	-0.0888*** (-3.97)	-0.0821*** (-3.54)	-0.0030** (-2.04)	0.0384** (2.05)	-0.0296 (-1.41)
<i>VIX</i>	-0.0002 (-1.37)	0.0049** (2.17)	-0.0143*** (-6.59)	0.0001 (0.87)	-0.0009 (-0.38)	-0.0144*** (-5.66)
<i>1 / Price</i>	-0.0091* (-1.76)	-0.3493 (-0.82)	0.3667*** (2.79)	-0.0083 (-1.61)	-0.6258*** (-3.98)	-0.2149 (-1.63)
<i>MarketCap</i>	0.0008 (0.65)	-0.0336* (-1.88)	-0.1063*** (-5.15)	0.0067*** (6.20)	-0.0186 (-1.07)	0.0454** (1.97)
<i>DollarVolume</i>	-0.0141*** (-20.71)	-0.4008*** (-33.50)	-0.1572*** (-13.58)	-0.0116*** (-15.38)	-0.3981*** (-32.81)	-0.2262*** (-12.96)
<i>Volatility</i>		-6.7584*** (-9.11)	-6.9182*** (-10.11)		-1.6781* (-1.91)	7.0452*** (5.52)
<i>QuotedSpread</i>	-0.0086*** (-7.42)	0.1405*** (5.76)	0.0129 (0.55)	-0.0046** (-2.16)	0.1214** (2.29)	-0.3427*** (-3.87)
<i>Inverted_Share</i>	0.0019 (0.21)	-1.7055*** (-9.98)	-1.8710*** (-9.50)	-0.0410*** (-3.54)	-1.6696*** (-9.04)	-2.4375*** (-10.33)
<i>Constant</i>	0.1446*** (5.65)	-3.6368*** (-9.22)	-2.9293*** (-6.38)	-0.0082 (-0.34)	-4.2016*** (-10.85)	-6.0882*** (-11.66)
<i>N</i>	390,157	331,849	378,814	411,540	382,074	406,659
<i>Adjusted R²</i>	0.0242	0.1287	0.0567	0.0156	0.0715	0.0148

Panel B: Regression results with stock and day fixed effects

	(1)	(2)	(3)	(4)	(5)	(6)
		Low TBC group			High TBC group	
	<i>VR</i>	<i>OIB_Predictability</i>	<i>Autocorrelation</i>	<i>VR</i>	<i>OIB_Predictability</i>	<i>Autocorrelation</i>
<i>Hidden_PT</i>	0.0645*** (5.21)	1.2103*** (6.21)	1.1403*** (5.64)	0.1187*** (6.94)	0.8551*** (3.56)	0.3195 (0.79)
<i>Dark_Share</i>	0.0082 (0.63)	-0.4241** (-2.07)	0.3056 (1.30)	0.0002 (0.02)	-0.4175** (-2.09)	0.1714 (0.65)
<i>Q x Event</i>	-0.0046 (-1.38)	0.0046 (0.10)	0.1082** (2.00)	0.0119*** (3.28)	-0.2694*** (-6.07)	-0.0949 (-1.17)
<i>T x Event</i>	-0.0046 (-1.50)	0.0150 (0.40)	-0.0524 (-1.15)	0.0035 (1.01)	-0.0220 (-0.62)	0.1148 (1.38)

<i>1 / Price</i>	-0.0202*	-0.8222*	-0.2091	-0.0040	-0.5015***	-0.2537***
	(-1.80)	(-1.96)	(-0.94)	(-0.88)	(-6.17)	(-3.54)
<i>MarketCap</i>	-0.0023	0.0535	-0.0017	-0.0002	-0.0207	0.0050
	(-1.14)	(1.58)	(-0.04)	(-0.09)	(-0.86)	(0.13)
<i>DollarVolume</i>	-0.0109***	-0.5172***	-0.2163***	-0.0072***	-0.4214***	-0.2293***
	(-13.59)	(-36.56)	(-15.07)	(-8.57)	(-32.53)	(-12.54)
<i>Volatility</i>		3.5978***	2.1296***		9.5863***	21.3205***
		(5.55)	(2.92)		(12.33)	(14.26)
<i>QuotedSpread</i>	-0.0097***	0.0437*	-0.1282***	-0.0049**	-0.0429	-0.5149***
	(-7.23)	(1.95)	(-5.80)	(-2.19)	(-1.02)	(-6.90)
<i>Inverted_Share</i>	-0.0119	-0.6350***	-1.1888***	-0.0184	-0.1300	-0.8191***
	(-1.14)	(-3.77)	(-6.53)	(-1.45)	(-0.65)	(-3.47)
<i>N</i>	390,157	331,849	378,814	411,540	382,074	406,659
<i>Adjusted R²</i>	0.0449	0.1469	0.0798	0.0381	0.0892	0.0443

Table D1

Effects of the Tick Size Pilot Program on hidden orders

This table presents the results of the following regression model: $Hidden_Orders_{i,t} = \beta_1 Q_i \times Event_t + \beta_2 T_i \times Event_t + \beta_3 TA_i \times Event_t + \beta_4 Hidden_Orders_{i,t-1} + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where $Hidden_Orders_{i,t}$ is the percentage of hidden orders for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (VIX), the inverse of closing price ($1 / Price$), the log of market capitalization ($MarketCap$), the log of trading volume in million U.S. dollar ($DollarVolume$), daily return volatility ($Volatility$), quoted spread ($QuotedSpread$), and the market share of inverted venues ($Inverted_Share$). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1) <i>Hidden_Orders</i>	(2) <i>Hidden_Orders</i>	(3) <i>Hidden_Orders</i>	(4) <i>Hidden_Orders</i>
<i>Q x Event</i>	-0.0066*** (-2.75)	-0.0199*** (-5.29)	-0.0124*** (-5.11)	-0.0233*** (-6.28)
<i>T x Event</i>	0.0033 (1.30)	0.0034 (0.78)	0.0029 (1.14)	0.0032 (0.73)
<i>TA x Event</i>	-0.0123*** (-4.15)	-0.0301*** (-6.18)	-0.0143*** (-4.83)	-0.0299*** (-6.22)
<i>Hidden_Orders (t – 1)</i>	0.7100*** (76.88)	0.4825*** (65.76)	0.6965*** (77.71)	0.4766*** (66.44)
<i>Q</i>	0.0018 (0.80)		0.0014 (0.64)	
<i>T</i>	-0.0037 (-1.37)		-0.0036 (-1.32)	
<i>TA</i>	0.0047* (1.66)		0.0047* (1.69)	
<i>Event</i>	0.0032* (1.72)		0.0021 (1.10)	
<i>VIX</i>	-0.0009** (-2.40)		-0.0010*** (-2.61)	
<i>1 / Price</i>	-0.0681*** (-10.80)	-0.0310** (-2.56)	-0.0677*** (-9.99)	-0.0359*** (-3.18)

<i>MarketCap</i>	-0.0139*** (-13.17)	-0.0095*** (-3.50)	-0.0146*** (-13.42)	-0.0087*** (-3.23)
<i>DollarVolume</i>	-0.0064*** (-11.71)	0.0034*** (5.35)	-0.0013** (-2.06)	0.0073*** (11.49)
<i>Volatility</i>	0.5542*** (14.82)	0.3936*** (10.08)	0.2939*** (7.19)	0.1235*** (3.14)
<i>QuotedSpread</i>			0.0200*** (17.82)	0.0201*** (17.29)
<i>Inverted_Share</i>	-0.0315*** (-4.61)	0.0893*** (13.64)	0.0014 (0.20)	0.0914*** (14.30)
Constant	0.4429*** (17.58)		0.4535*** (17.70)	
<i>N</i>	397,194	397,194	397,132	397,132
Stock and day F.E.	No	Yes	No	Yes
Adjusted R^2	0.6626	0.7106	0.6663	0.7127
<i>F</i> -statistic	1,343.75	613.43	1,183.43	591.93

Table D2

Effects of hidden orders on spreads and depth

This table presents the spreads and depth results from the following regression model: $Var_{i,t} = \beta_1 \widehat{Hidden_Orders}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is one of quoted spread (*QuotedSpread*), effective spread (*EffectiveSpread*), realized spread (*RealizedSpread*), quoted depth (*Depth*), or the proportion of executed shares that received price improvement (*PriceImprovement*), $\widehat{Hidden_Orders}_{i,t}$ is the predicted value of the percentage of hidden orders (*Hidden_Orders*) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), and the market share of inverted venues (*Inverted_Share*). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

Panel A: Regression results for spread measures

	(1) <i>QuotedSpread</i>	(2) <i>QuotedSpread</i>	(3) <i>EffectiveSpread</i>	(4) <i>EffectiveSpread</i>	(5) <i>RealizedSpread</i>	(6) <i>RealizedSpread</i>
<i>Hidden_Orders</i>	0.9385*** (15.43)	0.5587*** (11.08)	0.3217*** (14.37)	0.2261*** (9.12)	0.2886*** (16.09)	0.1571*** (7.93)
<i>Dark_Share</i>	-0.0259 (-0.30)	0.2936*** (3.27)	0.0178 (0.43)	0.0850* (1.81)	0.0978*** (3.09)	0.2035*** (5.38)
<i>Q x Event</i>	0.2923*** (13.85)	0.1523*** (7.11)	0.1839*** (17.90)	0.1423*** (13.53)	0.1064*** (13.58)	0.0650*** (7.89)
<i>T x Event</i>	0.0675*** (3.00)	0.0231 (0.99)	0.0305** (2.57)	0.0143 (1.15)	0.0223** (2.53)	0.0097 (1.05)
<i>Q</i>	0.0184 (0.77)		0.0061 (0.78)		0.0039 (0.63)	
<i>T</i>	-0.0311 (-1.15)		-0.0027 (-0.30)		-0.0033 (-0.47)	
<i>TA</i>	0.0471** (1.99)		-0.0001 (-0.02)		0.0026 (0.42)	
<i>Event</i>	0.0541*** (6.42)		0.0090*** (2.98)		0.0116*** (4.00)	
<i>VIX</i>	0.0042*** (3.66)		0.0022*** (4.50)		0.0029*** (5.77)	

<i>1 / Price</i>	0.0427 (0.38)	0.2349** (2.08)	0.4452*** (6.21)	0.3572*** (4.51)	0.2804*** (6.05)	0.2483*** (4.25)
<i>MarketCap</i>	0.0446*** (3.30)	-0.0186 (-0.91)	0.0036 (0.71)	-0.0334*** (-3.32)	0.0135*** (3.51)	-0.0078 (-1.06)
<i>DollarVolume</i>	-0.2508*** (-28.56)	-0.2058*** (-31.55)	-0.0768*** (-22.51)	-0.0538*** (-19.18)	-0.0625*** (-23.82)	-0.0467*** (-19.20)
<i>Volatility</i>	12.5597*** (23.18)	13.4554*** (36.80)	4.3956*** (22.06)	4.5749*** (35.92)	-0.2991** (-2.02)	-0.3591*** (-2.72)
<i>Inverted_Share</i>	-1.5856*** (-16.59)	0.0722 (1.09)	-0.4339*** (-11.27)	0.0713** (2.46)	-0.3825*** (-12.08)	0.0840*** (3.06)
Constant	-0.8868*** (-3.03)		-0.0998 (-0.88)		-0.3213*** (-3.74)	
<i>N</i>	397,132	397,132	397,160	397,160	395,490	395,490
Stock and day F.E.	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.5848	0.7726	0.4939	0.6106	0.3027	0.3926

Panel B: Regression results for depth and proportion of shares that received price improvement

	(1) <i>Depth</i>	(2) <i>Depth</i>	(3) <i>PriceImprovement</i>	(4) <i>PriceImprovement</i>
<i>Hidden_Orders</i>	74.3209*** (6.25)	142.4487*** (5.55)	0.3483*** (18.01)	0.1565*** (7.09)
<i>Dark_Share</i>	49.5431*** (3.76)	-34.1671 (-1.04)	-0.1856*** (-2.93)	0.0825** (3.06)
<i>Q x Event</i>	41.6354*** (6.58)	50.9826*** (9.18)	-0.0784*** (-7.79)	-0.0938*** (-19.13)
<i>T x Event</i>	3.9757 (0.74)	2.2958 (0.42)	-0.0101*** (-2.78)	0.0019 (0.45)
<i>Q</i>	0.2230 (0.20)		0.0005 (0.16)	
<i>T</i>	-2.7857 (-1.54)		0.0067 (1.59)	
<i>TA</i>	5.1211* (1.74)		-0.0121*** (-2.82)	

<i>Event</i>	1.7173 (0.58)		0.0024 (1.08)	
<i>VIX</i>	-0.1308 (-1.46)		0.0037*** (9.02)	
<i>1 / Price</i>	75.3871*** (3.79)	0.3998 (0.02)	-0.1597*** (-3.15)	-0.0527** (-2.57)
<i>MarketCap</i>	5.8613*** (2.90)	-14.0251* (-1.75)	0.0092*** (3.18)	0.0100*** (4.28)
<i>DollarVolume</i>	11.4187*** (4.79)	14.7813*** (2.65)	0.0117*** (6.26)	0.0033 (1.26)
<i>Volatility</i>	-643.4355*** (-3.25)	-660.3123** (-2.40)	-1.9314*** (-10.41)	-0.3756* (-1.89)
<i>QuotedSpread</i>			0.0626*** (15.63)	0.0551*** (12.10)
<i>Inverted_Share</i>	119.4172*** (5.85)	55.3179*** (4.66)	0.1711* (1.82)	0.2168*** (6.92)
Constant	-161.2136*** (-3.99)		-0.2248*** (-3.06)	
<i>N</i>	397,132	397,132	18,475	18,475
Stock and date F.E.	No	Yes	No	Yes
Adjusted R^2	0.0766	0.2341	0.5149	0.7992

Table D3

Effects of hidden orders on informed trading and price impact

This table presents the informed trading and price impact results from the following regression model: $Var_{i,t} = \beta_1 \widehat{Hidden_Orders}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is the probability of informed trading (PIN), the price nonsynchronicity measure based on the Fama-French 3 factors (PSI_FF3), the price nonsynchronicity measure based on the market and industry returns (PSI_BSW), or the price impact ($PriceImpact$), $\widehat{Hidden_Orders}_{i,t}$ is the predicted value of the percentage hidden orders ($Hidden_Orders$) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools ($Dark_Share$) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (VIX), the inverse of the closing price ($1 / Price$), the log of market capitalization ($MarketCap$), the log of trading volume in million U.S. dollar ($DollarVolume$), daily return volatility ($Volatility$), quoted spread ($QuotedSpread$), and the market share of inverted venues ($Inverted_Share$). N indicates the sample size. Standard errors are clustered by firm and date in columns 1, 2, 7, and 8, and by firm in columns 3, 4, 5, and 6. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	<i>PIN</i>	<i>PIN</i>	<i>PSI_FF3</i>	<i>PSI_FF3</i>	<i>PSI_BSW</i>	<i>PSI_BSW</i>	<i>PriceImpact</i>	<i>PriceImpact</i>
<i>Hidden_Orders</i>	0.0615*** (4.68)	0.0288 (1.33)	0.4599** (2.34)	2.2770*** (3.28)	0.9266*** (3.70)	0.7230 (0.79)	0.0332*** (4.50)	0.0686*** (5.90)
<i>Dark_Share</i>	-0.0966*** (-4.24)	-0.0703* (-1.93)	2.9014*** (7.17)	-3.3908*** (-3.48)	3.4995*** (6.72)	-1.9855 (-1.61)	-0.0794*** (-4.40)	-0.1170*** (-5.11)
<i>Q x Event</i>	0.0108** (2.02)	0.0054 (0.81)	0.2195*** (2.75)	0.6484*** (4.70)	0.1922* (1.92)	0.4205** (2.44)	0.0772*** (17.39)	0.0769*** (16.67)
<i>T x Event</i>	-0.0008 (-0.15)	-0.0026 (-0.42)	0.1829** (2.50)	-0.1135 (-1.40)	0.2588*** (2.85)	-0.0775 (-0.78)	0.0082* (1.73)	0.0047 (0.95)
<i>Q</i>	0.0031 (1.33)		-0.0518 (-0.90)		-0.0169 (-0.22)		0.0021 (0.78)	
<i>T</i>	-0.0001 (-0.03)		-0.0447 (-0.64)		-0.0648 (-0.72)		0.0007 (0.23)	
<i>TA</i>	-0.0035 (-0.93)		0.2640*** (4.47)		0.2826*** (3.84)		-0.0030 (-0.99)	
<i>Event</i>	0.0087** (1.99)		-0.2452*** (-7.18)		0.0907** (2.20)		-0.0023* (-1.75)	
<i>VIX</i>	0.0017* (1.89)		-0.1804*** (-14.83)		-0.0730*** (-4.72)		-0.0007*** (-3.12)	

<i>1 / Price</i>	0.0137 (1.44)	0.0378* (2.05)	0.1835 (0.73)	1.1977*** (3.78)	-0.0729 (-0.24)	0.4996 (1.34)	0.1647*** (5.82)	0.1088*** (3.86)
<i>MarketCap</i>	-0.0122*** (-6.17)	-0.0109** (-2.72)	-0.3624*** (-10.30)	-0.3787*** (-5.24)	-0.4193*** (-10.02)	-0.3571*** (-4.40)	-0.0099*** (-4.74)	-0.0257*** (-5.49)
<i>DollarVolume</i>	-0.0094*** (-6.36)	0.0033 (1.66)	0.2025*** (8.87)	0.3177*** (7.38)	0.1635*** (5.83)	0.3366*** (6.72)	-0.0141*** (-11.26)	-0.0071*** (-5.11)
<i>Volatility</i>	-1.6516*** (-14.06)	-1.3417*** (-7.15)					4.6903*** (38.79)	4.9338*** (45.65)
<i>QuotedSpread</i>	0.0282*** (10.98)	0.0276*** (6.41)	0.2758*** (6.29)	-0.0434 (-0.47)	0.2784*** (5.29)	0.0340 (0.33)		
<i>Inverted_Share</i>	-0.1484*** (-5.74)	-0.0753** (-2.66)	-3.3063*** (-6.13)	-4.7736*** (-4.66)	-3.0007*** (-4.61)	-2.8927** (-2.30)	-0.0509*** (-2.74)	-0.0116 (-0.67)
Constant	0.4071*** (7.82)		9.6889*** (11.65)		9.0868*** (9.14)		0.2223*** (4.75)	
<i>N</i>	18,373	18,373	5,060	5,057	4,961	4,954	395,490	395,490
Stock and date F.E.	No	Yes	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.2183	0.2882	0.3858	0.5644	0.3780	0.5708	0.2429	0.2810

Table D4

Effects of hidden orders on other measures of informed trading

This table presents the pricing efficiency results from the following regression model: $Var_{i,t} = \beta_1 \widehat{Hidden_Orders}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is one of variance ratios (VR), return predictability of order imbalance ($OIB_Predictability$), or return autocorrelation ($Autocorrelation$), $\widehat{Hidden_Orders}_{i,t}$ is the predicted value of the percentage of hidden orders ($Hidden_Orders$) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools ($Dark_Share$) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (VIX), the inverse of the closing price ($1 / Price$), the log of market capitalization ($MarketCap$), the log of trading volume in million U.S. dollar ($DollarVolume$), daily return volatility ($Volatility$), quoted spread ($QuotedSpread$), and the market share of inverted venues ($Inverted_Share$). Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1) VR	(2) VR	(3) <i>OIB_Predictability</i>	(4) <i>OIB_Predictability</i>	(5) <i>Autocorrelation</i>	(6) <i>Autocorrelation</i>
<i>Hidden_Orders</i>	0.0767*** (13.87)	0.0644*** (8.71)	0.9339*** (9.56)	-0.0255 (-0.25)	1.2089*** (13.16)	0.6864*** (5.10)
<i>Dark_Share</i>	-0.0072 (-0.80)	-0.0032 (-0.27)	-0.9442*** (-4.67)	-0.0157 (-0.08)	-0.3911* (-1.93)	0.0821 (0.33)
<i>Q x Event</i>	0.0011 (0.44)	0.0002 (0.07)	-0.0874** (-2.46)	-0.2162*** (-6.36)	-0.0286 (-0.57)	-0.1146** (-2.15)
<i>T x Event</i>	0.0006 (0.23)	0.0003 (0.12)	0.0071 (0.21)	-0.0277 (-0.86)	0.0946* (1.72)	0.0459 (0.84)
<i>Q</i>	0.0003 (0.13)		-0.0054 (-0.20)		-0.0127 (-0.44)	
<i>T</i>	-0.0004 (-0.17)		-0.0107 (-0.33)		-0.0258 (-0.71)	
<i>TA</i>	0.0004 (0.23)		0.0067 (0.21)		0.1149*** (3.08)	
<i>Event</i>	-0.0029* (-1.82)		0.0031 (0.18)		-0.0145 (-0.77)	
<i>VIX</i>	-0.0004 (-1.38)		0.0033 (1.23)		-0.0127*** (-4.44)	
<i>1 / Price</i>	-0.0273***	-0.0157**	-1.9409***	-0.6788***	-1.3158***	-0.4336***

	(-4.60)	(-2.57)	(-5.09)	(-3.15)	(-4.78)	(-2.64)
<i>MarketCap</i>	0.0039***	-0.0049***	-0.0583***	0.0601*	-0.0592***	0.0405
	(3.67)	(-2.63)	(-3.38)	(1.84)	(-3.01)	(0.99)
<i>DollarVolume</i>	-0.0121***	-0.0080***	-0.3680***	-0.4733***	-0.1523***	-0.1996***
	(-20.00)	(-11.49)	(-27.44)	(-37.10)	(-12.77)	(-13.89)
<i>Volatility</i>			-9.4485***	7.1514***	-3.6337***	12.3850***
			(-10.38)	(10.33)	(-3.70)	(11.86)
<i>QuotedSpread</i>	-0.0032***	-0.0031***	0.5071***	0.0844***	0.1683***	-0.1895***
	(-3.21)	(-3.00)	(22.58)	(4.11)	(6.24)	(-7.59)
<i>Inverted_Share</i>	-0.0255***	-0.0125	-1.6545***	0.1965	-2.6746***	-1.0900***
	(-2.99)	(-1.29)	(-9.02)	(1.22)	(-12.21)	(-5.62)
Constant	0.0500**		-3.3283***		-4.0205***	
	(2.09)		(-8.53)		(-8.90)	
<i>N</i>	396,646	396,646	359,909	359,909	388,500	388,500
Stock and day F.E.	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.0258	0.0523	0.1201	0.1479	0.0465	0.0804

Table D5

Effects of hidden orders on trading activity

This table presents the trading activity results from the following regression model: $Var_{i,t} = \beta_1 \widehat{Hidden_Orders}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is one of log of trading volume in million U.S. dollar (*DollarVolume*), log of trade size in U.S. dollars (*TradeSize*), or log of the number of trades (*Num_Trades*), $\widehat{Hidden_Orders}_{i,t}$ is the predicted value of the percentage of hidden orders (*Hidden_Orders*) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1) <i>DollarVolume</i>	(2) <i>DollarVolume</i>	(3) <i>TradeSize</i>	(4) <i>TradeSize</i>	(5) <i>Num_Trades</i>	(6) <i>Num_Trades</i>
<i>Hidden_Orders</i>	-0.9717*** (-8.17)	0.1325* (1.81)	0.7394*** (10.26)	0.1940*** (5.80)	-1.7110*** (-16.15)	-0.0615 (-1.10)
<i>Dark_Share</i>	2.0494*** (12.52)	1.5080*** (13.49)	0.3862*** (2.98)	0.7020*** (13.52)	1.6631*** (13.08)	0.8061*** (9.76)
<i>Q x Event</i>	-0.0457 (-1.29)	0.0156 (0.50)	0.2222*** (11.61)	0.1616*** (10.80)	-0.2679*** (-9.93)	-0.1460*** (-6.46)
<i>T x Event</i>	0.1246*** (3.59)	0.1486*** (4.67)	0.0972*** (6.46)	0.0767*** (5.28)	0.0274 (1.03)	0.0719*** (3.04)
<i>Q</i>	0.0204 (0.49)		0.0116 (0.47)		0.0088 (0.30)	
<i>T</i>	-0.0609 (-1.15)		-0.0469 (-1.56)		-0.0140 (-0.36)	
<i>TA</i>	0.0920* (1.78)		0.0931*** (3.28)		-0.0011 (-0.03)	
<i>Event</i>	0.1913*** (9.27)		0.0703*** (7.77)		0.1209*** (7.60)	
<i>VIX</i>	0.0022 (0.56)		-0.0034** (-2.36)		0.0056* (1.90)	
$1 / Price$	-1.4926***	-1.7246***	-2.4027***	-1.3230***	0.9101***	-0.4017**

	(-5.99)	(-4.64)	(-7.06)	(-5.66)	(5.13)	(-2.47)
<i>MarketCap</i>	1.0070***	0.2828***	0.3505***	0.1217***	0.6566***	0.1611***
	(43.96)	(8.14)	(22.86)	(5.86)	(37.94)	(7.33)
<i>Volatility</i>	33.5061***	29.6398***	-5.6716***	3.3634***	39.1778***	26.2763***
	(38.93)	(52.91)	(-11.70)	(18.57)	(55.33)	(58.28)
<i>QuotedSpread</i>	-0.7732***	-0.5217***	-0.0303**	-0.1099***	-0.7429***	-0.4118***
	(-30.86)	(-30.46)	(-2.53)	(-20.29)	(-37.07)	(-29.28)
<i>Inverted_Share</i>	-0.2989*	-1.0175***	-2.1496***	-1.2771***	1.8507***	0.2596***
	(-1.78)	(-10.33)	(-16.37)	(-23.26)	(13.88)	(4.00)
Constant	-19.9351***		0.6106*		-6.7302***	
	(-38.57)		(1.90)		(-16.95)	
<i>N</i>	397,132	397,132	397,132	397,132	397,132	397,132
Stock and day F.E.	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.7710	0.8916	0.5944	0.8406	0.7563	0.8940

Table D6

Effects of hidden orders on order aggressiveness

This table presents the order aggressiveness results from the following regression model: $Var_{i,t} = \beta_1 \widehat{Hidden_Orders}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is the proportion of shares executed using market (OT_MO), marketable limit (OT_MLO), inside-the-quote limit orders (OT_IQL), at-the-quote limit orders (OT_AQL), or near-the-quote limit orders (OT_NQL), respectively. $\widehat{Hidden_Orders}_{i,t}$ is the predicted value of the percentage of hidden orders (*Hidden_Orders*) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 2015 to September 2016, and the pilot period is from November 2016 to October 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

Panel A: Regression results without stock and month fixed effects					
	(1) OT_MO	(3) OT_MLO	(3) OT_IQL	(4) OT_AQL	(5) OT_NQL
<i>Hidden_Orders</i>	-0.0012*** (-4.20)	-0.0983*** (-4.04)	0.3613*** (14.79)	-0.2939*** (-8.98)	0.0321*** (3.68)
<i>Dark_Share</i>	0.0030*** (4.42)	0.0527* (1.66)	-0.3612*** (-3.60)	0.1482*** (3.21)	0.1573*** (4.76)
<i>Q x Event</i>	0.0005** (2.10)	0.0336*** (5.50)	-0.0500*** (-3.19)	0.0389*** (5.71)	-0.0230*** (-3.93)
<i>T x Event</i>	0.0002 (1.26)	0.0066** (1.99)	-0.0024 (-0.71)	-0.0113*** (-3.09)	0.0068** (2.42)
<i>Q</i>	0.0000 (0.01)	0.0002 (0.09)	-0.0019 (-0.51)	0.0024 (0.79)	-0.0008 (-0.37)
<i>T</i>	-0.0004** (-2.18)	-0.0041 (-1.28)	0.0025 (0.52)	0.0048 (1.08)	-0.0028 (-1.10)
<i>TA</i>	0.0003** (2.22)	0.0081*** (2.65)	-0.0014 (-0.26)	-0.0155*** (-2.74)	0.0085*** (3.85)
<i>Event</i>	-0.0008*** (-3.37)	0.0196*** (2.61)	0.0120** (2.34)	-0.0166*** (-3.56)	-0.0141*** (-5.87)
<i>VIX</i>	-0.0001* (-1.68)	0.0002 (0.11)	0.0009 (0.88)	-0.0009 (-0.72)	-0.0000 (-0.09)

<i>1 / Price</i>	0.0009 (1.18)	0.0878*** (2.63)	-0.3066*** (-3.97)	0.0475* (1.81)	0.1704*** (6.46)
<i>MarketCap</i>	-0.0005*** (-4.56)	-0.0088*** (-4.20)	-0.0032 (-0.81)	0.0102*** (4.09)	0.0024 (1.37)
<i>DollarVolume</i>	0.0010*** (9.71)	0.0037** (2.26)	0.0019 (0.71)	-0.0033** (-2.31)	-0.0033*** (-3.28)
<i>Volatility</i>	0.0356*** (3.08)	2.3004*** (16.42)	-1.5751*** (-6.61)	0.1487 (1.09)	-0.9097*** (-8.37)
<i>QuotedSpread</i>	0.0004** (2.46)	-0.0574*** (-11.65)	0.0646*** (16.34)	-0.0045 (-1.19)	-0.0031 (-1.36)
<i>Inverted_Share</i>	-0.0080*** (-4.09)	0.2033*** (4.37)	-0.4744*** (-2.90)	0.2739*** (3.72)	0.0052 (0.09)
Constant	0.0130*** (6.27)	0.5829*** (13.92)	0.2599** (2.30)	0.1509* (1.83)	-0.0067 (-0.14)
<i>N</i>	18,475	18,475	18,475	18,475	18,475
Adjusted <i>R</i> ²	0.0829	0.4998	0.5927	0.4566	0.3988

Panel B: Regression results with stock and month fixed effects

	(1) <i>OT_MO</i>	(2) <i>OT_MLO</i>	(3) <i>OT_IQL</i>	(4) <i>OT_AQL</i>	(5) <i>OT_NQL</i>
<i>Hidden_Orders</i>	0.0001 (0.11)	0.0269 (1.38)	0.0209 (1.08)	-0.0998*** (-4.90)	0.0519*** (3.39)
<i>Dark_Share</i>	-0.0033* (-2.06)	-0.1471** (-2.68)	-0.0613* (-1.94)	0.1603** (2.76)	0.0514** (2.61)
<i>Q x Event</i>	0.0008** (2.48)	0.0623*** (7.63)	-0.0900*** (-17.95)	0.0470*** (6.12)	-0.0201*** (-5.05)
<i>T x Event</i>	-0.0002 (-0.87)	0.0049 (1.18)	-0.0019 (-0.54)	-0.0043 (-0.99)	0.0016 (0.62)
<i>1 / Price</i>	0.0025* (1.99)	0.0423** (2.37)	-0.1036** (-2.45)	-0.0281 (-1.61)	0.0870*** (5.32)
<i>MarketCap</i>	-0.0004** (-2.91)	-0.0216*** (-5.20)	0.0161*** (3.64)	0.0044 (1.37)	0.0015 (0.76)
<i>DollarVolume</i>	0.0018***	0.0312***	-0.0107***	-0.0107***	-0.0116***

	(7.45)	(13.36)	(-4.70)	(-6.51)	(-8.85)
<i>Volatility</i>	0.0202*	0.8834***	1.1964***	-0.8925***	-1.2075***
	(1.85)	(4.19)	(8.74)	(-5.57)	(-11.64)
<i>QuotedSpread</i>	0.0008***	-0.0307***	0.0415***	-0.0097***	-0.0019
	(3.68)	(-7.87)	(13.06)	(-3.24)	(-0.57)
<i>Inverted_Share</i>	-0.0079***	-0.0739	-0.1227**	0.2165***	-0.0120
	(-3.48)	(-1.30)	(-2.52)	(3.80)	(-0.50)
<i>N</i>	18,475	18,475	18,475	18,475	18,475
<i>Adjusted R²</i>	0.1531	0.6761	0.8601	0.7018	0.6057

Table D7

Effects of hidden orders on cancellations, odd-lot volumes, and order-to-trade ratio

This table presents the algorithmic trading strategies results from the following regression model: $Var_{i,t} = \beta_1 \widehat{Hidden_Orders}_{i,t} + \beta_2 \widehat{Dark_Share}_{i,t} + \beta_3 Q_i \times Event_t + \beta_4 T_i \times Event_t + \beta_5 Q_i + \beta_6 T_i + \beta_7 TA_i + \beta_8 Event_t + Control_Variables + \varepsilon_{i,t}$, where Var is the ratio of the number of cancellation messages to the number of trades executed against displayed orders (*Cancellations*), the ratio of the odd-lot trading volume to the total trade volume (*OddLot*), or the ratio of order volumes to trade volumes that are executed against displayed orders (*OTR*), $\widehat{Hidden_Orders}_{i,t}$ is the predicted value of the percentage of hidden orders (*Hidden_Orders*) for stock i on day t , $\widehat{Dark_Share}_{i,t}$ is the predicted value of the percentage market share of dark pools (*Dark_Share*) for stock i on day t , Q_i equals 1 if stock i is subject to the quote rule during the pilot period and 0 otherwise, T_i equals 1 if stock i is subject to the trade rule during the pilot period and 0 otherwise, TA_i equals 1 if stock i is subject to the trade-at rule during the pilot period and 0 otherwise, and $Event_t$ equals 1 if date t is in the pilot period and 0 otherwise. The pre-pilot period is from October 1, 2015 to September 23, 2016, and the pilot period is from October 31, 2016 to October 31, 2017. Control variables include the CBOE Volatility Index (*VIX*), the inverse of the closing price ($1 / Price$), the log of market capitalization (*MarketCap*), the log of trading volume in million U.S. dollar (*DollarVolume*), daily return volatility (*Volatility*), quoted spread (*QuotedSpread*), and the market share of inverted venues (*Inverted_Share*). N indicates the sample size. Standard errors are clustered by firm and date. * denotes significance level at the 10% level, ** at the 5% level, and *** at the 1% level. t-statistics are reported in the parentheses.

	(1) <i>Cancellations</i>	(2) <i>Cancellations</i>	(3) <i>OddLot</i>	(4) <i>OddLot</i>	(5) <i>OTR</i>	(6) <i>OTR</i>
<i>Hidden_Orders</i>	-29.8297*** (-2.77)	-13.6476 (-0.79)	0.0750*** (6.44)	-0.0383*** (-6.11)	-482.5559** (-2.24)	-715.7446 (-1.54)
<i>Dark_Share</i>	70.8947* (1.69)	143.1931** (2.26)	-0.2572*** (-9.70)	0.0401*** (4.02)	554.2304 (1.07)	1,200.4611 (1.30)
<i>Q x Event</i>	-28.6204*** (-5.73)	-32.8834*** (-4.61)	-0.0009 (-0.24)	-0.0216*** (-7.81)	-189.3298** (-2.22)	-257.5879* (-1.86)
<i>T x Event</i>	0.6767 (0.13)	4.1534 (0.77)	-0.0159*** (-5.68)	-0.0023 (-0.89)	-70.4302 (-1.49)	-54.9657 (-1.12)
<i>Q</i>	2.0630 (1.04)		-0.0025 (-0.78)		-2.6603 (-0.15)	
<i>T</i>	0.3376 (0.10)		0.0095** (2.26)		61.8036 (1.29)	
<i>TA</i>	3.8845 (0.73)		-0.0149*** (-3.02)		-1.0890 (-0.02)	
<i>Event</i>	-5.1897*** (-4.09)		0.0298*** (15.41)		-29.5458* (-1.83)	
<i>VIX</i>	0.4689* (1.69)		-0.0005* (-1.74)		-5.4383 (-1.61)	

<i>1 / Price</i>	-126.3943*** (-5.28)	-71.8640*** (-3.76)	-0.3420*** (-5.82)	-0.1568*** (-4.69)	-822.9559** (-2.55)	-524.4347** (-2.06)
<i>MarketCap</i>	17.1650*** (5.78)	14.4759*** (3.22)	0.0400*** (16.12)	0.0459*** (15.59)	108.6049* (1.75)	89.1637** (2.06)
<i>DollarVolume</i>	-13.3701*** (-7.90)	-24.9212*** (-8.39)	-0.0307*** (-17.02)	-0.0517*** (-41.41)	-65.2108** (-1.96)	-136.9803** (-2.18)
<i>Volatility</i>	172.9354 (1.05)	825.4708*** (3.43)	-0.9599*** (-8.78)	0.1592*** (4.32)	930.4572 (0.35)	4,895.4206 (1.07)
<i>QuotedSpread</i>	34.5191*** (7.36)	26.3249*** (8.13)	0.0185*** (6.45)	0.0062*** (5.51)	305.6266*** (3.44)	273.3624*** (2.88)
<i>Inverted_Share</i>	53.8200* (1.91)	94.3707** (2.02)	-0.1279*** (-5.52)	-0.0067 (-0.70)	966.4022* (1.65)	1,528.9602 (1.50)
Constant	-324.4283*** (-4.61)		-0.4960*** (-9.21)		-2,086.5812 (-1.48)	
<i>N</i>	396,877	396,877	397,075	397,075	396,937	396,937
Stock and day F.E.	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.0086	0.0149	0.3749	0.6376	0.0017	0.0034